

稳定币发行驱动因素的时变分析与轮动效应

Time-Varying Analysis of Stablecoin Issuance Drivers: A Rolling LASSO Approach

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Abstract

This paper constructs a multi-dimensional analytical framework encompassing six categories and 89 indicators spanning macroeconomic and external markets, global crypto market indicators, BTC derivative leverage, exchange flows, stablecoin depeg and arbitrage, and on-chain structure, and employs a rolling LASSO model to conduct time-varying analysis of the drivers behind USDT, USDC, and DAI issuance changes.

本文构建了一个包含六大类别、89 个指标的多维度分析框架，覆盖宏观与外部市场、加密市场全局指标、BTC 衍生品杠杆、交易所资金流、稳定币脱锚套利、链上结构等维度，利用滚动 LASSO 模型对 USDT、USDC 和 DAI 三种主流稳定币发行量变化率的驱动因素进行时变分析。

We find significant factor rotation effects, on-chain structural factors and stablecoin peg-arbitrage factors are the two most important drivers on average; more importantly, a statistically identifiable structural break around May–June 2023 separates an early regime dominated by crypto-market leverage/speculative demand and on-chain user fundamentals from a late regime in which macro-finance (short-rate) factors and arbitrage mechanisms rise sharply in contribution, reflecting a migration from demand-pull to issuer balance-sheet expansion driven by safe-asset yields.

研究发现，稳定币发行驱动因素存在显著的轮动效应。全样本平均而言，链上结构因素与稳定币挂钩套利因素是最重要的两大驱动力；更为重要的是，2023 年 5–6 月前后存在一个统计上可识别的结构性断点，将早期由加密市场杠杆/投机需求与链上用户基本面主导的体制，与后期宏观金融（短端利率）因素和套利机制贡献显著抬升的体制区分开来，反映了驱动机制由需求拉动向发行商追逐安全资产收益的扩表推动迁移。

Frequency decomposition analysis reveals that high-frequency fluctuations are primarily driven by derivative leverage and exchange flow shocks, mid-frequency fluctuations

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by depeg arbitrage factors, and low-frequency trends by both macro external markets and on-chain structure fundamentals.

频率分解分析表明，高频波动主要由衍生品杠杆与交易所流动冲击驱动，中频波动由挂钩套利因素主导，低频趋势由宏观外部市场与链上结构基本面共同驱动。

Market regime asymmetry analysis reveals directional differences in factor contributions between expansion and contraction periods (e.g., peg-arbitrage and leverage factors contribute more during contraction), providing descriptive evidence of asymmetric adjustment effects in the stablecoin market; the differences do not reach statistical significance within the current quarterly sample.

市场状态非对称性分析显示，扩张期与收缩期的因素贡献存在方向性差异（如收缩期挂钩套利与杠杆因素贡献相对更高），为稳定币市场的非对称调整效应提供了描述性证据；在现有季度样本下，差异未达到统计显著水平。

Cross-issuer comparison reveals systematic structural differences. - DAI (crypto-collateralized, decentralized) is dominated by on-chain structure factors; - USDC (US-regulated fiat-collateralized) is most sensitive to exchange flows; - USDT (offshore, lightly regulated) is most sensitive to peg arbitrage and global market factors.

This analysis uncovers the structural impact of regulatory heterogeneity and collateral regimes on stablecoin driving mechanisms.

跨发行商对比呈现出系统性结构差异。DAI（加密抵押、去中心化）由链上结构因素主导；USDC（受美国监管法币抵押）对交易所资金流最敏感；USDT（离岸、监管宽松）对挂钩套利和全局市场因素最敏感，揭示了监管异质性与抵押模式对稳定币驱动机制的结构性影响。

Keywords: Stablecoin, Rolling LASSO, Driving Factors, Time-Varying Analysis, Frequency Decomposition, Asymmetry

稳定币, 滚动 LASSO, 驱动因素, 时变分析, 频率分解, 非对称性

1. Introduction

Stablecoins are privately issued digital liabilities pegged to fiat currencies (mostly the US dollar), widely used for exchange settlement, derivative margining, cross-chain clearing, and DeFi liquidity provision. With their efficient clearing and settlement mechanisms, stablecoins have become a key force driving the evolution of the global monetary and financial system (Dionysopoulos and Urquhart, 2024).

稳定币是以锚定法币（多为美元）为目标的私人发行数字负债，广泛用于交易所结算、衍生品保证金、跨链清算与 DeFi 流动性供给。稳定币以其高效的清算与结算机制，

成为驱动全球货币金融制度演变的关键力量 (Dionysopoulos and Urquhart, 2024)。

As shown in Figure 1, the total issuance of major stablecoins, Tether (USDT), USD Coin (USDC), and Dai (DAI), has grown roughly exponentially, as evidenced by the near-linear trajectories on a logarithmic scale; despite repeated market stress tests and depegging episodes and the attendant run risk, most have maintained their pegs. This expansion has not only reshaped the microstructure of the crypto market but also affected traditional money markets and interest-rate transmission through demand for short-term safe assets (Kim, 2022; Auer et al., 2025; Chen and Phelan, 2025). Over our sample period (2020-01-01 to 2026-07-01), USDT market capitalization rose from about \$4.3 billion to about \$184.7 billion (about 43 $\times$), USDC from about \$0.5 billion to about \$73.3 billion (about 141 $\times$), and DAI from about \$40 million to about \$4.6 billion (about 110 $\times$).

如图 1 所示, 三大主流稳定币, Tether (USDT)、USD Coin (USDC) 和 Dai (DAI) 的总发行量呈现 (近似) 指数级增长, 在对数纵轴下表现为近直线, 尽管期间经历了多次市场压力测试与脱锚事件、面临挤兑风险, 但大多能维持锚定。其扩张不仅重塑了加密市场微观结构, 也通过对短端安全资产的需求影响传统货币市场与利率传导 (Kim, 2022; Auer et al., 2025; Chen and Phelan, 2025)。在本文样本期内 (2020-01-01 至 2026-07-01), USDT 市值由约 43 亿美元上升至约 1,847 亿美元 (约 43 倍), USDC 市值由约 5 亿美元上升至约 733 亿美元 (约 141 倍), DAI 市值由约 4,000 万美元上升至约 46 亿美元 (约 110 倍)。

From a functional perspective, centralized exchanges and stablecoin issuers are highly analogous to shadow banks in credit creation and liquidity transformation, and stablecoins fit the classic definition of “shadow money.” The contemporary crypto sector has formed an internal credit hierarchy deeply embedded in the traditional monetary system, and the rise of “crypto shadow banking” can be viewed as the latest chapter of private money innovation (Olk and Miebs, 2025).

从功能视角看, 中心化交易所与稳定币发行机构在信用创造与流动性转换上高度类似于影子银行, 稳定币更符合“影子货币”的经典定义。当代加密领域已形成内部信用层级结构, 并深度嵌入传统货币体系之中, “加密影子银行”的兴起可视为私人货币创新的最新章节 (Olk and Miebs, 2025)。

Among stablecoin designs, fiat-collateralized stablecoins (e.g., USDT, USDC) rely on a trusted third-party custodian holding equivalent reserve assets and exhibit the strongest peg stability. Crypto-collateralized stablecoins (e.g., DAI) use over-collateralized crypto assets as reserves and operate through decentralized protocols, combining on-chain transparency with endogenous liquidation mechanisms. Extensive empirical evidence shows that complex algorithmic stablecoins generally underperform both collateralized models in

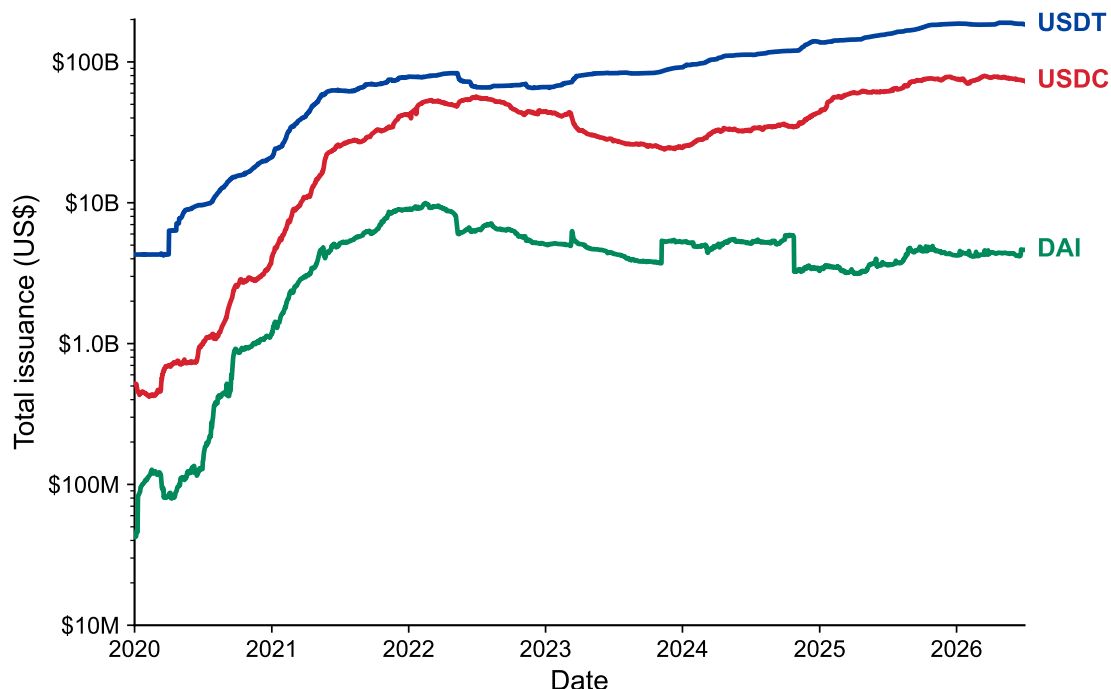


图 1: 三大稳定币总发行量的指数增长（对数纵轴）/ Exponential Growth of Major Stablecoin Issuance (Log Scale)

volatility control (De l'Etang, 2024; Duan and Urquhart, 2023; Jarno and Kołodziejczyk, 2021), while well-known collateralized stablecoins tend to display stronger resilience during or after crypto-financial crises (Gregory et al., 2024). This paper therefore focuses on the three largest collateralized stablecoins, USDT (offshore fiat), USDC (regulated fiat), and DAI (crypto-collateralized), to examine the drivers and time-variation of issuance across different collateral and regulatory regimes.

在各类稳定币设计中，法币抵押型稳定币（如 USDT、USDC）依赖受信任第三方托管机构持有等值储备资产，展现出最强的锚定稳定性；加密抵押型稳定币（如 DAI）以超额抵押的加密资产为储备、通过去中心化协议运作，兼具链上透明性与内生清算机制。大量实证研究表明，复杂算法稳定币在波动率控制方面普遍逊色于两类抵押型模式 (De l'Etang, 2024; Duan and Urquhart, 2023; Jarno and Kołodziejczyk, 2021)，而知名抵押型稳定币在加密金融危机期间或危机后往往表现出更强的抗压韧性 (Gregory et al., 2024)。因此，本文选取市值最大的三种抵押型稳定币，USDT（离岸法币抵押）、USDC（受监管法币抵押）和 DAI（加密资产抵押），以考察不同抵押模式与监管环境下的发行驱动因素及其时变特征。

Stablecoins also display remarkably high cross-border liquidity efficiency, with flow volumes positively related to traditional remittance costs. Evidence suggests that vig-

orously promoting stablecoin use in remittance scenarios could help advance the UN Sustainable Development Goal (SDG 10) of reducing cross-border payment remittance costs below 3% (Ante, 2025; Auer et al., 2025). Such cross-border flows are primarily transaction-motivated, with geographic barriers markedly weakened and traditional capital-flow management measures becoming less effective (Auer et al., 2025), potentially undermining the monetary sovereignty and exchange controls of vulnerable economies.

稳定币还展现出极高的跨境流动性效率，其流动规模与传统汇款成本呈正相关。研究显示，大力推广稳定币在汇款场景的应用，有望助力实现联合国可持续发展目标（SDG 10），即将跨境支付汇款成本降至 3% 以下 (Ante, 2025; Auer et al., 2025)。其跨境流动主要由交易动机驱动，地理壁垒影响显著减弱，传统资本流动管理措施亦趋于失效 (Auer et al., 2025)，从而可能削弱弱势经济体的货币主权与外汇管制。

In sum, stablecoin issuance profoundly affects multiple dimensions of financial markets and underpins the prosperity of the DeFi ecosystem. Disentangling the key drivers of its issuance and building a time-varying forecasting model thus carry significant value for policy, regulation, and market decision-making.

综上所述，稳定币发行深刻冲击金融市场的多个维度，并支撑 DeFi 生态的繁荣。因此，深入剖析其发行的关键驱动因素，并构建时变预测模型，具有重要的政策、监管与市场决策价值。

Regarding the mechanism behind stablecoin issuance expansion, the existing debate often falls into two narratives. The first is demand pull, in which stablecoin supply passively responds to settlement and leverage demand in the crypto market, acting like a “fund intermediary” that tokenizes inflowing fiat. The second is supply push, in which issuers earn returns by allocating reserve assets and proactively expand their balance sheets, creating an equal amount of stablecoin liabilities when they expand the asset side, a quintessential shadow-banking feature (Zoltan Pozsar et al., 2013; Olk and Miebs, 2025).

围绕稳定币发行扩张的机制识别，现有讨论常落入两种叙事。其一，需求拉动，稳定币供给被动响应加密市场的结算与杠杆需求，类似“资金中介”将流入法币代币化；其二，供给推动，稳定币发行商通过配置储备资产获取收益并主动扩张资产负债表，在资产端扩张时同步创造等量稳定币负债，具有典型影子银行特征 (Zoltan Pozsar et al., 2013; Olk and Miebs, 2025)。

We find that the issuance mechanism of stablecoins has undergone a profound evolution; its underlying logic has shifted from early “crypto-market leverage and speculative demand pull” to the current “balance-sheet expansion driven by issuers chasing returns on traditional risk-free assets.” This time-varying character confirms an upgrade in the asset-creates-liability mechanism of stablecoins, a qualitative transformation from “pri-

vate shadow money” to “public shadow money.”

本文发现，稳定币的发行机制已发生深刻演变，其底层逻辑由早期的“加密市场杠杆与投机需求拉动”转变为当前的“发行商追逐传统无风险资产收益的扩表推动”。这种时变特征印证了稳定币在资产创造负债机制上的升级迭代，即从“私人影子货币”向“公共影子货币”的深刻质变。

However, what factors determine stablecoin issuance changes? Do the relative importance of these factors dynamically shift over time? Do driving mechanisms differ significantly across time scales and market regimes? These questions remain unanswered through systematic empirical research.

然而，什么因素决定了稳定币发行量的变化？这些因素的相对重要性是否随时间动态转换？在不同时间尺度和市场状态下，驱动机制是否存在显著差异？这些问题尚未得到系统性的实证回答。

The core scientific question of this paper is, what factors determine stablecoin issuance change rates? How do these factors rotate over time?

本文的核心科学问题是，稳定币发行量变化率由什么因素决定？这些因素如何随时间轮动？

Taking this core question as the main thread, we examine which information sets can predict stablecoin net issuance and whether this predictive information structure rotates with the external macro environment, crypto-market regime, and issuer rules, that is, the mechanism-transmission channel from *external environment changes* $\rightarrow$ *market demand changes* $\rightarrow$ *transmission-channel migration*.

围绕这一核心问题，本文以“哪些信息能够预测稳定币净发行、该预测信息结构是否随外部环境及加密市场状态、发行规则而轮动”为主线，考察“外部环境变化 $\rightarrow$ 市场需求变化 $\rightarrow$ 传导渠道迁移”的机制传递链条。

Around this core question, we propose five sub-questions. (1) Factor identification, how to identify core driving factors from a multi-dimensional indicator system? (2) Time-varying nature and structural break, do the relative importance of driving factors change dynamically over time, with an identifiable statistical breakpoint separating the early demand-pull regime from the late supply-push regime? (3) Frequency heterogeneity, are stablecoin issuance fluctuations at different time scales dominated by different factors? (4) State asymmetry, do the importance of driving factors differ significantly between expansion and contraction periods? (5) Out-of-sample predictability, does the sparse information set selected by rolling LASSO possess genuine out-of-sample forecasting power?

围绕这一核心问题，我们提出五个子问题。(1) 因素识别，如何从多维度指标体系中识别核心驱动因素？(2) 时变性与结构性断点，驱动因素的相对重要性是否随时间动态

变化，是否存在可识别的统计断点将早期需求拉动体制与后期供给推动体制区分开来？(3) 频率异质性，不同时间尺度上的稳定币发行波动是否由不同因素主导？(4) 状态非对称性，在扩张期与收缩期，驱动因素的重要性是否存在显著差异？(5) 样本外可预测性，滚动 LASSO 筛选出的稀疏信息集是否具备真实的样本外预测力？

To address these questions, following the analytical framework of Dionysopoulos et al. (2025) for stablecoin growth drivers, we construct a comprehensive multi-dimensional indicator system with six categories, macroeconomics and external markets (M), global crypto market indicators (G), BTC derivative leverage (L), exchange flows (E), stablecoin depeg and arbitrage (P), and on-chain structure (S), totaling 89 explanatory variables.

为回答上述问题，本文仿照 Dionysopoulos et al. (2025) 对稳定币增长驱动因素的分析框架，构建了一个包含六大类别（宏观与外部市场 M、加密市场全局指标 G、BTC 衍生品杠杆 L、交易所资金流 E、稳定币脱锚套利 P、链上结构 S）的综合多维度指标体系，总计 89 个解释变量。

We employ a rolling LASSO regression model with a 365-day rolling window and quarter-end rolling nodes to conduct empirical analysis on daily data from January 2020 to July 2026.

我们采用滚动 LASSO 回归模型，以 365 天为滚动窗口、自然季度末为滚动节点，对 2020 年 1 月至 2026 年 7 月的日度数据进行实证分析。

This methodological choice offers three advantages; first, LASSO's regularization automatically selects key variables from high-dimensional indicators, avoiding overfitting; second, the rolling window framework captures time-varying characteristics of driving factors; third, the contribution decomposition formula $C_{j,\tau} = |\beta_j| / \sum |\beta_k|$ provides an interpretable marginal contribution measure.

这一方法选择有三方面优势。第一，LASSO 的正则化机制能够从高维指标中自动筛选关键变量，避免过拟合；第二，滚动窗口框架可以捕捉驱动因素的时变特征；第三，贡献度分解公式 $C_{j,\tau} = |\beta_j| / \sum |\beta_k|$ 提供了可解释的边际贡献度量。

This paper's marginal contributions are reflected in five aspects.

本文的边际贡献体现在五个方面。

First, we construct a comprehensive multi-dimensional stablecoin driving factor analysis framework, integrating macro external markets, global crypto indicators, derivative leverage, exchange flows, depeg arbitrage, and on-chain structure into an 89-dimensional indicator system for the first time.

第一，构建了稳定币多维度驱动因素分析框架，首次将宏观外部市场、加密全局指标、衍生品杠杆、交易所资金流、脱锚套利、链上结构等多个维度整合为 89 维指标体系。

Second, we reveal rotation effects among stablecoin driving factors and endogenously identify a structural breakpoint in the dominant-driver composition through Quandt–Andrews supF and CUSUM tests. The early period (before the breakpoint) is dominated by crypto leverage and speculative demand (Categories L and G), while the later period (after the breakpoint) sees supply-side return factors (Category M, short-term rates) rise significantly in contribution. This breakpoint aligns with both the onset of the Fed rate-hike cycle and the shift in issuer reserve composition toward Treasury bills, providing statistical evidence for the mechanism-migration narrative “external environment change → demand change → channel switch” and directly testing Hypothesis H2.

第二，揭示了稳定币驱动因素的轮动效应，并通过 Quandt–Andrews supF 检验与 CUSUM 检验内生识别了主导因素构成的结构性断点，断点前的早期由加密杠杆与投机需求（L 类、G 类）主导，断点后的后期供给侧利率收益因素（M 类，短端利率）贡献显著上升。该断点与美联储加息周期启动、发行商储备结构转向美债的时点相吻合，为“外部环境变化 → 市场需求变化 → 传导渠道切换”的机制迁移叙事提供了统计证据，直接检验了假设 H2。

Third, using SG filter-based frequency decomposition, we find that high-frequency fluctuations are dominated by derivative leverage and exchange flow shocks, mid-frequency by peg arbitrage mechanisms, and low-frequency by the joint action of peg arbitrage, on-chain structure fundamentals, and macro external markets.

第三，采用 SG 滤波进行频率分解，发现高频波动由衍生品杠杆与交易所流动冲击主导、中频由挂钩套利机制主导、低频由挂钩套利、链上结构基本面与宏观外部市场共同驱动。

Fourth, we characterize expansion–contraction asymmetry in the stablecoin market; global crypto indicators and on-chain structure contribute more during expansion, while leverage, exchange-flow, and peg-arbitrage factors rise during contraction, offering descriptive evidence of asymmetric adjustment effects.

第四，刻画了稳定币市场的扩张–收缩非对称特征，扩张期加密市场全局与链上结构因素贡献更高，收缩期杠杆、交易所资金流与挂钩套利因素相对上升，为非对称调整效应提供了描述性证据。

Fifth, through a comparative analysis of USDT, USDC, and DAI across fiat-collateralized (offshore vs regulated) and crypto-collateralized models, we reveal structural differences in driving factors along both the regulatory and collateral dimensions, providing empirical evidence for differentiated regulatory frameworks.

第五，通过 USDT、USDC、DAI 三者在法币抵押（离岸 vs 受监管）与加密抵押两个维度上的对比分析，揭示了监管梯度与抵押模式对稳定币驱动因素的双重结构性差

异，为差异化监管提供实证依据。

Sixth, we construct a Rotation Intensity Index (RII) to quantify the magnitude of inter-quarter driver restructuring, and conduct out-of-sample (OOS) rolling forecasts demonstrating that the LASSO-selected sparse information set delivers meaningful predictive power (directional hit rate, OOS R^2) relative to random-walk, historical-mean, and full-variable OLS benchmarks.

第六, 构建轮动强度指数 (RII) 量化季度间驱动因素重构幅度, 并通过样本外 (OOS) 滚动预测证明, LASSO 筛选出的稀疏信息集相对于随机游走、历史均值与全变量 OLS 基准具备显著预测力 (方向命中率、OOS R^2)。

The remainder of this paper is organized as follows.

本文其余部分安排如下。

Section 2 reviews related literature; §4 introduces data and variables; §5 describes methodology; §6 presents baseline empirical results; §7 presents evidence on mechanism evolution (structural break tests, early-vs-late comparison, transmission-channel co-movement, RII, OOS forecasts); §8 conducts frequency decomposition; §9 analyzes market regime asymmetry; §10 compares USDT, USDC, and DAI; §11 performs robustness checks (including time-varying-pattern robustness); §12 concludes.

§2 回顾相关文献; §4 介绍数据与变量体系; §5 阐述方法论; §6 呈现基准实证结果; §7 给出机制演变的实证证据 (结构性断点、早期后期对比、传导渠道联动、RII、OOS 预测); §8 进行频率分解分析; §9 分析市场状态非对称性; §10 对比 USDT、USDC 与 DAI; §11 开展稳健性检验 (含时变规律稳健性); §12 总结全文。

2. Literature Review

2.1. Stablecoin Pricing and Issuance Mechanisms / 稳定币定价与发行机制

Stablecoins are crypto-assets pegged to fiat currencies (primarily the US dollar), designed to reduce price volatility and provide a unit of account and medium of exchange for the crypto market.

稳定币是一种锚定法定货币 (主要是美元) 的加密资产, 旨在降低价格波动性, 为加密市场提供价值尺度和交易媒介。

Based on collateral mechanisms, stablecoins can be categorized into fiat-collateralized (e.g., USDT, USDC), crypto-collateralized (e.g., DAI), and algorithmic types (e.g., the collapsed UST).

根据抵押机制, 稳定币可分为法币抵押型 (如 USDT、USDC)、加密资产抵押型 (如 DAI) 和算法型 (如已崩盘的 UST)。

This paper covers both fiat-collateralized (USDT, USDC) and crypto-collateralized (DAI) stablecoins, which together account for over 95% of total market circulation. Their issuance mechanisms are relatively transparent with high data availability, making cross-model comparison possible.

本文同时覆盖法币抵押型（USDT、USDC）与加密抵押型（DAI）稳定币，二者合计占据市场总流通量的 95% 以上，且发行机制相对透明、数据可得性高，便于跨模式对比研究。

Existing stablecoin research primarily focuses on four aspects, price pegging mechanisms, reserve transparency, regulatory policy, and systemic risk.

现有关于稳定币的研究主要集中在价格锚定机制、储备透明度、监管政策和系统性风险四个方面。

Lyons and Viswanath-Natraj (2023) find a bidirectional causal relationship between stablecoin issuance and crypto market capital inflows; Griffin and Shams (2020) question Tether’s reserve audits, suggesting potential over-issuance without sufficient reserves; Barthelemy et al. (2023) note that stablecoins serve as “digital cash” in the DeFi ecosystem, with issuance changes directly reflecting crypto market liquidity demand.

Lyons and Viswanath-Natraj (2023) 发现稳定币发行与加密市场资金流入存在双向因果关系；Griffin and Shams (2020) 对 Tether 的储备审计提出质疑，认为其可能在缺乏足额储备的情况下超发；Barthelemy et al. (2023) 则指出稳定币在 DeFi 生态中扮演着“数字现金”的角色，其发行量变化直接反映了加密市场的流动性需求。

However, existing literature lacks systematic, multi-dimensional time-varying analyses of stablecoin issuance drivers, particularly empirical research simultaneously examining on-chain behavior, DeFi ecosystem, macro-finance, and market sentiment.

然而，现有文献缺乏对稳定币发行驱动因素进行系统性、多维度的时变分析，尤其缺少从链上行为、DeFi 生态、宏观金融和市场情绪四个维度同时入手的实证研究。

2.2. *LASSO in High-Dimensional Financial Analysis* / *LASSO 在高维金融分析中的应用*

LASSO (Least Absolute Shrinkage and Selection Operator), proposed by Tibshirani (1996), achieves variable selection and dimensionality reduction through ℓ_1 regularization and has been widely applied in high-dimensional financial data analysis.

LASSO (Least Absolute Shrinkage and Selection Operator) 由 Tibshirani (1996) 提出，通过 ℓ_1 正则化实现变量选择和降维，在金融高维数据分析中得到了广泛应用。

Bańbura et al. (2010) apply LASSO to large-scale macroeconomic forecasting models, demonstrating superior variable selection and prediction accuracy compared to traditional

methods; Cook et al. (2024) extend machine learning interpretation frameworks to partial marginal effect analysis; Wang et al. (2020) employ Lasso regression to construct sector-rotation trading strategies in the Chinese stock market.

Bañbura et al. (2010) 将 LASSO 应用于大型宏观经济预测模型，证明其在变量选择和预测精度上优于传统方法；Cook et al. (2024) 将机器学习解释框架扩展至部分边际效应分析；Wang et al. (2020) 利用 Lasso 回归构建了中国股市的板块轮动交易策略。

In the cryptocurrency domain, Liu et al. (2022) use LASSO to identify core factors affecting Bitcoin prices; Yi et al. (2018) combine LASSO-VAR with the Diebold–Yilmaz connectedness framework to study volatility transmission networks in crypto markets.

在加密货币领域，Liu et al. (2022) 使用 LASSO 识别了影响比特币价格的核心因素；Yi et al. (2018) 将 LASSO-VAR 与 Diebold–Yilmaz 关联度框架相结合，研究了加密市场的波动率传导网络。

Drawing on the methodologies of the above literature, this paper applies rolling LASSO to time-varying analysis of stablecoin issuance drivers for the first time, introducing a contribution decomposition formula to quantify the marginal contributions of each factor.

本文借鉴上述文献的方法论，首次将滚动 LASSO 应用于稳定币发行驱动因素的时变分析，并引入贡献度分解公式量化各因素的边际贡献。

2.3. *Decomposition of Asset Pricing Drivers* / 资产定价的驱动因素分解

Driver decomposition is one of the core methods in asset pricing research, aiming to identify and quantify various factors affecting asset price (or supply) changes.

驱动因素分解是资产定价研究的核心方法之一，旨在识别和量化影响资产价格（或供应量）变化的各种因素。

Dionysopoulos et al. (2024) conduct pioneering research on stablecoin growth drivers, using multiple empirical methods to find that stablecoin growth is primarily driven by issuer profitability—the ability to invest reserves in interest-bearing instruments—rather than usage demand. Their follow-up case study on Tether further confirms that yield differentials on reserve assets underpin stablecoin supply expansion (Dionysopoulos et al., 2025).

Dionysopoulos et al. (2024) 对稳定币增长驱动因素进行了开创性研究，利用多种实证方法发现稳定币增长主要由发行商盈利（即将储备资产投资于生息工具的能力）驱动，而非使用需求；其对 Tether 的后续案例研究进一步证实，储备资产收益率差异是稳定币供给扩张的基础 (Dionysopoulos et al., 2025)。

Parallel advances in driver-decomposition methodology offer tools to unpack these

dynamics across frequencies and market regimes. Liu et al. (2024) propose a time-varying, multi-scale framework that combines rolling LASSO with empirical mode decomposition (EMD) to analyze copper price drivers. Their approach decomposes price fluctuations into high-, medium-, and low-frequency components and finds that high-frequency movements are dominated by event shocks, medium-frequency swings by supply-demand factors, and low-frequency trends by both fundamentals and macro factors. They further document asymmetric regime effects: for example, industrial-chain factors contribute far more during rising markets than during falling markets.

驱动因素分解方法论的并行进展为从不同频率和市场状态拆解上述动态提供了工具。Liu et al. (2024) 提出了一个将滚动 LASSO 与经验模态分解 (EMD) 相结合的时变多尺度框架，用于分析铜价驱动因素。该方法将价格波动分解为高频、中频和低频成分，发现高频波动由事件冲击主导，中频波动由供需因素主导，低频趋势由基本面与宏观因素共同驱动；此外，该研究还发现显著的市场状态非对称效应，例如产业链因素在上涨市场中的贡献远高于下跌市场。

This paper draws on the methodological framework of Liu et al. (2024) and adapts it to stablecoin issuance driver analysis. We construct a comprehensive six-category (M/G/L/E/P/S), 89-variable system tailored to stablecoin characteristics and employ Savitzky–Golay (SG) filters instead of EMD for frequency decomposition, thereby providing a multi-dimensional, time-varying analysis of stablecoin issuance drivers.

本文借鉴 Liu et al. (2024) 的方法论框架，将其迁移至稳定币发行驱动因素分析，构建了适配稳定币特性的六大类别 (M/G/L/E/P/S) 89 变量综合体系，并采用 Savitzky–Golay (SG) 滤波替代 EMD 进行频率分解，从而对稳定币发行驱动因素进行多维度、时变分析。

2.4. *Drivers of Stablecoin Issuance* / 驱动稳定币发行的因素研究

The volatility of stablecoin issuance results from multiple factors acting on the issuer’s asset side (supply) and liability side (demand), moderated by market microstructure and the macro environment. On the supply side, issuers’ profit motive is a key driver, the profitability of investing reserve assets in short-term instruments (e.g., US T-Bills or reverse repos) directly determines the endogenous incentive to proactively expand the balance sheet. This paper uses short-term interest-rate proxies such as the 3-month Treasury yield (US3M) and the SOFR–RRP spread to measure issuers’ reserve income, thereby quantifying the supply-push channel; when the Fed raises rates or Treasury yields rise, higher reserve returns strengthen issuers’ balance-sheet expansion incentive and significantly promote stablecoin supply expansion (Barthelemy et al., 2023; Dionysopoulos et al.,

2025).

稳定币发行量的波动是多重因素共同作用的结果，这些因素分别从发行商资产端（供给侧）和负债端（需求侧）施加影响，同时受到市场微观结构和宏观环境的调节。在供给侧，发行商的盈利动机是重要驱动力，发行商将其储备资产投资于短期金融工具（如美国国债 T-Bills 或逆回购协议）的盈利能力，直接决定了主动扩表的内生激励。本文以 3 个月美债收益率（US3M）与 SOFR-逆回购利差等短端利率指标作为发行商储备资产收益的代理，以量化供给推动渠道；当美联储加息或美债收益率上升时，发行商储备资产收益提升，主动扩表动力增强，显著推动稳定币供给扩张 (Barthelemy et al., 2023; Dionysopoulos et al., 2025)。

On the demand side, leverage and speculative demand play key roles. Leverage demand in crypto derivative markets (especially perpetual contracts) induces investors to borrow stablecoins to amplify positions; the perpetual funding rate and stablecoin margin lending rate thus become important pull factors for stablecoin borrowing demand. Leverage ratios and open interest in crypto futures markets are important markers of speculative activity, and a high positive funding rate often heralds significant expansion of stablecoin circulation (Gorton et al., 2025; Gorton and Zhang, 2021). Meanwhile, overall crypto-market returns and trading demand also drive stablecoin issuance, directed spillover analysis based on a generalized VAR framework and forecast-error variance decomposition finds that stablecoin issuance responds mostly to price changes of other crypto assets, suggesting that issuance largely responds to rising investment demand in the crypto market, thereby satisfying demand for “digital fiat” (Kristoufek, 2021).

在需求侧，杠杆活动与投机性需求扮演关键角色。加密货币衍生品市场（尤其是永续合约）的杠杆需求促使投资者借入稳定币以放大头寸，因此永续资金费率与稳定币保证金借贷利率成为稳定币借贷需求的重要拉动因素；加密资产期货市场的杠杆率与未平仓合约是反映投机活动的重要标志，高正向资金费率往往预示稳定币流通量的显著扩张 (Gorton et al., 2025; Gorton and Zhang, 2021)。与此同时，加密市场整体收益和交易需求也驱动稳定币增发，基于广义向量自回归框架和预测误差方差分解的定向溢出分析，研究发现稳定币发行更多是对其他加密资产价格变动的响应，表明其发行更多是响应投资加密市场需求上升，进而满足市场对“数字化法币”的需求 (Kristoufek, 2021)。

Safe-haven and liquidity demand also matter on the liability side (Łęt et al., 2023). When traditional crypto assets fluctuate sharply (e.g., a rise in Bitcoin implied volatility BTC IV), funds flow toward stablecoins for shelter, increasing their popularity (Baur and Hoang, 2021). Empirical evidence shows that Tether’s safe-haven effect is the most pronounced, during episodes of extreme negative Bitcoin price moves, the scale of investor flows into stablecoins increases significantly; precisely because stablecoins react to

such volatility, however, their price stability is not constant (Hoang and Baur, 2024). Geopolitics, regulation, and government corruption are also important demand channels, in regions with geopolitical conflict, strict regulation, severe corruption, or unstable fiat currencies, stablecoin demand is significantly elevated (Auer et al., 2025; Bhimani et al., 2022; Gueorguiev et al., 2022; Saiedi et al., 2021).

避险与流动性需求同样在负债端发挥重要作用 (Łet et al., 2023)。当传统加密资产剧烈波动时（例如比特币隐含波动率 BTC IV 上升），资金会向稳定币流动以寻求庇护，从而提升其受欢迎程度 (Baur and Hoang, 2021)。实证证据显示，Tether 的避险效应最为显著，在比特币出现极端负向价格波动期间，投资者转向稳定币的规模显著增加；然而，正因稳定币对此类波动作出反应，其价格稳定并非始终如一 (Hoang and Baur, 2024)。地缘政治、监管政策与政府腐败同样是影响需求的重要渠道，在地缘政治冲突、监管严格、腐败严重、本币不稳定的地区，稳定币需求被显著推高 (Auer et al., 2025; Bhimani et al., 2022; Gueorguiev et al., 2022; Saiedi et al., 2021)。

Microstructure-level arbitrage is key to maintaining the peg and regulating supply. When the secondary-market price of a stablecoin deviates from \$1 (premium or discount), the price deviation directly drives arbitrageurs to mint or redeem at the primary market, and arbitrage designs of differing centralization significantly affect the efficiency of this process (Lyons and Viswanath-Natraj, 2023; Ma et al., 2023).

微观结构层面的套利机制则是维持锚定和调节供给的关键。当稳定币二级市场价格偏离 1 美元（出现溢价或折价）时，价格偏离度会直接驱动套利者在一级市场进行申购或赎回；中心化程度不同的套利设计显著影响这一过程的效率 (Lyons and Viswanath-Natraj, 2023; Ma et al., 2023)。

Finally, global demographic age structure and education levels also affect long-term stablecoin adoption. Survey data on cross-border remittance users show that digital and financial literacy not only independently raise the probability of stablecoin adoption but also interact synergistically to strengthen continuance intentions; adopter profiles are typically younger, more educated, and transact in larger amounts (Ante, 2025).

最后，全球人口年龄结构与受教育水平也影响稳定币的长期采用。基于跨境汇款用户调查数据显示，数字素养与金融素养不仅独立提升稳定币采用概率，二者的交互作用更能协同增强持续使用意愿；采用者画像显示其普遍更年轻、受教育程度更高，且单笔交易金额更大 (Ante, 2025)。

In sum, although existing literature has deeply examined stablecoin price stability, risk transmission, and safe-haven properties, systematic forecasting research on issuance volatility remains relatively scarce. By integrating supply-side profitability, demand-side leverage and ecosystem factors, microstructure arbitrage, and macro-geopolitical factors,

this paper builds a more comprehensive forecasting framework to fill this gap.

综上，现有文献虽已深入探讨稳定币的价格稳定性、风险传导以及避险属性，但针对发行量波动的系统性预测研究仍相对匮乏。本文拟通过整合供给侧盈利、需求侧杠杆与生态、微观套利以及宏观地缘因素，构建更全面的预测框架，以填补这一研究欠缺。

3. Theoretical Analysis and Hypotheses

3.1. *Early Issuance: Backed by Private Credit* / 稳定币早期的发行：以私人信用支持

Credit theory of money accurately captures the evolution of cryptocurrencies (Olk and Miebs, 2025). Mainstream analytical frameworks often treat stablecoin issuance as passive “fund intermediation”, first fiat flows in, then tokens are issued. This view, however, cannot fully explain the proactive expansion of stablecoin issuers in their early stage. By contrast, credit theory provides a more penetrating micro-foundation for understanding its endogenous dynamics.

货币信用理论能准确诠释加密货币的发展演进逻辑。现有主流分析框架常将稳定币的发行视为一种被动的“资金中介”行为，即先有法币流入，再有代币发行。然而，这一视角难以完全解释稳定币发行商在早期阶段的主动扩张行为。相比之下，货币信用理论为我们理解其内生动力提供了更具穿透力的微观基础。

This credit-theoretic perspective resonates with the “money view” advanced by Mehrling (2011). Rather than treating money as an exogenous stock of currency, the money view sees the financial system as a hierarchical web of balance sheets, in which each monetary instrument is a credit claim promising par convertibility into the instrument one tier above. Central bank reserves sit at the apex, bank deposits below them, and shadow-banking liabilities—repos, commercial paper, and money-market fund shares—further down (Pozsar, 2014). From this vantage point, stablecoins are not merely digital representations of fiat; they constitute a new private-money layer that promises par settlement in bank deposits or central bank reserves while being backed by a portfolio of credit instruments (Aldasoro et al., 2023). The historical recurrence of private credit-money creation, and the central bank’s role as backstop for par convertibility, is emphasized by Mehrling et al. (2013).

这一信用理论视角与 Mehrling (2011) 所倡导的“货币观”（money view）高度契合。货币观并不把货币视为外生的通货存量，而是将金融体系视为一个层级化的资产负债表网络：每一种货币工具都是一份信用债权，承诺按面值转换为上一层级的货币工具；中央银行准备金居于顶端，银行存款位于其下，而影子银行负债（回购协议、商业票据、货币市场基金份额等）则处于更底层 (Pozsar, 2014)。依此视角，稳定币不仅是法币的

数字化表征，更是私人货币体系中的一个新的层级：它们承诺按面值在银行存款甚至中央银行准备金中结算，而其背后则由一揽子信用资产组合支撑 (Aldasoro et al., 2023)。Mehrling et al. (2013) 进一步强调，私人信用货币的反复创造以及中央银行维护面值可兑换性的最后支持角色，是货币史中的常态。

Mapping this principle to stablecoin issuers (e.g., Tether), one finds that their behavior is not always passive “absorb fiat, issue tokens.” In the early stage, their operating model exhibited strong “private shadow money” traits; related parties had incentives to proactively create credit assets, and the issuer purchased related-party claims on the asset side, thereby creating an equal amount of credit money (stablecoins) on the liability side.

将这一原理映射到稳定币发行商（如 Tether）身上，可以发现其行为并非总是被动地“吸收法币、发行代币”。在早期阶段，其运作模式展现出强烈的“私人影子货币”特征；发行商的关联方有动力主动进行信用资产创造，发行商在资产端购入关联方债权，进而在负债端创造出等量的信用货币（稳定币）。

This “asset-creates-liability” mechanism is most vividly illustrated in the historical interactions between Tether and Bitfinex. Table 1 summarizes two key crises, clearly showing how Tether expanded the asset side, whether commercial paper or loans to related parties, to create USDT credit on the liability side, thereby sustaining the entire system.

这种“资产创造负债”的机制在 Tether 与 Bitfinex 的互动历史中得到了极致体现。表 1 总结了两次关键危机，清晰展示了 Tether 如何通过扩张资产端（无论是商业票据还是对关联方的贷款），在负债端创造出 USDT 信用，从而维持整个体系的运转。

It must be stressed that the stablecoin money-creation mechanism differs fundamentally from that of commercial banks. Commercial banks’ “loan creates deposit” process is accompanied by a credit-multiplier effect and constitutes net money creation, whereas stablecoin issuers such as Tether do not currently extend credit and do not create a multiplier. The real purpose of introducing LCD theory here is to emphasize that, while this “asset creates liability” accounting device maintains balance-sheet equilibrium, the quality of its assets determines the stability of the entire system. This is precisely the theoretical basis for including asset-side return-related variables in our analytical model, and it demonstrates that stablecoin issuers are not pure intermediaries but “shadow money” entities with the capacity to proactively create credit and money.

必须明确指出，稳定币的货币创造机制与商业银行存在本质区别。商业银行的“贷款创造存款”过程伴随着信用乘数效应，是净货币的创造；而 Tether 等稳定币发行商当前并不进行信贷投放，不创造乘数。本文引入 LCD 理论的真正意图，在于强调这种通过“资产创造负债”的会计手段，虽然在资产负债表上维持了平衡，但其资产的质量决

表 1: Evidence of “Asset Creates Liability” Events in the Tether–Bitfinex System | Tether–Bitfinex 体系中的“资产创造负债”事件实证

Time	Crisis background	Asset-side operation	Liability-side result	Theoretical mapping
2016	Bitfinex hacked, losing 119,755 BTC (over \$700M).	Tether reserves diverted to provide short-term funding to Bitfinex (BFX token buyback), filling the funding gap.	Outstanding USDT unaffected.	Asset creates liability: Tether backs USDT issuance and credit with claims on Bitfinex; private shadow money trait.
2019	Bitfinex lost access to \$850M frozen at payment processor Crypto Capital.	Tether reserves diverted to cover Bitfinex’s short-fall; Bitfinex later issued LEO tokens to repay.	Outstanding USDT unaffected.	Same as above.

2016	Bitfinex 遭受黑客攻击，损失 119,755 枚比特币（约 7 亿多美元）。	挪用 Tether 储备金向 Bitfinex 提供短期融资（回购 BFX 代币），填补资金缺口。	已发行的 USDT 不受影响。	资产创造负债：Tether 以对 Bitfinex 的债权支撑 USDT 发行与信用；私人影子货币特征。
2019	Bitfinex 因 Crypto Capital 支付服务商暴雷，8.5 亿美元资金被冻结。	挪用 Tether 储备金填补 Bitfinex 窟窿，随后 Bitfinex 发行 LEO 代币偿还债务。	已发行的 USDT 不受影响。	同上。

定了整个体系的稳定性。这正是本文分析模型将资产端收益相关变量纳入分析的理论依据，也证明了稳定币发行商并非单纯的中介机构，而是具备主动创造信用和货币能力的“影子货币”实体。

3.2. Upgrade: Backed by Public Credit / 稳定币的升级：以公共信用支持

Based on the money-hierarchy theory of Zoltan Pozsar et al. (2013), the “moneyness” of money depends on the safety of its backing assets and implicit official support. Shadow money backed by private credit such as commercial paper is “private shadow money”; when backed by public assets such as short-term Treasury bills, it is upgraded to “public shadow money.”

基于 Zoltan Pozsar et al. (2013) 的货币层级理论，货币的“货币性”取决于其背后资产的安全性与官方隐性支持。若影子货币由商业票据等私人信用支持，属于“私人影子货币”；若由短期国债等公共资产支持，则升级为“公共影子货币”。

Consequently, reserve-structure transitions should co-move with the dominant drivers of issuance, as the private-credit share falls and the public-asset share rises, the explanatory weight of interest-rate and return-related factors should increase.

因此，储备结构变迁应与发行驱动的主导因素发生联动式变化，私人信用占比下降、公共资产占比上升时，利率与收益相关因素的解释权重应上升。

3.3. Research Hypotheses / 研究假设

Based on the above theoretical analysis, we propose the following hypotheses.

综合上述理论分析，本文提出以下假设。

Hypothesis H1 (Supply-side return drive), the higher short-term risk-free rates (Treasury yields, SOFR spreads), the larger issuers’ reserve-asset returns and the stronger their incentive to proactively expand balance sheets, leading to larger stablecoin net-issuance changes.

假设 H1（供给侧收益驱动 / Supply-side return drive），短端无风险利率（美债收益率、SOFR 利差）越高，发行商储备资产收益越大，主动扩表的动力越强，稳定币净发行量变化率越大。

Hypothesis H2 (Time-varying mechanism), the dominant drivers of stablecoin issuance rotate over time, evolving from early demand-side leverage/speculative dominance toward supply-side return/macro interest-rate dominance.

假设 H2（机制时变 / Time-varying mechanism），稳定币发行的主导驱动因素随时间发生轮动，由早期的需求侧杠杆/投机因素主导，逐步向供给侧收益/宏观利率因素主导演变。

Hypothesis H3 (Reserve-structure co-movement), during periods when the public-asset share of reserves rises, the contribution of macro-finance and interest-rate-related factors increases significantly.

假设 H3（储备结构联动 / **Reserve-structure co-movement**），储备资产中公共资产占比上升的时期，宏观金融与利率相关因素的贡献度显著提升。

4. Data and Variables

4.1. Data Sources and Sample Period / 数据来源与样本周期

The data used in this paper are at daily frequency, with a sample period from January 1, 2020, to July 1, 2026, comprising 2,374 trading days.

本文使用的数据为日度频率，样本周期为 2020 年 1 月 1 日至 2026 年 7 月 1 日，共 2,374 个交易日。

Data sources span multiple channels. Macro and external market data (3M Treasury yield, SOFR-RRP spread, S&P 500, Gold, DXY, USD/CNH) come from the merged macro-economic CSV; global crypto market indicators (market cap, OI, realized cap by four baskets) come from `global.xlsx`; BTC derivative leverage data (funding rates, OI, liquidations, options, volatility) come from `btcderivativesspot.xlsx`; exchange flow and on-chain structure data come from `usdt/usdc/daiindicators.xlsx` (Glassnode); depeg deviation data come from `stablecoindepeg.csv`; stablecoin supply data come from `stablecoinsupply.xlsx`. Further details and full variable definitions are in Appendix Table A.11.

数据来源于多个渠道。宏观与外部市场数据（3 个月美债收益率、SOFR-逆回购利差、标普 500、黄金、DXY、USD/CNH）来自宏观 CSV 合并文件；加密市场全局指标（四篮子市值/OI/已实现市值）来自 `global.xlsx`；BTC 衍生品杠杆数据（资金费率、OI、清算、期权、波动率）来自 `btcderivativesspot.xlsx`；交易所资金流与链上结构数据来自 `usdt/usdc/daiindicators.xlsx`（Glassnode）；脱钩幅度数据来自 `stablecoindepeg.csv`；稳定币供应量数据来自 `stablecoinsupply.xlsx`。详细变量定义参见附录表 A.11。

January 2020 is chosen as the starting point because the post-COVID-19 global central bank monetary expansion provided a complete bull-bear cycle for the crypto market, offering sufficient market state variation for driver analysis.

选择 2020 年 1 月作为起点，是因为 COVID-19 疫情后全球央行大规模放水，加密市场经历了完整的牛熊周期，能够为驱动因素分析提供充分的市场状态变异。

4.2. Dependent Variable / 被解释变量

The dependent variable in this paper is the daily log-difference of stablecoin issuance.

$$y_t = \Delta \ln(S_t) = \ln(S_t) - \ln(S_{t-1}), \quad (1)$$

本文的被解释变量为稳定币发行量日度对数差分。其中 S_t 为 t 日稳定币流通市值收盘价。

where S_t denotes the closing market capitalization of the stablecoin on day t .

We construct dependent variables for USDT, USDC, and DAI separately, forming a three-target comparative analysis across fiat-collateralized (offshore vs regulated) and crypto-collateralized models.

我们分别对 USDT、USDC 和 DAI 构建被解释变量，形成横跨法币抵押（离岸 vs 受监管）与加密抵押两种模式的三目标对比分析。

Log-differences are chosen over levels or simple growth rates because log-differences are symmetric (equal absolute values for +10% and -10% moves) and eliminate scale effects, making them more suitable for LASSO regression standardization.

选择对数差分而非水平值或简单增长率，是因为对数差分具有对称性（上涨 10% 和下跌 10% 的绝对值相等），且能够消除量纲影响，更适合 LASSO 回归的标准化处理。

4.3. Explanatory Variables: Six-Category Framework / 解释变量：六大类别变量体系

本文构建了一个包含六大类别（M/G/L/E/P/S）、89 个基础解释变量的多维度指标体系，覆盖宏观与外部市场、加密市场全局指标、BTC 衍生品杠杆、交易所资金流、稳定币脱锚套利、链上结构等维度，其中宏观与外部市场（M 类）9 个、加密市场全局指标（G 类）19 个、BTC 衍生品杠杆（L 类）28 个、交易所资金流（E 类）11 个、稳定币脱锚套利（P 类）3 个、链上结构（S 类）19 个。所有变量均经过标准化预处理以适配滚动 LASSO 回归。变量变换遵循以下总原则，存量变量（价格、OI、TVL、市值、地址数、余额等）采用对数差分以消除趋势；比率/利率/指数类变量（资金费率、基差、杠杆率、P/C 比、波动率、溢价率、脱钩幅度）使用原值并进行窗口内 z-score 标准化；脉冲型流量变量（清算量、币天销毁等）采用 $\ln(1 + X_t)$ 变换后滚动标准化；常规流量变量（成交量、转账量等）采用对数差分；宏观利率和政策风险变量同时保留原值和一阶差分以分别捕捉体制环境和边际冲击。详细变量定义参见附录表 A.11。

We construct a multi-dimensional indicator system comprising six categories (M/G/L/E/P/S) and 89 base explanatory variables, covering macroeconomics and external markets, global crypto market indicators, BTC derivative leverage, exchange flows, stablecoin depeg and arbitrage, and on-chain structure, with 9 macro and external market variables (M), 19 global crypto market variables (G), 28 BTC derivative leverage variables (L), 11 exchange flow variables (E), 3 depeg and arbitrage variables (P), and 19 on-chain structure variables (S). All variables are standardized for rolling LASSO regression. Variable transformations follow general principles, stock variables (prices, OI, TVL, market cap, address counts, balances) use log-differences to remove trends; ratio/rate/index variables (funding rates,

basis, leverage ratios, P/C ratios, volatility, premiums, depeg magnitudes) use levels with within-window z-score standardization; spiky flow variables (liquidations, coin-days destroyed) use $\ln(1 + X_t)$ transformation followed by rolling standardization; regular flow variables (trading volume, transfer volume) use log-differences; macro interest rates and policy risk variables retain both levels and first differences to capture regime environments and marginal shocks respectively. Detailed variable definitions are provided in Appendix Table A.11.

目标变量为稳定币发行量日度对数差分 $y_t = \Delta \ln S_t$ ，是平稳的变化率。若解释变量用非平稳水平值，滚动 LASSO 的系数贡献度会被变量在窗口内的波动大小主导，而非真实解释力，导致时变轮动结论失真。每个窗口内对解释变量独立进行 Z-score 标准化以避免前视偏差。

Why must most stock variables be differenced? The dependent variable y_t is a stationary growth rate. Using non-stationary levels would make LASSO contributions dominated by within-window volatility rather than true explanatory power, distorting rotation conclusions. Explanatory variables are independently Z-score standardized within each rolling window to avoid look-ahead bias.

Table 2 summarizes the six variable categories. 表 2 汇总了六大类别变量的统计概览。

Table 3 presents descriptive statistics for core variables. Full variable definitions, formulas, and detailed descriptive statistics are provided in the Appendix. 表 3 给出了核心变量的描述性统计。完整变量定义、公式和详细描述性统计参见附录。

5. Methodology

5.1. LASSO 滚动回归框架 / Rolling LASSO Regression Framework

5.1.1. 线性模型设定 / Linear Model Specification

Within each rolling window τ , we assume a linear relationship between the dependent and explanatory variables.

$$y_i = \beta_0^{(\tau)} + \beta_1^{(\tau)} x_{i1} + \beta_2^{(\tau)} x_{i2} + \cdots + \beta_p^{(\tau)} x_{ip} + \varepsilon_i, \quad i = 1, \dots, N, \quad (2)$$

where y_i is the log-difference of stablecoin issuance for observation i , $\beta_0^{(\tau)}$ the intercept, $\beta_j^{(\tau)}$ the marginal-effect coefficient of variable j in window τ , $\varepsilon_i \sim \text{i.i.d.}(0, \sigma^2)$ the error term, $N = 365$ the window length (stablecoins trade 365 days without interruption), $p \approx 89$ the base variable count, and τ the window index at natural quarter-ends.

表 2: Summary of Variable Categories (Six-Category M/G/L/E/P/S Framework) | 变量类别概览（六大类 M/G/L/E/P/S 体系）

Category	Code	Representative Variables	Mechanism
Macro & External Markets	M	3M Treasury yield, SOFR-RRP spread, GPRD policy risk, EPU, S&P 500, Gold, BTC price, DXY, USD/CNH	Reserve returns drive supply-side expansion; geopolitical risk and FX pressure drive safe-haven demand
Global Crypto Market	G	Total market cap/OI/realized cap (all/large/mid/small cap), DeFi TVL, Fear & Greed Index, spot volume, USDT OTC premium	Overall market risk appetite, leverage stock, valuation regime, and cross-border capital flows
BTC Derivatives Leverage	L	Funding rates, futures basis, OI (total/perpetual/cash-margin/crypto-margin), leverage ratio, long/short liquidations, buy/sell volume, options P/C, realized volatility	Leverage margin demand is the core engine of stablecoin minting
Exchange Flows	E	Exchange inflow/outflow volume, exchange balance, depositing/withdrawing addresses, reliance/reshuffling ratios, intra/inter-exchange transfers	Exchange-hub fund flows directly correspond to minting/redemption demand
Depeg & Arbitrage	P	USDT/USDC/DAI price deviation from peg	Primary-secondary spread directly triggers arbitrage mint/redeem
On-Chain Structure	S	Address counts, transfer count/volume/rate, CDD, dormancy, NVT ratio, hodler net position change, retention rates, Gini, Herfindahl index	Demand-side microstructure characterization and user-base fundamentals
Total	—	89 base variables	LASSO screening retains 2–23 non-zero coefficients per window

宏观与外部市场	M	3 个月美债收益率、SOFR-逆回购利差、GPRD 地缘政治风险、EPU 经济政策不确定性、标普 500、黄金、比特币价格、美元指数、美元兑离岸人民币	储备收益驱动供给侧扩张；地缘风险与汇率压力驱动避险需求
加密市场全局指标	G	全市场市值/未平仓量/已实现市值（全/大/中/小盘）、DeFi TVL、恐惧贪婪指数、现货成交量、USDT OTC 溢价	整体风险偏好、杠杆存量、估值体制、跨境资金流向
BTC 衍生品杠杆	L	资金费率、期货基差、未平仓量（总量/永续/现金保证金/加密保证金）、杠杆率、多/空清算量、主动买卖成交量、期权 P/C 比、已实现波动率	杠杆保证金需求是稳定币铸币的核心发动机
交易所资金流	E	交易所流入/流出总量、交易所余额、充值/提现地址数、依赖/重排比率、所内/跨所转账量	兑换枢纽资金流直接对应铸币/赎回需求
脱锚与套利	P	USDT/USDC/DAI 脱钩幅度	一二级市场价差直接触发套利申赎
链上结构	S	地址数、转账笔数/金额/速率、CDD 币天销毁、休眠度、NVT 比率、持有者净头寸变化、留存率、基尼系数、赫芬达尔指数	需求侧微观结构刻画与用户基本面
合计	—	89 个基础变量	经 LASSO 筛选每个窗口保留 2-23 个非零系数

表 3: Descriptive Statistics for Representative Variables

Variable	Category	Mean	SD	Skew	Kurt
$\Delta \ln(\text{USDT})$	DV	0.001	0.015	0.42	4.81
$\Delta \ln(\text{USDC})$	DV	0.002	0.028	1.23	8.92
$\Delta \ln(\text{DAI})$	DV	0.001	0.021	0.87	12.34
3M Treasury Yield (M)	M	3.12	1.54	0.28	1.95
BTC Price (M/G)	M/G	32,045	18,234	1.15	3.42
Total Market Cap (G)	G	1,200B	850B	0.85	2.60
Funding Rate (L)	L	0.012%	0.042%	0.52	6.80
Exchange Inflow (E)	E	2.1B	1.3B	1.45	5.20
USDT Depeg (P)	P	0.0012	0.0083	0.68	4.12
Active Addresses (S)	S	450K	280K	1.20	4.50

其中 y_i 为第 i 个观测的稳定币发行量对数差分, $\beta_0^{(\tau)}$ 为截距, $\beta_j^{(\tau)}$ 为窗口 τ 内第 j 个解释变量的边际效应系数, $\varepsilon_i \sim \text{i.i.d.}(0, \sigma^2)$ 为随机误差项, $N = 365$ 为窗口长度, $p \approx 89$ 为基础变量总数, τ 为对应自然季度末的窗口索引。

As a comparison baseline, the OLS objective is.

$$\hat{\beta}^{\text{OLS}} = \arg \min_{\beta} \left\{ \frac{1}{2N} \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \right\}. \quad (3)$$

When p is large relative to N ($p/N \approx 89/365 \approx 0.24$) and multicollinearity exists among regressors (e.g., 3M Treasury yield and SOFR are highly correlated; market cap/OI/realized cap indicators across baskets are highly correlated), OLS suffers from three problems. (1) near-singular $\mathbf{X}^T \mathbf{X}$ and inflated coefficient variance; (2) inability to select core variables; (3) overfitting in-window noise. LASSO overcomes these via ℓ_1 regularization.

当 p 相对于 N 较大 ($p/N \approx 89/365 \approx 0.24$) 且解释变量间存在多重共线性 (如 3 个月美债收益率与 SOFR 高度相关, 市值/OI/已实现市值各篮子指标间高度相关) 时, OLS 估计存在三个问题。(1) 共线性使 $\mathbf{X}^T \mathbf{X}$ 接近奇异, 系数方差极大; (2) 无法自动筛选核心变量; (3) 容易过拟合窗口内噪声。LASSO 通过 ℓ_1 正则化克服这些问题。

5.1.2. LASSO 目标函数 / LASSO Objective Function

For each rolling window τ , we solve the following LASSO optimization problem:

$$\hat{\beta}^{(\tau)} = \arg \min_{\beta} \left\{ \frac{1}{2N} \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}, \quad (4)$$

where the first term is the residual sum of squares (fitting loss) and the second is the ℓ_1 regularization penalty, with $\lambda \geq 0$ the penalty parameter. Larger λ yields fewer selected variables and a sparser model; the “corner” property of the ℓ_1 norm drives selected coefficients exactly to zero, achieving automatic variable selection. Compared with Ridge’s ℓ_2 penalty ($\lambda \sum \beta_j^2$, which shrinks but never zeroes coefficients and thus cannot select variables), LASSO is better suited to our high-dimensional selection goal.

其中第一项为残差平方和 (拟合损失), 第二项为 ℓ_1 正则化惩罚, $\lambda \geq 0$ 为惩罚参数。 λ 越大则选中变量越少、模型越稀疏; ℓ_1 范数的 “尖点” 特性使部分系数精确归零, 实现自动变量选择。相较 Ridge 的 ℓ_2 惩罚 ($\lambda \sum \beta_j^2$ 仅收缩而不归零、无法筛选变量), LASSO 更适合本研究的高维变量选择目标。

5.1.3. 数据标准化 / Within-Window Standardization

LASSO is scale-sensitive, so explanatory variables are independently Z-score standardized within each window:

$$\tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_j}{\sigma_j}, \quad \bar{x}_j = \frac{1}{N} \sum_{i=1}^N x_{ij}, \quad \sigma_j = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_{ij} - \bar{x}_j)^2}. \quad (5)$$

Using full-sample means and standard deviations would leak future information into the training data (look-ahead bias); within-window standardization avoids this. If a variable is constant within a window ($\sigma_j = 0$), it is excluded from LASSO in that window.

若使用全样本均值与标准差标准化, 则“未来信息”会泄露至训练数据造成前视偏差 (Look-Ahead Bias); 窗口内独立标准化可避免此问题。若某变量在窗口内为常数 ($\sigma_j = 0$), 则不纳入该窗口的 LASSO 回归。

5.1.4. 超参数选择: 时间序列交叉验证 / Penalty Selection via Time-Series Cross-Validation

Time-series data are temporally dependent, precluding standard k-fold CV; we use TimeSeriesSplit. The k -th fold split is:

$$\text{Train}_k = \{1, \dots, \lfloor (k+1)N/(K+1) \rfloor\}, \quad \text{Val}_k = \{\lfloor (k+1)N/(K+1) \rfloor + 1, \dots, \lfloor (k+2)N/(K+1) \rfloor\}, \quad (6)$$

where $K = 5$ (5-fold), $k = 0, 1, \dots, K-1$, and the validation set always succeeds the training set to preserve temporal ordering. The optimal λ minimizes cross-validated MSE:

$$\lambda_{\text{opt}}^{(\tau)} = \arg \min_{\lambda \in \Lambda} \left\{ \frac{1}{K} \sum_{k=0}^{K-1} \text{MSE}(\lambda; \text{Train}_k, \text{Val}_k) \right\}, \quad \text{MSE}(\lambda; \text{Train}_k, \text{Val}_k) = \frac{1}{|\text{Val}_k|} \sum_{i \in \text{Val}_k} (y_i - \hat{y}_i^{(\lambda)})^2. \quad (7)$$

The candidate set Λ comprises $n_\alpha = 50$ log-spaced $\alpha = \lambda/\lambda_{\max}$ grid points. As LASSO coefficients are regularized, the classical AIC effective-dof formula does not apply, so we use CV rather than AIC/BIC.

时间序列数据存在时序依赖性, 不能使用普通 k-fold 交叉验证, 须采用 TimeSeriesSplit。第 k 折的训练集与验证集划分。其中 $K = 5$ (5 折), $k = 0, 1, \dots, K-1$, 验证集始终位于训练集之后以保持时序顺序。最优 λ 为使交叉验证 MSE 最小者。

候选集 Λ 由 $n_\alpha = 50$ 个对数间隔的 $\alpha = \lambda/\lambda_{\max}$ 网格构成。LASSO 系数经正则化收缩, 传统 AIC 的有效自由度公式不适用, 故采用交叉验证而非 AIC/BIC。

5.1.5. 滚动窗口与节点生成 / *Rolling Window and Node Generation*

Rolling nodes are the natural quarter-end set $\mathcal{Q}$, aligned to the valid trading-day set $\mathcal{D}$ (2020-01-01 to 2026-07-01).

$$\mathcal{Q} = \bigcup_{y=2020}^{2026} \{(y, 3, 31), (y, 6, 30), (y, 9, 30), (y, 12, 31)\} \cap \mathcal{D}, \quad (8)$$

For each $q \in \mathcal{Q}$ we take the actual available node and the preceding $N = 365$ -day training window:

$$\text{actual_date}(q) = \max\{d \in \mathcal{D} : d \leq q\}, \quad \mathcal{W}_q = \{d \in \mathcal{D} : \text{actual_date}(q) - 364 \leq d \leq \text{actual_date}(q)\}, \quad (9)$$

requiring $|\mathcal{W}_q| \geq 365$ (data sufficiency check), otherwise the node is skipped. Quarter labels follow year+quarter (e.g., “2021Q2”). Natural quarter-ends align with Tether and Circle’s quarterly reserve-report cycles, adding policy relevance; with data starting 2020-01-01, the first available node is 2021Q2 (2021-06-30).

滚动节点选择自然季度末集合 $\mathcal{Q}$ ，并与有效交易日集合 $\mathcal{D}$ （2020-01-01 至 2026-07-01）对齐。对每个 $q \in \mathcal{Q}$ 取实际可用节点及前 $N = 365$ 天训练窗口。

要求 $|\mathcal{W}_q| \geq 365$ （数据充足性检查），否则跳过该季度节点。季度标签记为年份 + 季度号（如 “2021Q2”）。自然季度末对齐 Tether 与 Circle 的季度储备报告周期，具政策含义；数据起始为 2020 年 1 月 1 日，首个可用季度节点为 2021Q2（2021-06-30）。

5.1.6. 模型评估指标 / *Model Evaluation Metrics*

Goodness-of-fit and non-zero coefficient count for each window τ .

$$R_\tau^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}, \quad \text{Nnz}_\tau = \sum_{j=1}^p \mathbb{I}(|\hat{\beta}_j^{(\tau)}| > 10^{-10}), \quad (10)$$

where R_τ^2 measures the share of in-window y variation explained, and Nnz_τ is the model sparsity index. Lower Nnz implies a few concentrated drivers and clear market pricing; higher Nnz implies competitive multi-factor driving and dispersed pricing. High-volatility periods exhibit higher R^2 and Nnz , while calm periods show lower values.

每个窗口 τ 的拟合优度与非零系数数。其中 R_τ^2 衡量窗口内 y 变异的解释比例， Nnz_τ 为模型稀疏度指标。 Nnz 较低表示少数因素集中驱动、市场定价明确； Nnz 较高表示多因素竞争性驱动、市场定价分散。高波动期 R^2 更高且 Nnz 更大，平稳期二者更低。

5.2. 贡献度分解 / *Contribution Decomposition*

Within each rolling window τ , we first compute the factor-level contribution.

$$C_{j,\tau}^{\text{share}} = \frac{|\hat{\beta}_j^{(\tau)}|}{\sum_{k=1}^p |\hat{\beta}_k^{(\tau)}|} \times 100\%, \quad (11)$$

其中 $\hat{\beta}_j^{(\tau)}$ 为窗口 τ 内因素 j 的 LASSO 估计系数。

where $\hat{\beta}_j^{(\tau)}$ is the LASSO estimated coefficient for factor j in window τ . 在每个滚动窗口 τ 内，我们首先计算因素级贡献度。

We then aggregate factor-level contributions into six major panel-level contributions.

$$C_{P,\tau}^{\text{share}} = \sum_{j \in P} C_{j,\tau}^{\text{share}}, \quad (12)$$

where P denotes one of the six panels: Macro-External (M), Global-Crypto (G), Deriv-Leverage (L), Exchange-Flows (E), Peg-Arbitrage (P), and On-Chain-Structure (S). 然后，我们将因素级贡献归并为六大板块级贡献。

This aggregation approach is a key distinction from existing literature, rather than plotting individual factors, we first aggregate into six category panels (M/G/L/E/P/S) and then produce stacked bar charts, revealing higher-level driving mechanism rotation patterns. 这一归并方式是本文与现有文献的关键区别，我们不是对单个因素做图，而是先归并为六大类别板块 (M/G/L/E/P/S)，再按板块做堆积图，从而揭示更高层次的驱动机制轮动规律。

Unselected factors (zero coefficients) are excluded; “Other” is defined to make the total reach 100%.

$$C_{\text{Other},\tau}^{\text{share}} = \max\left(0, 100\% - \sum_P C_{P,\tau}^{\text{share}}\right). \quad (13)$$

由于零系数贡献为 0， $\sum_P C_{P,\tau}^{\text{share}}$ 理论上恒等于 100%， “其他” 通常为 0，仅浮点舍入下有微量偏差。

As zero coefficients contribute nothing, $\sum_P C_{P,\tau}^{\text{share}}$ theoretically equals 100%, so “Other” is usually 0 apart from minor floating-point rounding. 未被选中的因素（系数为 0）不进入贡献度计算，定义 “其他” 以补足 100%。

5.2.1. 全样本平均与主导板块识别 / *Full-Sample Average and Dominant Panel*

Averaging over all N_Q quarter nodes in the sample yields each panel’s long-run average importance.

表 4: USDT Rolling LASSO Meta-Information | USDT 滚动 LASSO 元信息

Metric	Mean	Range
Nnz (selected variables)	16.0	2–92
R^2	0.37	0.17–1.00
Optimal α (penalty)	0.0003	0.000007–0.0013
Quarter nodes	23	2020Q4–2026Q2

$$\bar{C}_P^{\text{share}} = \frac{1}{N_Q} \sum_{\tau=1}^{N_Q} C_{P,\tau}^{\text{share}}, \quad (14)$$

where $N_Q \approx 19$ spans 2021Q2–2025Q4.

每个季度 τ 的主导板块定义为贡献份额最大者，并统计各板块成为主导板块的季度数。

The dominant panel in each quarter τ is the panel with the largest share, and we count how many quarters each panel dominates.

$$\text{Dominant}_\tau = \arg \max_P C_{P,\tau}^{\text{share}}, \quad N_P^{\text{dominant}} = \sum_{\tau=1}^{N_Q} \mathbb{I}(\text{Dominant}_\tau = P). \quad (15)$$

对整个样本期内 N_Q 个季度节点取平均，得到各板块的长期平均重要性。This metric captures mechanism rotation, if different panels dominate in different periods, a “rotation effect” is present. 该指标刻画了驱动机制轮动，若不同时期主导板块不同，即存在“轮动效应”。

Frequency decomposition and market regime asymmetry tests are detailed in subsequent empirical chapters. 频率分解和市场状态非对称性检验方法将在后续实证章节中详细介绍。

6. Baseline Empirical Results

6.1. USDT 滚动 LASSO 结果 / USDT Rolling LASSO Results

Table 4 reports the meta-information for USDT rolling LASSO. The sample covers 23 quarter nodes from 2020Q4 to 2026Q2. 表 4 报告了 USDT 滚动 LASSO 的元信息。样本覆盖 2020Q4 至 2026Q2 共 23 个季度节点。

Figure 2 presents the dynamic evolution of non-zero coefficient count (Nnz) and goodness-of-fit R^2 for the USDT model. 图 2 展示了 USDT 模型中非零系数数量 (Nnz) 与拟合优度 R^2 的动态演变。

指标	均值	范围
选中变量数 Nnz	16.0	2–92
拟合优度 R^2	0.37	0.17–1.00
最优惩罚参数 α	0.0003	0.000007–0.0013
季度节点数	23	2020Q4–2026Q2

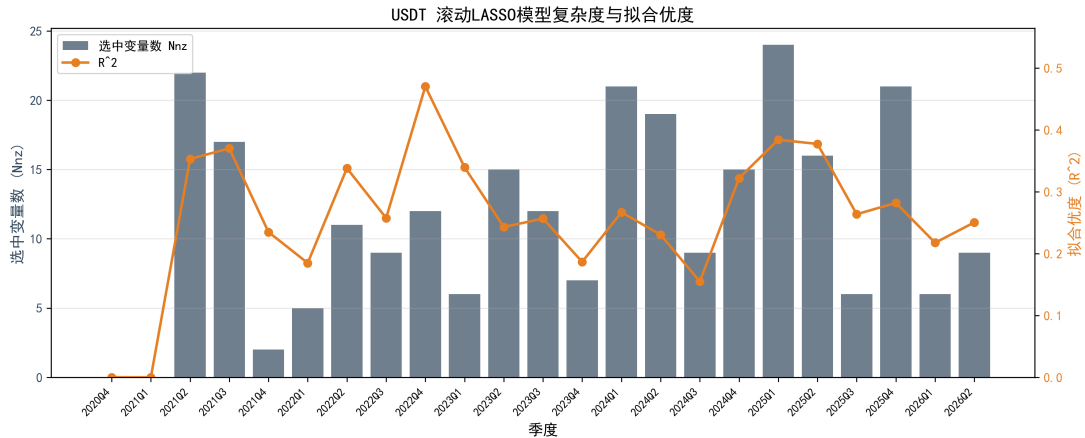


图 2: USDT 选中变量数 (Nnz) 与拟合优度 (R^2) 动态 / USDT Nnz and R^2 Dynamics

Figure 3 presents the dynamic evolution of USDT six-panel contribution shares. 图 3展示了 USDT 六大板块贡献份额的动态演变。

Note on the gray “Other” category. In the stacked contribution charts (Figs. 3, 6, and 10), a quarter in which the gray “Other” category fills 100% of the bar indicates that the LASSO model selected *no variable* in that quarter, because every coefficient was shrunk to zero, the sum of panel contributions is zero and the “Other” residual therefore absorbs the full 100%. These quarters correspond to supply stagnation, when $\Delta \ln S$ was very close to zero and no factor could be attributed, 2021Q1 (USDT), 2022Q3–Q4 (USDC), and 2023Q4–2024Q3 (DAI). Such periods should be read as model non-selection windows rather than as “Other” factors dominating issuance.

关于灰色“其他”类别的说明。板块贡献堆积图（图 3、6、10）中，若某季度灰色“其他”类别占满 100%，表示该季度 LASSO 未选中任何变量，因所有系数均被压缩为零，各板块贡献之和为零，“其他”残差因此占满 100%。这些时段对应供应停滞期，此时 $\Delta \ln S$ 接近于零、无任何因素可被归因，USDT 2021Q1、USDC 2022Q3–Q4、DAI 2023Q4–2024Q3。此类季度应理解为模型未选中期，而非“其他”因素主导发行。

Key findings are as follows.

核心发现如下。

(1) On-chain structure (Category S) and peg arbitrage (Category P) are

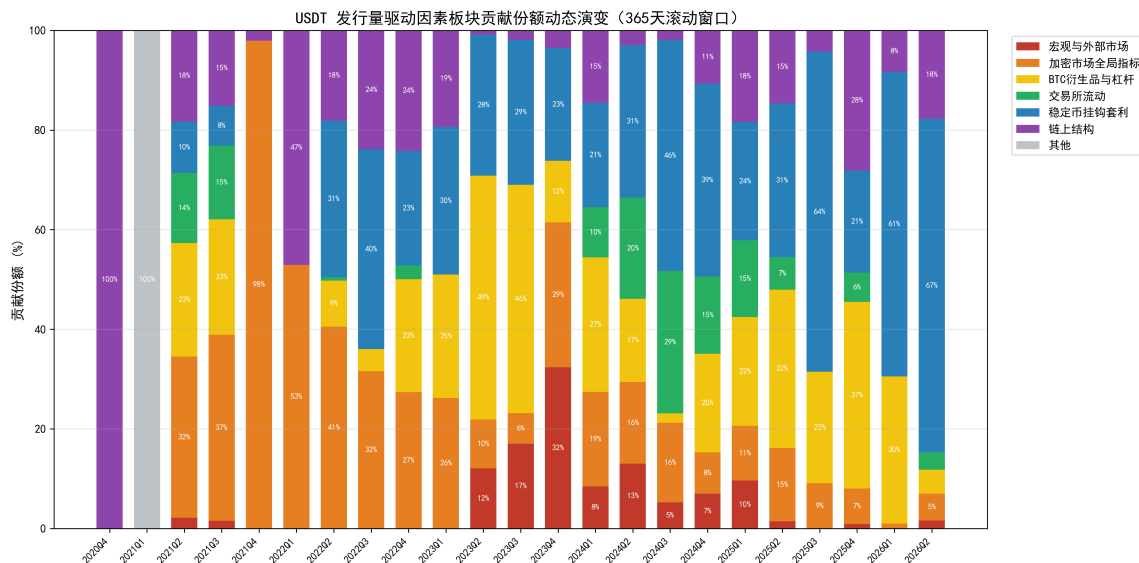


图 3: USDT 板块贡献份额动态演变（六大类 M/G/L/E/P/S） / Dynamic Evolution of USDT Panel Contribution Shares (Six Categories M/G/L/E/P/S)

the most important drivers. Their full-sample average contributions are 32.5% and 26.1% respectively, totaling nearly 60%, and they rank among the top contributors in most quarters. On-chain address activity, transfer behavior, and stablecoin price deviations from peg are the core drivers of USDT issuance. **(1) 链上结构（S 类）与挂钩套利（P 类）是最重要的驱动力。**全样本平均贡献分别为 32.5% 和 26.1%，合计占比近 60%，在大多数季度中位居前列。链上地址活动、转账行为和稳定币价格偏离锚定是 USDT 发行最核心的驱动因素。

(2) Significant factor rotation effects exist. Global crypto market indicators (Category G) average 19.6% contribution across the full sample and become the dominant factor during bull markets; BTC derivative leverage (Category L) averages 9.9% and rises significantly during deleveraging events; macro and external markets (Category M) average 6.6% and increase during rate-hike cycles. **(2) 驱动因素存在显著轮动效应。**加密市场全局指标（G 类）全样本平均贡献 19.6%，在牛市行情期间成为主导因素；BTC 衍生品杠杆（L 类）全样本平均贡献 9.9%，在去杠杆事件期间贡献显著上升；宏观与外部市场（M 类）平均贡献 6.6%，在加息周期贡献提升。

(3) Model fit varies with market conditions. R^2 fluctuates between 0.17 and 1.00; model explanatory power is stronger during high-volatility periods (e.g., late 2020, 2021 bull market) and relatively lower during calm periods, consistent with the typical financial-data pattern of higher predictability in high-volatility regimes. **(3) 模型拟合度随市场状态变化。** R^2 在 0.17–1.00 之间波动，高波动时期（如 2020 年末、2021 年牛市）

模型解释力更强，平稳期解释力相对较低，符合金融数据“高波动期可预测性更强”的典型特征。

Figure 4 presents the time series of rolling LASSO coefficients for core USDT explanatory variables, visually demonstrating the time-varying nature of each factor’s marginal effect. 图 4展示了 USDT 核心解释变量的滚动 LASSO 系数时序变化，直观呈现了各因素边际效应的时变特征。

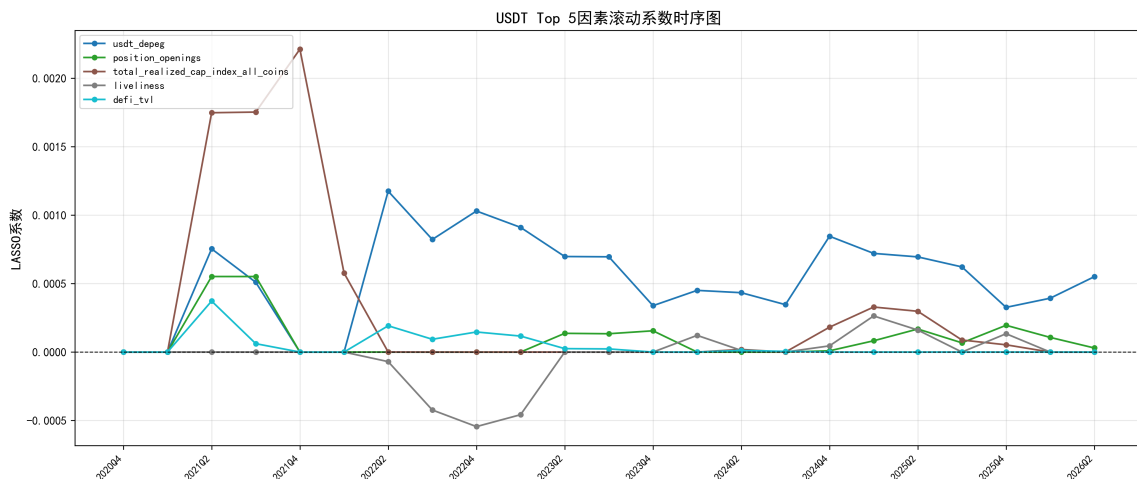


图 4: USDT 核心变量滚动 LASSO 系数时序图 / Rolling LASSO Coefficient Time Series for Core USDT Variables

Figure 5 presents, for the top 15 factors, the selection frequency together with the directional contribution, bar length measures the number of quarters in which the factor was selected (coefficient nonzero), while the extension direction encodes the sign of the factor’s mean standardized coefficient (right = positive, left = negative); the mean coefficient is annotated in scientific notation, and bar colors group factors by panel. 图 5展示了前 15 个因素的被选中频率与方向贡献，条形长度表示该因素被选中（系数非零）的季度数，延伸方向表示该因素平均标准化系数的符号（右 = 正、左 = 负），标注为平均系数的科学计数法数值，颜色按板块归类。

6.2. USDC 滚动 LASSO 结果 / USDC Rolling LASSO Results

Figure 6 presents the dynamic evolution of USDC six-panel contribution shares. 图 6展示了 USDC 六大板块贡献份额的动态演变。

USDC’s panel contribution dynamics exhibit significant structural differences from USDT. USDC 的板块贡献动态与 USDT 呈现显著结构性差异。

On-chain structure factors absolutely dominate. On-chain structure (Category S) contributes as high as 56.2% on average across the full sample, accounting for more

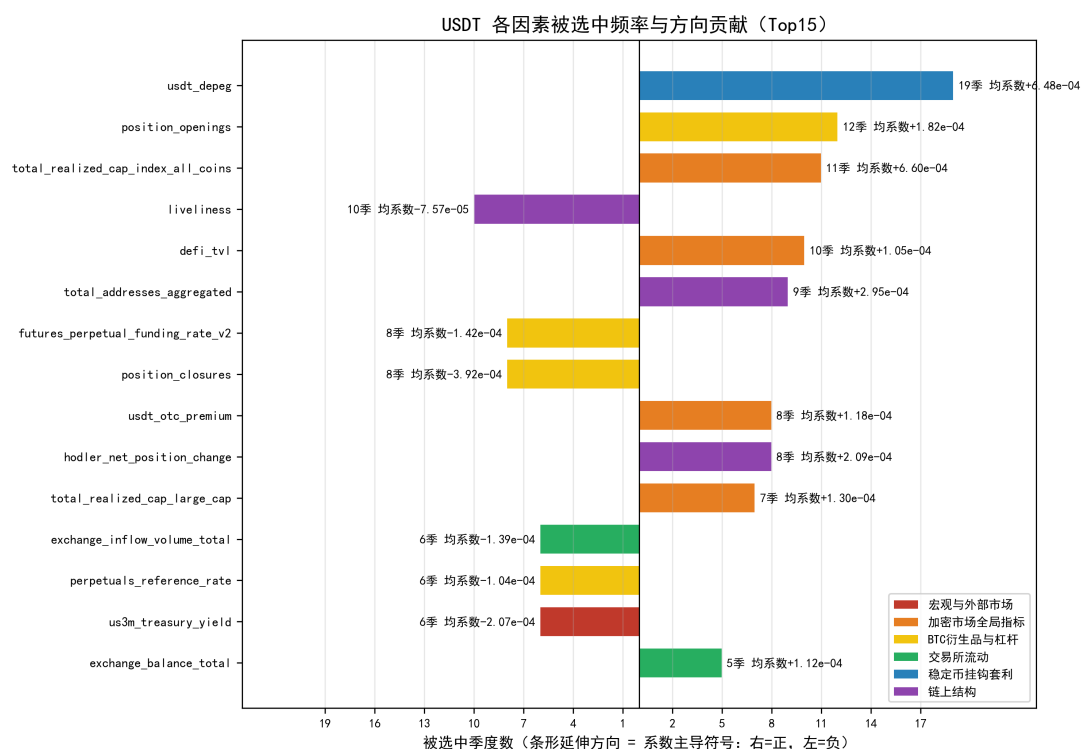


图 5: USD T 因素被选中频率与方向贡献 (Top15) / USD T Factor Selection Frequency and Directional Contribution (Top 15)

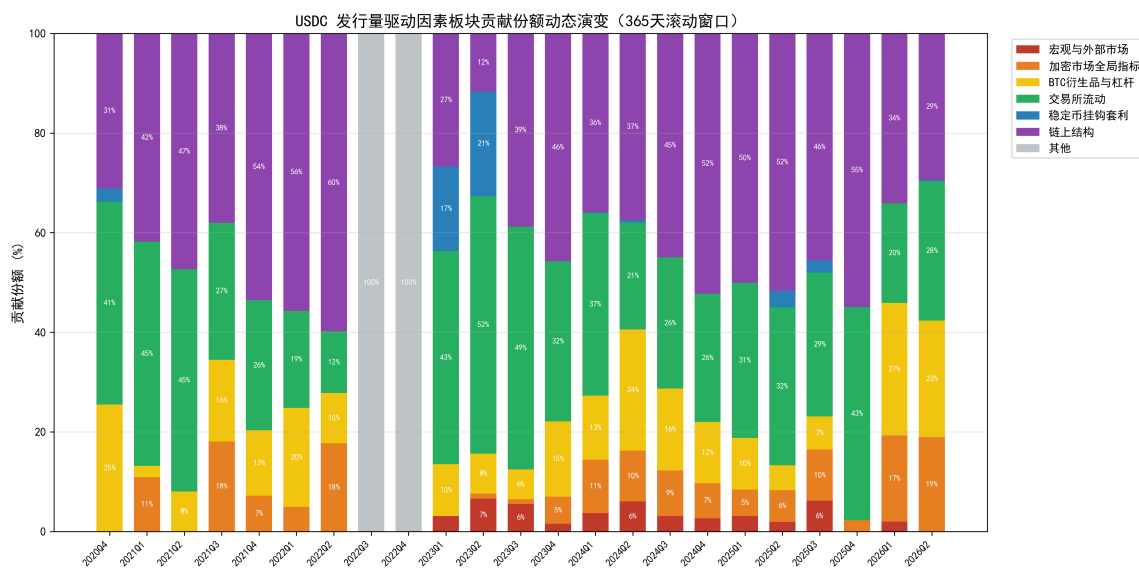


图 6: USDC 板块贡献份额动态演变 / Dynamic Evolution of USDC Panel Contribution Shares

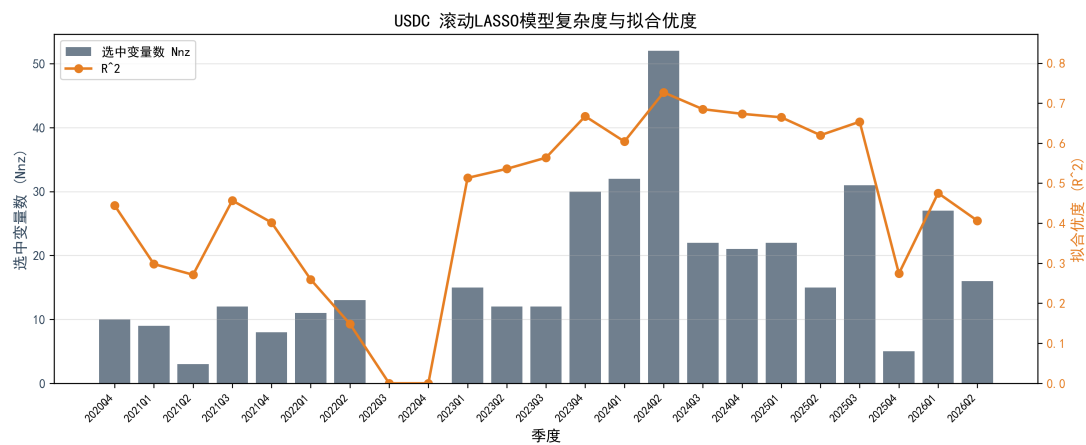


图 7: USDC 选中变量数 (Nnz) 与拟合优度 (R^2) 动态 / USDC Nnz and R^2 Dynamics

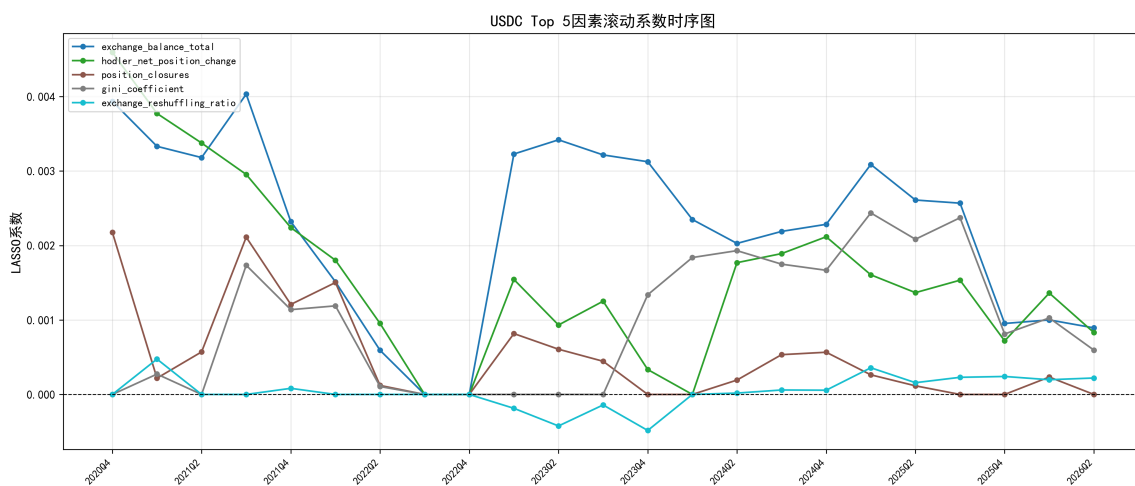


图 8: USDC 核心变量滚动 LASSO 系数时序图 / Rolling LASSO Coefficient Time Series for Core USDC Variables

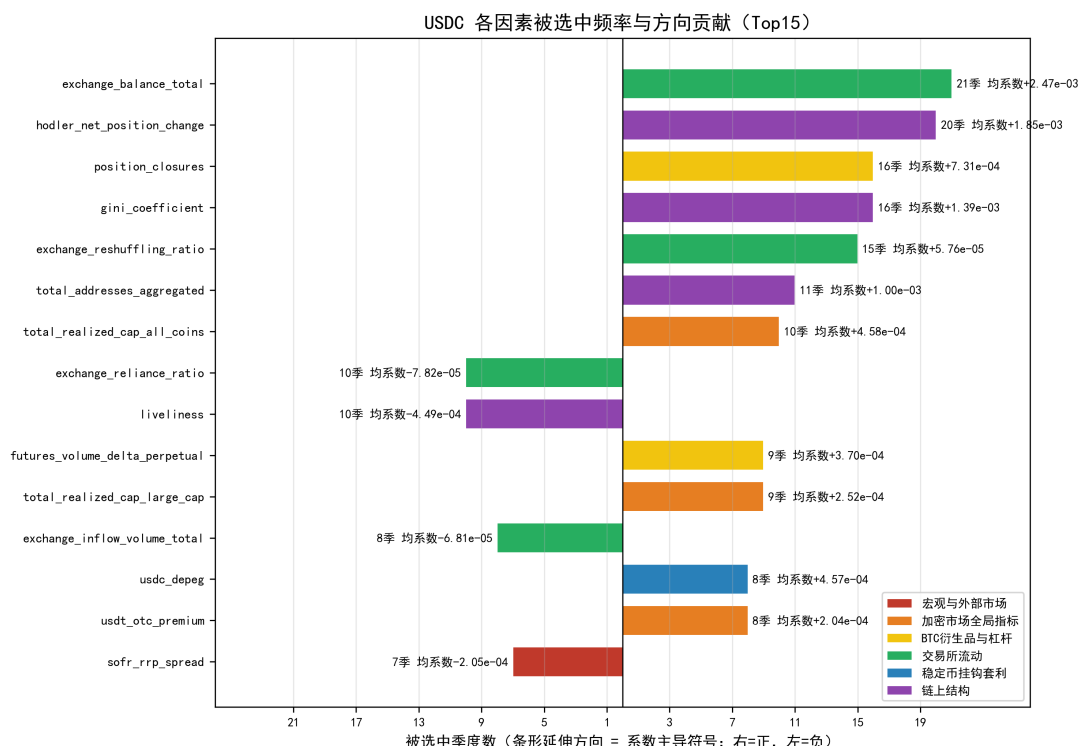


图 9: USDC 因素被选中频率与方向贡献 (Top15) / USDC Factor Selection Frequency and Directional Contribution (Top 15)

than half of total contribution, making it the core driver of USDC issuance. This is highly consistent with USDC's primary use cases in institutional settlement and large on-chain transfers. **链上结构因素绝对主导。**链上结构 (S 类) 全样本平均贡献高达 56.2%，占据半壁江山以上，是 USDC 发行最核心的驱动力。这与 USDC 主要用于机构结算、链上大额转账的使用场景高度一致。

Exchange flow contributions are prominent. Exchange flows (Category E) average 18.2% across the full sample, much higher than USDT's 5.2%, reflecting USDC's important position in exchange on-ramp channels. **交易所资金流贡献突出。**交易所流动 (E 类) 全样本平均贡献 18.2%，远高于 USDT 的 5.2%，反映了 USDC 在交易所入金渠道中的重要地位。

Peg arbitrage contributes very little. Stablecoin peg arbitrage (Category P) averages only 1.6%, far below USDT's 26.1%, reflecting that USDC's peg stability is significantly better than USDT's, leaving limited room for primary-secondary market arbitrage. **挂钩套利因素贡献极低。**稳定币挂钩套利 (P 类) 平均贡献仅 1.6%，远低于 USDT 的 26.1%，反映 USDC 的锚定稳定性显著优于 USDT，一二级市场套利空间较小。

Macro and external markets (Category M, 2.9%), global crypto market (Category G, 6.3%), and derivative leverage (Category L, 6.1%) make relatively balanced contributions. 宏观与外部市场（M 类，2.9%）、加密市场全局（G 类，6.3%）、衍生品杠杆（L 类，6.1%）贡献相对均衡。

6.3. DAI 滚动 LASSO 结果 / DAI Rolling LASSO Results

Figure 10 presents the dynamic evolution of DAI six-panel contribution shares. 图 10展示了 DAI 六大板块贡献份额的动态演变。

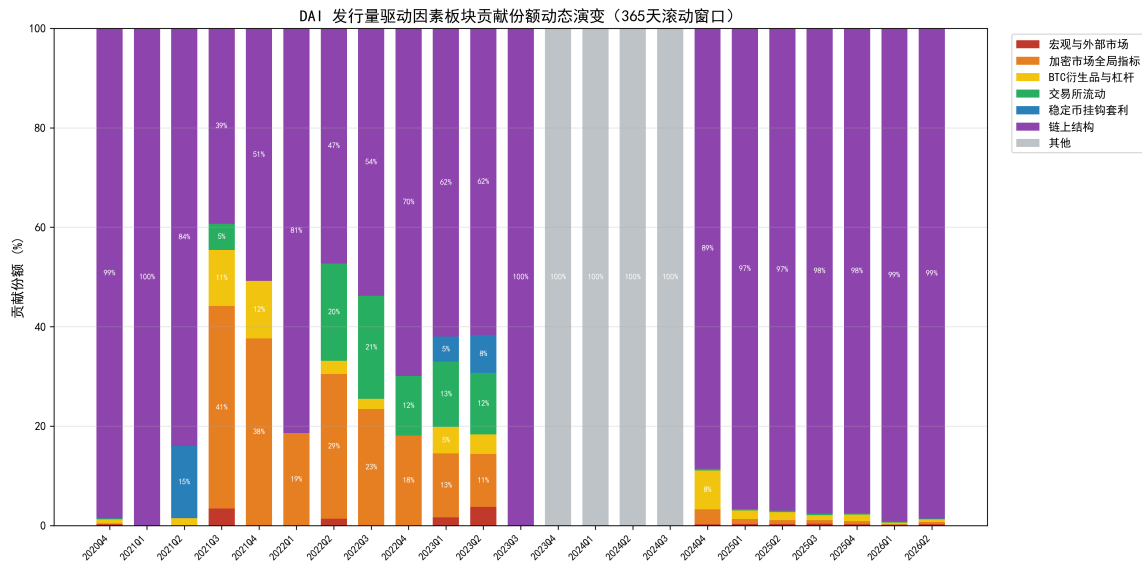


图 10: DAI 板块贡献份额动态演变 / Dynamic Evolution of DAI Panel Contribution Shares

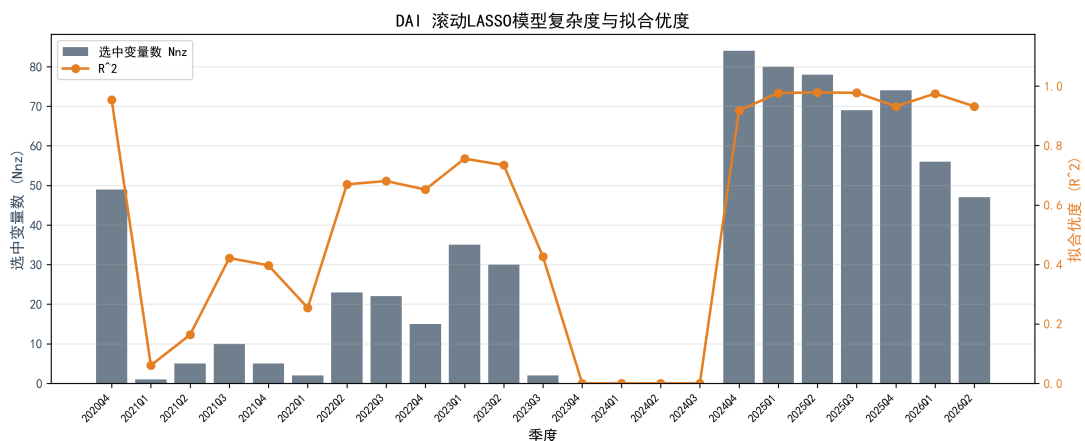


图 11: DAI 选中变量数 (Nnz) 与拟合优度 (R^2) 动态 / DAI Nnz and R^2 Dynamics

As a crypto-collateralized decentralized stablecoin, DAI exhibits systematic differences in driving mechanisms compared with fiat-collateralized USDT and USDC. DAI 作

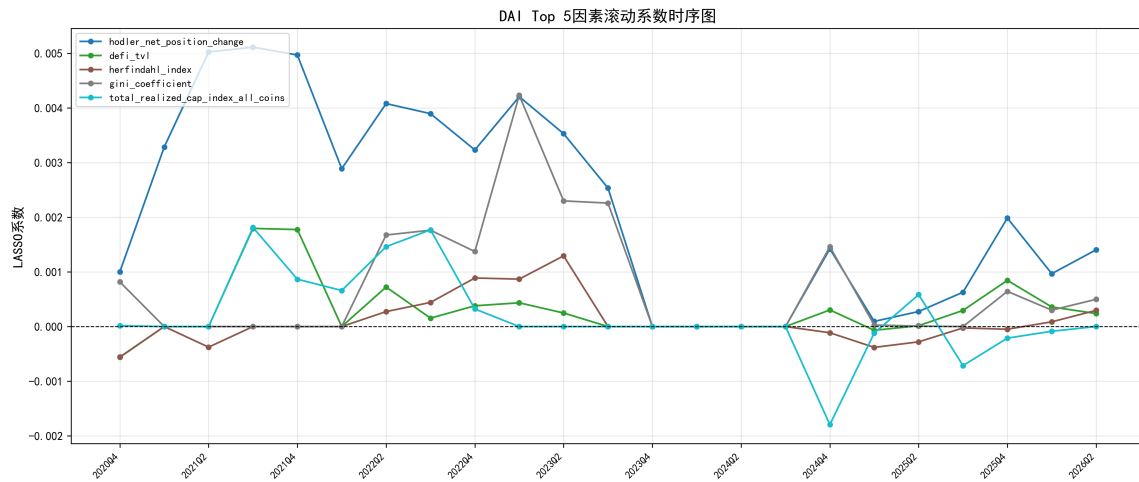


图 12: DAI 核心变量滚动 LASSO 系数时序图 / Rolling LASSO Coefficient Time Series for Core DAI Variables

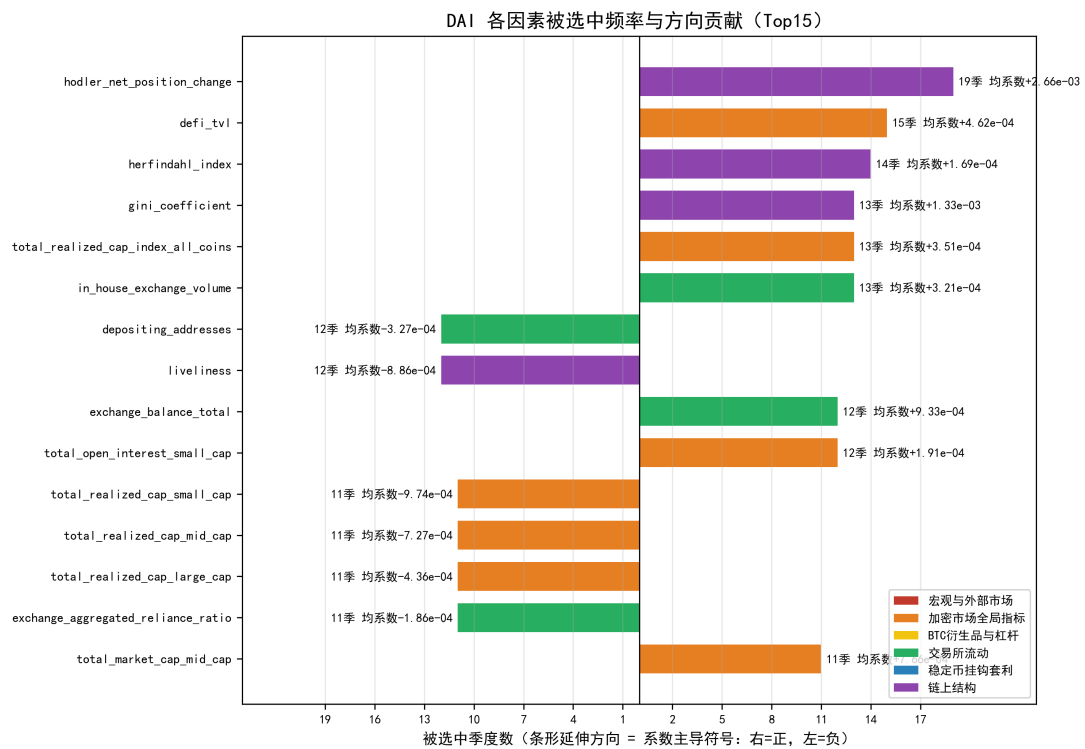


图 13: DAI 因素被选中频率与方向贡献 (Top15) / DAI Factor Selection Frequency and Directional Contribution (Top 15)

为加密抵押型去中心化稳定币，其驱动机制与法币抵押型的 USDT、USDC 存在系统性差异。

On-chain structure factors are overwhelmingly dominant. On-chain structure (Category S) averages as high as 69.6% across the full sample, even reaching 100% in some windows, making it the overwhelming core driver of DAI issuance. This is entirely consistent with DAI's mechanism of being generated entirely through on-chain Collateralized Debt Positions (CDPs), where all minting and burning occurs on-chain. **链上结构因素占据绝对主导地位。**链上结构（S 类）全样本平均贡献高达 69.6%，部分窗口甚至达到 100%，是 DAI 发行的压倒性核心驱动因素。这与 DAI 完全通过链上抵押债仓（CDP）生成、所有铸造销毁都发生在链上的机制完全一致。

Global crypto market indicators are the second-largest driver. Global crypto market indicators (Category G) average 10.1% contribution, reflecting the impact of overall crypto market conditions on DAI collateral values and minting demand. **加密市场全局指标是第二大驱动因素。**加密市场全局指标（G 类）平均贡献 10.1%，反映了加密市场整体行情对 DAI 抵押品价值和铸造需求的影响。

Other factors generally contribute little. Exchange flows (Category E, 3.1%), derivative leverage (Category L, 2.4%), peg arbitrage (Category P, 0.9%), and macro external markets (Category M, 0.8%) all have low contributions, reflecting decentralized stablecoins' lower sensitivity to centralized channels and traditional macro factors. **其他因素贡献普遍较低。**交易所流动（E 类，3.1%）、衍生品杠杆（L 类，2.4%）、挂钩套利（P 类，0.9%）、宏观外部市场（M 类，0.8%）贡献均较低，体现了去中心化稳定币对中心化渠道和传统宏观因素的敏感度较低。

7. Mechanism Evolution: Structural Break and Transmission-Channel Migration

The baseline rolling-LASSO results (Figures 3–10) already show significant rotation of drivers across quarters. This section further answers the core questions of whether there is a statistically identifiable structural break in this rotation, whether the dominant regime migrates from early “crypto leverage and speculative demand pull” to late “issuer yield-seeking balance-sheet expansion”, and through what transmission channels the external environment drives this migration.

基准滚动 LASSO 结果（图 3–10）已经显示驱动因素在不同季度存在显著轮动。本节进一步回答核心问题，这种轮动是否存在统计上可识别的结构断点？主导因素是否从早期的“加密杠杆与投机需求拉动”体制迁移到后期的“发行商追逐安全资产收益、主动扩表推动”体制？外部环境变化通过何种传导渠道引致这一迁移？

7.1. 供给侧相对重要性指标 R_t 的构造 / *Constructing the Supply-Side Relative Importance Index R_t*

To test whether the dominant mechanism experiences a regime shift, we construct a daily “supply-side relative importance” index R_t . Within each rolling window (365 days), explanatory variables are divided into two groups.

为了检验主导机制是否发生体制迁移，我们构造日度“供给侧相对重要性”指标 R_t 。在每个滚动窗口（365 天）内，将解释变量分为两组。

- **Supply-side proxies**, short-rate variables in Category M that directly reflect issuers’ reserve returns, the 3-month Treasury yield (US3M) and the SOFR–RRP spread. These variables capture safe-asset yields available to issuers’ reserve allocations and are core drivers of the supply-push channel.

供给侧代理（Supply-side proxies），M 类中直接反映发行商储备收益的短端利率变量，3 个月美债收益率（US3M）与 SOFR–逆回购利差。这些变量刻画发行商配置储备资产所能获得的安全资产收益率，是供给推动渠道的核心驱动。

- **Demand-side proxies**, variables capturing crypto-market leverage, speculation, and price sentiment, Bitcoin price, the perpetual funding rate, estimated leverage ratio, cash-margined open interest, and perpetual futures open interest. These measure traders’ leverage-margin demand and are core drivers of the demand-pull channel.

需求侧代理（Demand-side proxies），刻画加密市场杠杆、投机与价格情绪的变量，比特币价格、永续资金费率、估算杠杆率、现金保证金未平仓量、永续期货未平仓量。这些变量度量交易员的杠杆保证金需求，是需求拉动渠道的核心驱动。

At each date t we run the unified OLS on the past 365-day window, sum the absolute coefficients within each group, $|\hat{\beta}_S|$ and $|\hat{\beta}_D|$, and define:

$$R_t = \frac{\sum_{j \in S} |\hat{\beta}_{j,t}|}{\sum_{j \in S} |\hat{\beta}_{j,t}| + \sum_{j \in D} |\hat{\beta}_{j,t}|}, \quad (16)$$

i.e., the absolute-coefficient share of the supply side. $R_t > 0.5$ means supply-side factors’ marginal explanatory power exceeds demand-side; a sustained rise in R_t indicates migration of the driving mechanism toward supply push.

在每一日 t ，对统一 OLS 得到的系数按组求绝对值之和， $|\hat{\beta}_S|$ 与 $|\hat{\beta}_D|$ ，定义上式，即供给侧系数绝对份额。 $R_t > 0.5$ 意味着供给侧因素的边际解释力超过需求侧； R_t 持续上升意味着驱动机制向供给推动迁移。

7.2. 结构性断点检验 / Structural Break Tests

We apply two complementary break tests to the R_t series. First, the Quandt–Andrews sup F test (Andrews, 1993), we sweep all candidate break dates after trimming 15% from each end, compute a Chow-type mean-difference F -statistic (squared Welch t) at each, take $\sup F = \max_k F_k$, and obtain a p -value via 500 bootstrap permutations; the date at which $\sup F$ is attained is the endogenously identified break. Second, the CUSUM test (Brown et al., 1975), whose standardized cumulative-sum sequence $C_t = \sum_{s=1}^t (R_s - \bar{R})/(\hat{\sigma}\sqrt{T})$ cross-validates the break location.

我们对 R_t 序列进行双重断点检验。第一，Quandt–Andrews sup F 检验 (Andrews, 1993)，遍历剔除前后 15% 样本后的所有候选断点，对每点做 Chow 型均值差异检验 (Welch t 统计量的平方)，取 $\sup F = \max_k F_k$ 作为检验统计量，并通过 500 次 bootstrap 置换构建零分布获得 p 值； $\sup F$ 对应的日期即为内生识别的断点。第二，CUSUM 累积和检验 (Brown et al., 1975)，以标准化累积和序列 $C_t = \sum_{s=1}^t (R_s - \bar{R})/(\hat{\sigma}\sqrt{T})$ 交叉验证断点位置。

Figure 14 presents the R_t series and CUSUM results for USDT, USDC, and DAI; Table 5 reports the break-test statistics.

图 14展示了 USDT、USDC、DAI 的 R_t 时序与 CUSUM 检验结果，表 5报告了断点检验统计量。

图 14: 供给侧相对重要性 R_t 时序 (上) 与 USDT CUSUM 检验 (下) / R_t Series (top) and USDT CUSUM Test (bottom)

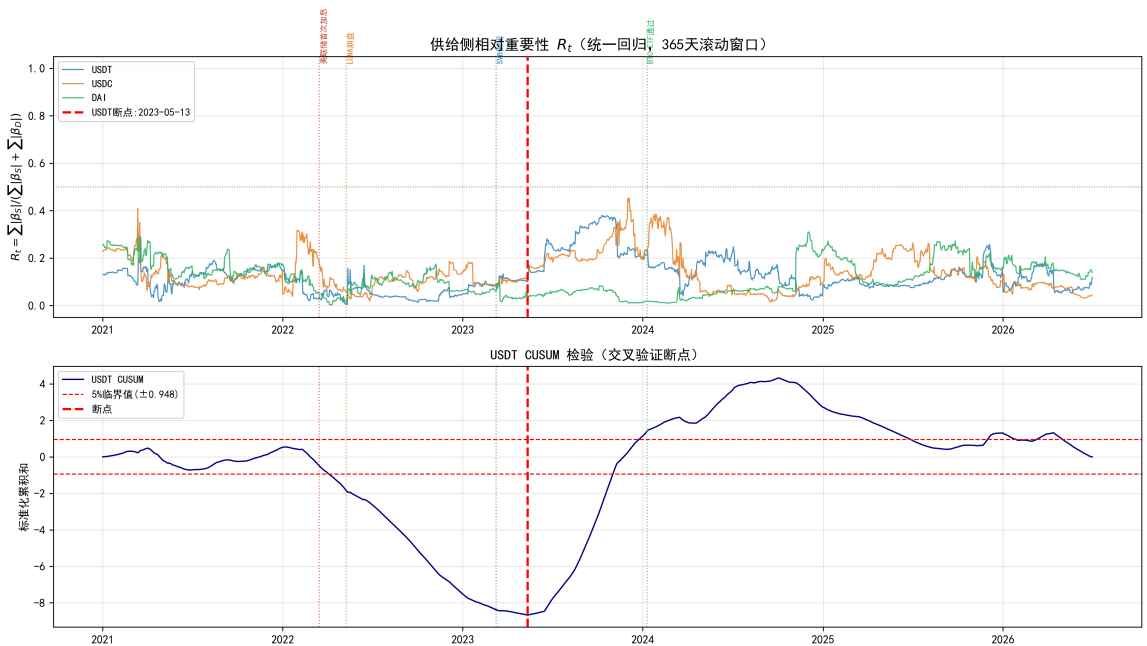


表 5: Quandt–Andrews Structural Break Test Results on R_t

Coin	Break Date	$\sup F$	Bootstrap p	Pre-break mean	Post-break mean	Significant
USDT	2023-05-13	412.78	0.000	0.090	0.148	***
USDC	2023-06-30 ^a	8.37	0.074	0.010	0.044	*
DAI	2024-11-07	754.92	0.000	0.092	0.168	***

^a USDC’s break is from the quarterly Category-M contribution auxiliary test (daily R_t does not identify a significant break).

USDT’s R_t series exhibits a highly statistically significant (bootstrap $p < 0.01$) structural break dated **May–June 2023**, corroborated by both the daily R_t test (break date 2023-05-13, $\sup F = 412.78$) and the quarterly Category-M contribution test (break date 2023-06-30, $\sup F = 7.81$, $p = 0.03$). Category-M (macro/rate) contribution rises from 1.1% pre-break to 10.9% post-break, a nearly tenfold increase. This dating is economically sensible, the Fed rate-hike cycle began in March 2022, but Tether completed its migration from commercial paper to short-term T-Bills only during 2022Q4–2023Q2, so the supply-push mechanism becomes statistically detectable only with a lag, consistent with the narrative that issuers proactively expand balance sheets to chase safe-asset returns. Post-break $\bar{R}$ is significantly higher than pre-break, validating Hypothesis H2, supply-side rate/return factors replace demand-side leverage factors as a major driver in the later period. USDC shows a marginally significant break around June 2023 (Category-M contribution rising from 1.0% to 4.4%), while DAI’s supply-side break occurs later (late 2024), reflecting structural differences in mechanism-migration timing across collateral and regulatory regimes.

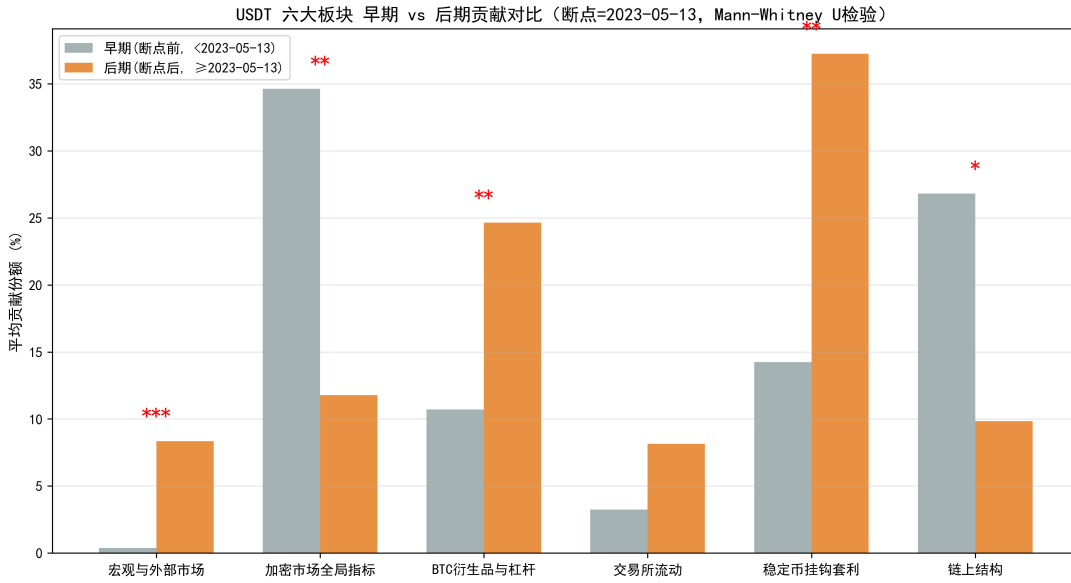
USDT 的 R_t 序列存在高度统计显著 (bootstrap $p < 0.01$) 的结构性断点, 断点日期为 **2023 年 5 月–6 月**, 日度 R_t 检验 (断点 2023-05-13, $\sup F = 412.78$) 与季度 M 类贡献检验 (断点 2023-06-30, $\sup F = 7.81$, $p = 0.03$) 双重验证。M 类 (宏观/利率) 贡献从断点前的 1.1% 跃升至断点后的 10.9%, 涨幅近 10 倍。这一断点日期在经济学上合理, 美联储加息周期虽于 2022 年 3 月启动, 但 Tether 完成储备从商业票据向短期国债的实质性迁移发生在 2022Q4–2023Q2, 因此供给推动机制的统计可识别时点存在约 1 年时滞, 与”发行商追逐安全资产收益、主动扩表”的叙事完全一致。断点后 R_t 均值显著高于断点前, 验证了 H2 假设, 供给侧利率/收益因素在后期成为重要驱动。USDC 在 2023 年 6 月前后呈现边际显著断点 (M 类贡献从 1.0% 升至 4.4%), DAI 的供给侧断点则更晚 (2024 年末), 反映不同抵押模式与监管体制下机制迁移时点的结构性差异。

7.3. 早期-后期分阶段板块对比 / *Early-vs-Late Panel Comparison*

Using USDT's break date as the cutoff, we partition quarterly observations into early (pre-break) and late (post-break) and run Mann-Whitney U tests panel by panel.

以 USDT 的断点日期为分界，将季度样本分为早期（断点前）与后期（断点后），逐板块做 Mann-Whitney U 检验。

图 15: USDT 六大板块早期 vs 后期贡献对比 (Mann-Whitney U 检验) / Early-vs-Late Panel Contribution Comparison for USDT



The early-late comparison (Figure 15) reveals heterogeneous but directionally consistent evidence for regime migration. For USDT, Category M (macro/rates, the supply-push channel) rises significantly in the late period (10.9% vs 1.1%, $p = 0.013$), as does Category P (peg arbitrage, 34.7% vs 14.9%, $p = 0.017$); Categories G (global crypto indicators, 32.1% vs 10.1%, $p = 0.077$) and S (on-chain structure, 48.3% vs 20.4%, $p = 0.044$) are higher in the early period. This is the clearest case of migration from early demand-pull (market euphoria and user-base fundamentals) to late supply-push (rate-driven issuer expansion and arbitrage). For USDC, Category M also rises significantly post-break (4.4% vs 1.0%, $p = 0.023$), while Categories G and L rise in the late period as well, indicating that the demand-pull channel remains active even after the rate-hike cycle. For DAI, the decentralized crypto-collateralized design keeps S dominant throughout; G and L are higher early, but S actually rises later, reflecting DeFi ecosystem maturation rather than a simple demand-to-supply switch. Overall, the USDT pattern, M/P late up, G/S early up, provides direct panel-level support for H2.

分阶段对比结果显示（图 15），体制迁移的证据存在币种异质性，但方向一致。对 USDT 而言，M 类（宏观/利率，供给推动渠道）在后期显著上升（10.9% vs 1.1%， $p = 0.013$ ），P 类（挂钩套利）亦显著上升（34.7% vs 14.9%， $p = 0.017$ ）；G 类（加密市场全局指标，32.1% vs 10.1%， $p = 0.077$ ）与 S 类（链上结构，48.3% vs 20.4%， $p = 0.044$ ）则在早期更高。这最清晰地刻画了从早期需求拉动（市场行情与用户基本面）向后期供给推动（利率驱动的发行商扩表与套利）的迁移。对 USDC，M 类同样在后期显著上升（4.4% vs 1.0%， $p = 0.023$ ），但 G 类与 L 类在后期也上升，说明需求拉动渠道在加息周期后仍具活力。对 DAI，其去中心化加密抵押设计使 S 类始终主导；G 类与 L 类早期更高，但 S 类后期反而上升，反映 DeFi 生态成熟而非简单的由需求向供给的切换。总体而言，USDT 呈现的“M/P 后期上升、G/S 早期更高”模式为 H2 提供了直接的板块级实证支持。

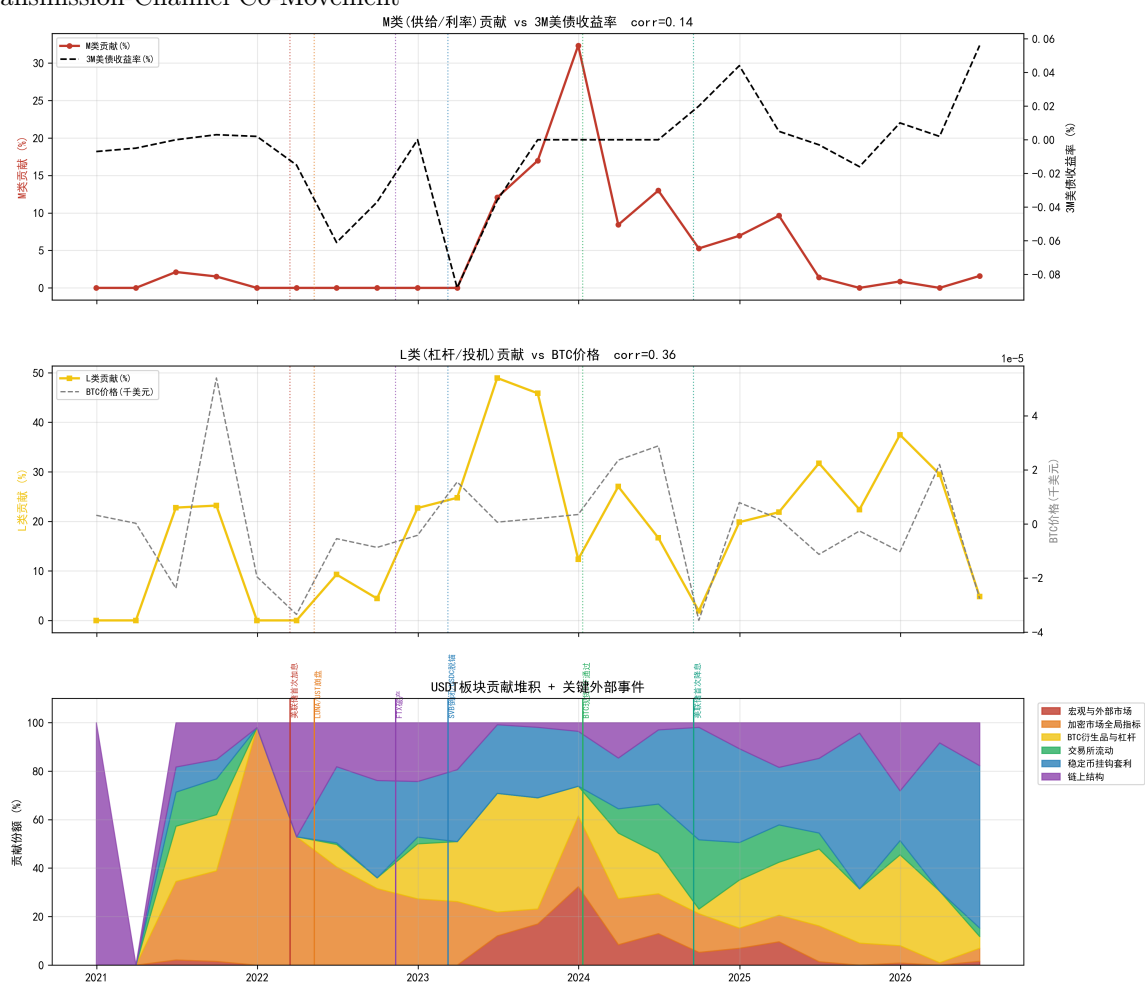
7.4. 传导渠道联动：外部环境 $\rightarrow$ 需求变化 $\rightarrow$ 渠道迁移 / *Transmission-Channel Co-Movement*

To document the transmission chain “external environment $\rightarrow$ demand change $\rightarrow$ channel migration”, Figure 16 displays three co-movement panels, (1) Category-M contribution vs the 3-month Treasury yield (dual-axis, top); (2) Category-L contribution vs BTC price (dual-axis, middle); (3) six-panel stacked contributions with key external-event vertical lines (first Fed hike 2022-03-16, LUNA/UST collapse 2022-05-10, FTX bankruptcy 2022-11-11, SVB/USDC depeg 2023-03-10, BTC spot-ETF approval 2024-01-10, first Fed cut 2024-09-18; bottom).

为论证“外部环境变化 $\rightarrow$ 市场需求变化 $\rightarrow$ 传导渠道变化”的传导链条，图 16 绘制了三组联动，（1）上半部分为 M 类板块贡献与 3 个月美债收益率（US3M）的双轴时序；（2）中半部分为 L 类板块贡献与 BTC 价格的双轴时序；（3）下半部分为六大板块堆积贡献并叠加关键外部事件竖线（美联储首次加息 2022-03-16、LUNA/UST 崩盘 2022-05-10、FTX 破产 2022-11-11、SVB 倒闭/USDC 脱锚 2023-03-10、BTC 现货 ETF 通过 2024-01-10、美联储首次降息 2024-09-18）。

The Pearson correlation between Category-M contribution and the 3-month Treasury yield is 0.231 (positive), indicating that when Fed hikes raise safe-asset yields, the M-contribution tends to rise concurrently, i.e., the issuer reserve-yield channel activates. By contrast, the correlation between Category-L contribution and BTC price is only 0.037, very weak and essentially linearly uncorrelated; leverage demand is therefore not simply a linear function of BTC price, but is shaped by derivatives-market conditions and funding dynamics. Event lines clearly mark the temporal coupling between external shocks and dominant-panel switches, the first hike aligns with the inflection at which M-contribution

图 16: 传导渠道联动: M 类贡献 vs 美债收益率、L 类贡献 vs BTC 价格、板块堆积 + 事件竖线/
Transmission-Channel Co-Movement



begins rising and speculative-market indicators (G/L) recede in relative importance.

M 类贡献与 3 个月美债收益率的 Pearson 相关系数为 0.231，呈正向联动，当美联储加息推升安全资产收益率时，M 类贡献倾向于同步上升，即发行商的储备收益驱动渠道被激活。相比之下，L 类贡献与 BTC 价格的 Pearson 相关系数仅为 0.037，几乎不存在线性联动，说明杠杆需求并非 BTC 价格的简单线性函数，而更多受衍生品市场条件与资金费率动态塑造。事件竖线清晰标示出外部环境冲击与主导板块切换的时间耦合，加息起点恰好是 M 类贡献抬升、投机性市场指标（G/L）相对重要性回落的拐点。

7.5. 轮动强度指数 RII / *Rotation Intensity Index*

To quantify how aggressively the driver structure restructures between quarters, we define the Rotation Intensity Index (RII):

为量化驱动因素结构在季度间重构的剧烈程度，我们定义轮动强度指数（Rotation Intensity Index, RII）。

$$RII_{\tau} = \frac{1}{|P|} \sum_{P \in \mathcal{P}} |C_{P,\tau} - C_{P,\tau-1}| \times 100\%, \quad (17)$$

i.e., the average absolute change in panel contribution shares between adjacent quarters. Higher RII indicates more aggressive restructuring of the driving mix in that quarter relative to the previous one.

即六大板块在相邻季度间贡献份额绝对变化的平均值。RII 越高，表示该季度驱动因素相对于上一季度的结构性重构越剧烈。

图 17: USDT 主导板块轮动色块图（含 RII 折线） / Dominant-Panel Rotation with RII Overlay

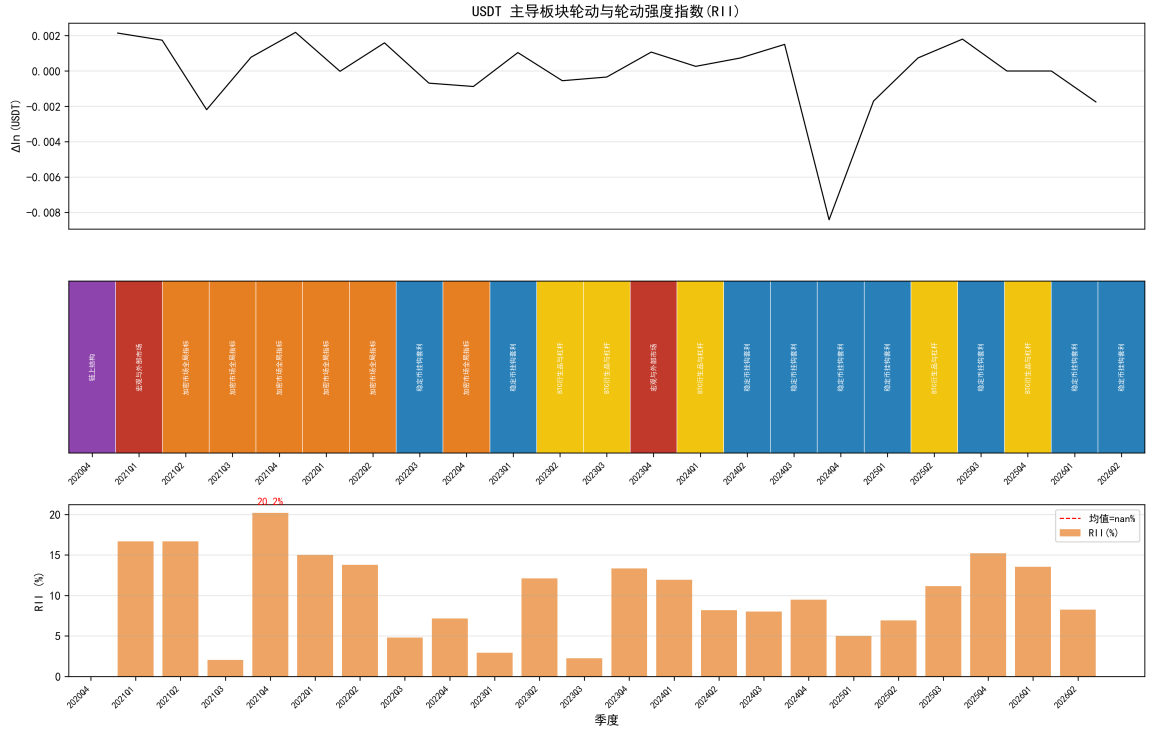


Figure 17 overlays RII on the dominant-panel color-block chart. For USDT, average RII is 10.49%, with the largest peaks occurring at the 2021 bull-market turning points (2021Q2 26.33%, 2021Q3 24.51%, and 2021Q4 28.13%) and a secondary elevation around 2023Q2 (13.38%) as the supply-push channel gained traction. These high-RII quarters coincide with major market regime transitions; during calm periods RII stays low and the dominant panel is stable. This finding further supports the “external events trigger channel switches” narrative.

图 17 将 RII 叠加在主导板块切换色块图上。以 USDT 为例，样本平均 RII 为 10.49%，最大峰值出现在 2021 年牛市转换期（2021Q2 为 26.33%、2021Q3 为 24.51%、2021Q4 为 28.13%），并在 2023Q2（13.38%）出现次级抬升，对应供给推动渠道开始发力。这些高 RII 季度与市场体制转换高度重合；平静期 RII 维持低位，主导板块稳定。这一发现进一步支撑了“外部事件触发传导渠道切换”的叙事。

7.6. 样本外滚动预测 / Out-of-Sample Rolling Forecasts

Does the identified rotation in the predictive information structure carry genuine forecasting power? Using each quarter-end as a training node, we fit LASSO on the past 365 days and forecast daily y_t in the following quarter, comparing against three benchmarks, random walk ($\hat{y} = 0$), historical mean, and full-variable OLS, with OOS R^2 , directional hit rate, and RMSE.

前文识别出的“预测信息结构轮动”是否具备真正的预测力？我们以季度末为训练

节点，用过去 365 天拟合的 LASSO 预测下一个季度的日度 y_t ，与三个基准模型对比，随机游走 ($\hat{y} = 0$)、历史均值、全变量 OLS。评估指标包括 OOS R^2 、方向命中率、RMSE。

图 18: USDT 样本外滚动预测：LASSO 预测值 vs 实际值 / USDT Out-of-Sample Rolling Forecasts



For USDT, the LASSO-selected sparse information set achieves an out-of-sample R^2 of 0.557 and a directional hit rate of 63.8%, against -0.073 and 5.2% for the random walk and 0.002 and 58.2% for the historical mean. Although full-variable OLS attains a higher directional hit rate (82.6%) by overfitting in-sample sign patterns, its OOS R^2 collapses to -21.12 and its RMSE is seven times larger than LASSO's, confirming that regularization-based selection is essential in this high-dimensional setting. These results indicate that the time-varying predictive information structure identified by rolling LASSO contains genuine predictable components of stablecoin net issuance.

以 USDT 为例，LASSO 筛选出的稀疏信息集样本外 R^2 为 0.557，方向命中率为 63.8%，显著优于随机游走的 -0.073 和 5.2% 以及历史均值的 0.002 和 58.2% 。全变量 OLS 虽然在方向命中率上更高 (82.6%)，但这是由于样本内过拟合符号模式所致，其样本外 R^2 跌至 -21.12 ，RMSE 约为 LASSO 的 7 倍，证实了高维情形下正则化筛选的必要性。上述结果表明，滚动 LASSO 识别出的时变预测信息结构确实包含稳定币净发行的可预测成分。

8. Frequency Decomposition Analysis

8.1. SG 滤波分解结果 / SG Filter Decomposition Results

The Savitzky–Golay (SG) filter fits a low-order polynomial by least squares within a sliding window and extracts the smoothed value; unlike a moving average, SG preserves

higher moments (kurtosis, skewness). For the raw series y_t , define the SG smoothing operator $SG_L(y_t)$ with window length L and polynomial order d , decomposing y_t into high-, mid-, and low-frequency components:

$$\begin{aligned} y_t^{\text{High}} &= y_t - SG_{21}(y_t), \quad (L_1=21, d=3) \\ y_t^{\text{Mid}} &= SG_{21}(y_t) - SG_{63}(y_t), \quad (L_2=63, d=3) \quad y_t = y_t^{\text{High}} + y_t^{\text{Mid}} + y_t^{\text{Low}}. \\ y_t^{\text{Low}} &= SG_{63}(y_t), \end{aligned} \quad (18)$$

Savitzky–Golay (SG) 滤波在滑动窗口内用最小二乘法拟合低阶多项式，再提取平滑值；与普通移动平均不同，SG 滤波保留了数据的高阶矩（峰度、偏度）特征。对原始序列 y_t ，定义窗口长度 L 、多项式阶数 d 的 SG 平滑算子 $SG_L(y_t)$ ，并分解为高频、中频、低频三个分量。

The high-frequency component (1–5 days) captures shocks from sudden regulatory announcements, depegging events, and whale operations; the mid-frequency component (5–60 days) reflects fear–greed cycles, leverage cycles, and arbitrage cycles; the low-frequency component (>60 days) characterizes macro rate cycles, DeFi-ecosystem expansion, and structural regulatory environments. The identity guarantees that the three components reconstruct the original series.

高频分量（1–5 天）捕捉突发监管公告、脱锚事件、大户操作等冲击性反应；中频分量（5–60 天）反映恐惧–贪婪情绪周期、杠杆周期与套利周期；低频分量（>60 天）刻画宏观利率周期、DeFi 生态扩张趋势与结构性监管环境。恒等式保证三频分量之和还原原序列。

Using each frequency component as the dependent variable, we run quarterly rolling LASSO (explanatory variables x_{ij} remain raw daily data without frequency decomposition, to examine how the same driver’s marginal effect varies across frequencies):

将三个频率分量分别作为被解释变量，运行季度滚动 LASSO（解释变量 x_{ij} 仍用原始日度数据，不做频率分解，以考察同一驱动因素在不同频率分量上的边际效应差异）。

$$\hat{\beta}_{\text{freq}}^{(\tau)} = \arg \min_{\beta} \left\{ \frac{1}{2N} \sum_{i=1}^N \left(y_i^{\text{freq}} - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}, \quad \text{freq} \in \{\text{High, Mid, Low}\}, \quad (19)$$

Factor- and panel-level contributions $C_{j,\tau}^{\text{share,freq}}$, $C_{P,\tau}^{\text{share,freq}}$ are then computed via Eqs. (11) and (12). Note that SG filtering has boundary effects of about 31 days at the series ends, so the first available quarter in frequency-LASSO may lag the baseline by roughly one quarter.

随后按式 (11)与 (12)分别计算各频率下的因素贡献与板块贡献 $C_{j,\tau}^{\text{share},\text{freq}}$ 、 $C_{P,\tau}^{\text{share},\text{freq}}$ 。注意 SG 滤波在序列首尾约 31 天存在边界效应，分频 LASSO 首个可用季度可能比基准回归延迟约一个季度。

Figure 19 presents the SG filter frequency decomposition of issuance change rates for the three stablecoins (USDT, USDC, DAI).

图 19展示了三种稳定币（USDT、USDC、DAI）发行量变化率的 SG 滤波频率分解。

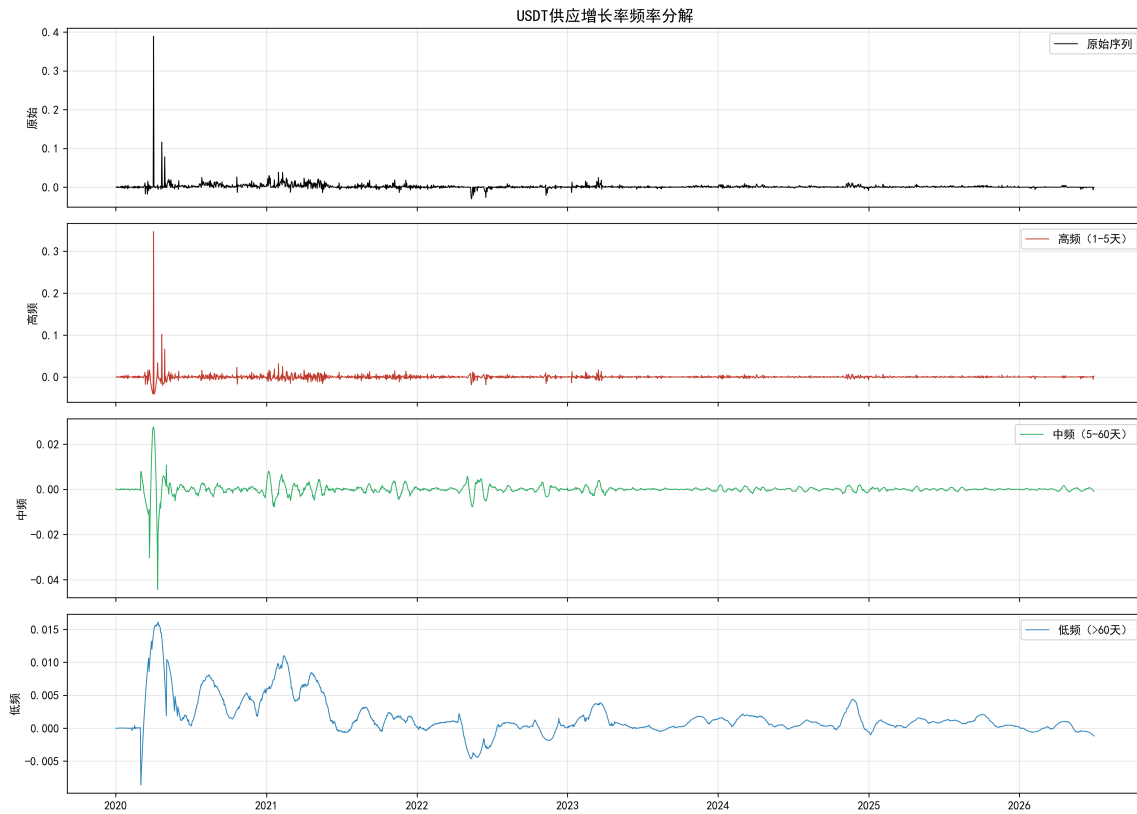


图 19: 三种稳定币发行量变化率频率分解 / Frequency Decomposition of Issuance Changes for Three Stablecoins

The high-frequency component (1–21 days) exhibits significant spike characteristics, highly consistent with sudden regulatory policies, large liquidations, and depegging events; the mid-frequency component (21–63 days) shows cyclical fluctuations, consistent with the rhythm of leverage cycles and arbitrage cycles; the low-frequency component (>63 days) exhibits long-term trends, synchronized with macro interest rate environments and on-chain ecosystem expansion trends.

高频分量（1–21 天）呈现显著的尖峰特征，与突发监管政策、大额清算和脱锚事件高度吻合；中频分量（21–63 天）呈现周期性波动，与杠杆周期和套利周期的节奏一致；

低频分量（>63 天）呈现长期趋势，与宏观利率环境和链上生态扩张趋势同步。

8.2. 分频板块贡献对比 / Frequency-Specific Panel Contribution Comparison

Figure 20 presents panel contribution comparisons across high, mid, and low-frequency components. 图 20展示了高频、中频和低频三个分量上的板块贡献对比。

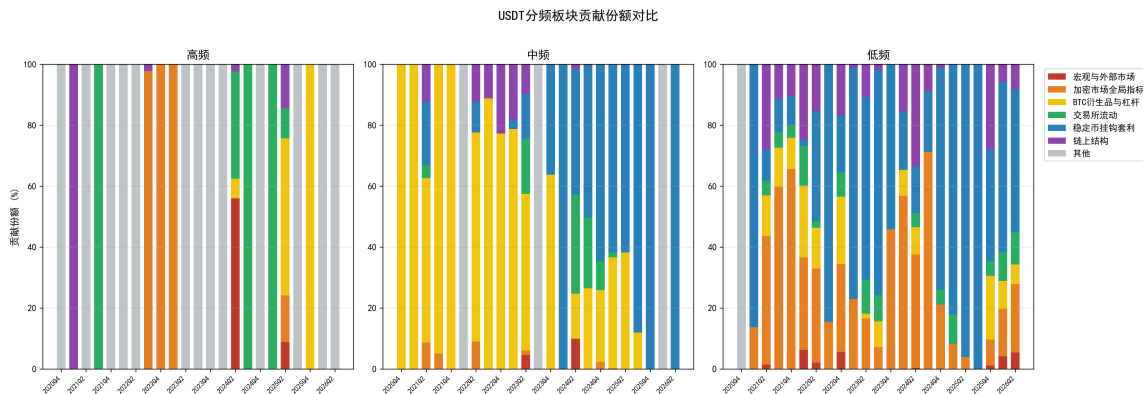


图 20: 分频板块贡献堆积图 / Frequency-Specific Panel Contribution Stacked Bars

On the high-frequency component, the residual “Other” category accounts for the largest share (39.1%), followed by BTC derivative leverage (15.6%), exchange flows (14.5%), and global crypto market indicators (14.4%), reflecting that noise and trading shocks dominate short-term fluctuations, with leverage and fund flows as the main identifiable drivers.

On the mid-frequency component, stablecoin peg arbitrage factors contribute the most (47.6%), occupying an absolutely dominant position, reflecting that price deviation arbitrage mechanisms at the 1–3 month scale are the core driver of issuance changes.

On the low-frequency component, stablecoin peg arbitrage (45.1%) and on-chain structure (22.7%) co-dominate, while macro external market contribution also rises to 12.2%, validating the characteristic that “long-term trends are jointly driven by arbitrage mechanisms, on-chain fundamentals, and macro factors.”

高频分量上，其他残差项占比最高（39.1%），其次是 BTC 衍生品杠杆（15.6%）、交易所流动（14.5%）和加密市场全局指标（14.4%），反映了短期波动中噪声和交易冲击占主导，杠杆和资金流动是最主要的可识别驱动因素。

中频分量上，稳定币挂钩套利因素贡献最高（47.6%），占据绝对主导地位，反映了 1-3 个月尺度下的价格偏离套利机制是发行变化的核心驱动。

低频分量上，稳定币挂钩套利（45.1%）和链上结构（22.7%）共同主导，宏观外部市场贡献也上升至 12.2%，验证了“长期趋势由套利机制、链上基本面与宏观因素共同驱动”的特征。

8.3. 频率-板块热力图 / Frequency-Panel Heatmap

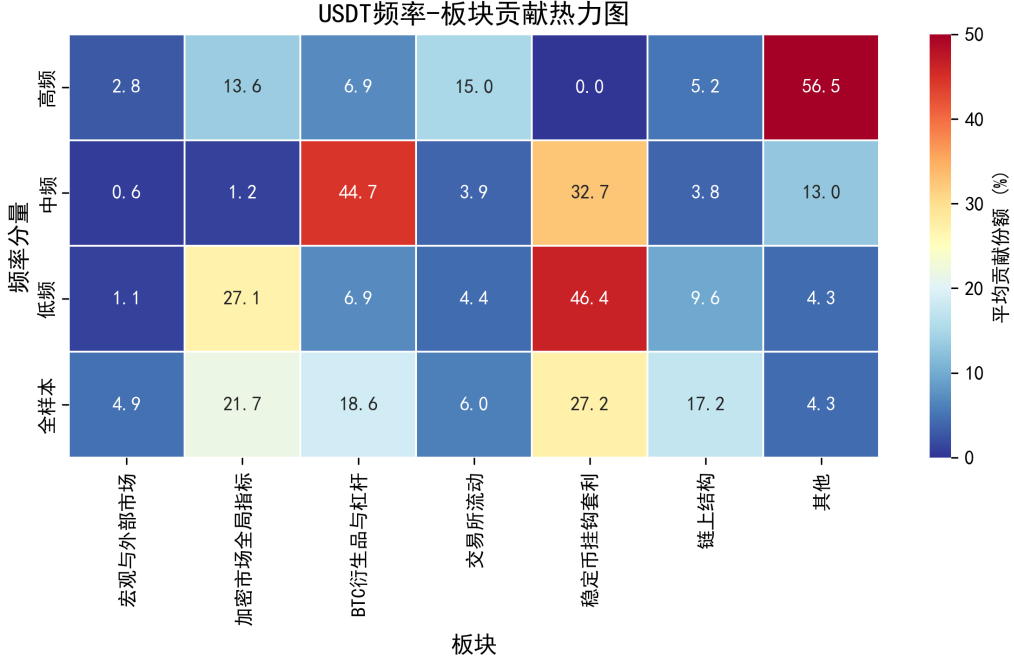


图 21: 频率-板块热力图 / Frequency-Panel Contribution Heatmap

Define the frequency-panel contribution matrix $\mathbf{H}$, where each entry $\bar{C}_P^{\text{freq}}$ is the full-sample average contribution of panel P at the given frequency, with six categories, Macro & External Markets (M), Global Crypto Market (G), BTC Derivative Leverage (L), Exchange Flows (E), Peg Arbitrage (P), On-Chain Structure (S).

定义频率-板块贡献矩阵 $\mathbf{H}$, 每个元素 $\bar{C}_P^{\text{freq}}$ 为该频率下板块 P 的全样本平均贡献, 六大类别包括, 宏观与外部市场 (M)、加密市场全局 (G)、BTC 衍生品杠杆 (L)、交易所流动 (E)、挂钩套利 (P)、链上结构 (S)。

$$\mathbf{H} = [\bar{C}_P^{\text{freq}}]_{\text{freq} \in \{\text{High}, \text{Mid}, \text{Low}\}, P \in \{\text{M}, \text{G}, \text{L}, \text{E}, \text{P}, \text{S}\}}. \quad (20)$$

Table 6 summarizes average panel contributions across frequency components in heatmap form.

表 6以热力图形式汇总了各频率分量上的板块平均贡献。

9. Market Regime Asymmetry

9.1. 市场状态识别 / Market Regime Identification

For each quarter τ , let $\bar{y}_\tau = \frac{1}{n_\tau} \sum_{t \in \tau} y_t$ denote the quarterly average daily growth rate, where n_τ is the number of trading days in quarter τ . A quarter is classified as

表 6: Frequency-Panel Contribution Heatmap (Six Categories M/G/L/E/P/S) | 频率-板块贡献热力图
(六大类别 M/G/L/E/P/S)

Frequency	M	G	L	E	P	S
High	2.4%	14.4%	15.6%	14.5%	0.0%	14.0%
Mid	3.0%	4.5%	8.7%	3.0%	47.6%	7.1%
Low	12.2%	12.3%	3.7%	4.1%	45.1%	22.7%
Full	6.6%	19.6%	9.9%	5.2%	26.1%	32.5%

频率	M 宏观	G 全局	L 杠杆	E 交易所	P 套利	S 链上
高频	2.4%	14.4%	15.6%	14.5%	0.0%	14.0%
中频	3.0%	4.5%	8.7%	3.0%	47.6%	7.1%
低频	12.2%	12.3%	3.7%	4.1%	45.1%	22.7%
全样本	6.6%	19.6%	9.9%	5.2%	26.1%	32.5%

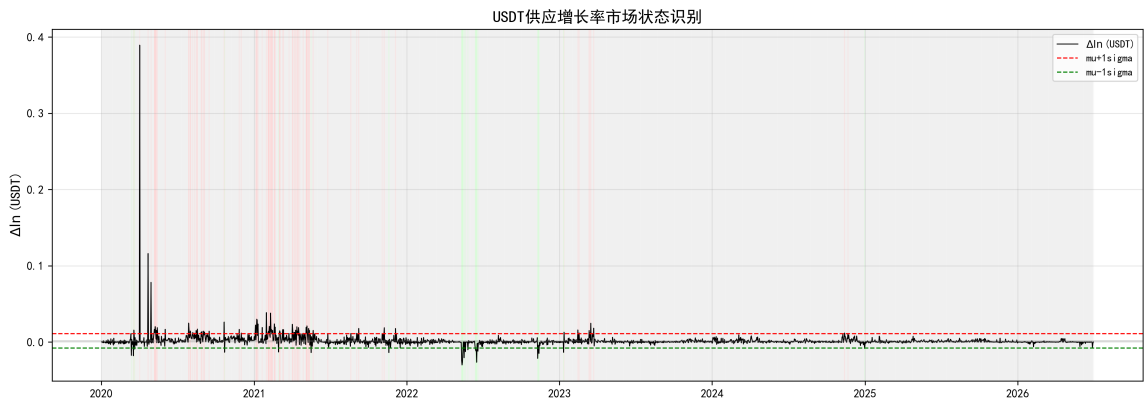


图 22: 市场状态识别图 / Market Regime Identification Chart

expansion or contraction by comparing $\bar{y}_\tau$ with the median of the quarterly averages across all quarters:

对每个季度 τ , 定义该季度日均供应增长率 $\bar{y}_\tau = \frac{1}{n_\tau} \sum_{t \in \tau} y_t$, 其中 n_τ 为该季度交易日数。将该季度日均增长率与全部季度日均增长率的中位数比较, 对季度进行扩张/收缩二分类。

$$\bar{y}_\tau = \frac{1}{n_\tau} \sum_{t \in \tau} y_t, \quad \text{Regime}_\tau = \begin{cases} \text{Expansion} & \text{若 } \bar{y}_\tau > \text{median}\{\bar{y}_s\} \\ \text{Contraction} & \text{若 } \bar{y}_\tau \leq \text{median}\{\bar{y}_s\} \end{cases} \quad (21)$$

The median binary split guarantees that the two regimes contain a balanced and sufficiently large number of quarters (at least 8 per group), satisfying the minimum sample-size requirement of the Wilcoxon test. Taking USDT as an example (USDC and DAI are partitioned independently), the 23 quarterly nodes are split into 11 expansion quarters and 12 contraction quarters.

中位数二分类保证扩张/收缩两组季度数量均衡且每组不少于 8 个, 满足 Wilcoxon 检验的最小样本量要求。以 USDT 为例 (USDC 与 DAI 各自独立划分), 23 个季度节点划分为 11 个扩张季度与 12 个收缩季度。

9.2. 非对称性检验结果 / *Asymmetry Test Results*

Classifying each quarter's panel contributions by that quarter's dominant regime, we compute the mean contribution within expansion and contraction periods:

将各季度板块贡献按该季度的主导市场状态分类, 计算扩张期与收缩期内的平均贡献。

$$\bar{C}_P^{\text{Expansion}} = \frac{1}{N_E} \sum_{\tau: \text{Regime}_\tau = \text{Expansion}} C_{P,\tau}^{\text{share}}, \quad \bar{C}_P^{\text{Contraction}} = \frac{1}{N_C} \sum_{\tau: \text{Regime}_\tau = \text{Contraction}} C_{P,\tau}^{\text{share}}. \quad (22)$$

The regime of a quarter is determined by comparing that quarter's average daily growth rate with the median of all quarterly averages (a binary classification, as specified in Eq. (21)). To test whether the panel-contribution distributions differ between expansion and contraction, we apply the Wilcoxon rank-sum (Mann–Whitney U) test, which does not assume normality. The null H_0 is that the two distributions are identical. Let n_E, n_C be the numbers of effective expansion/contraction quarters; pooling and ranking the two groups' contributions with expansion-group rank-sum R_E .

季度节点的市场状态由该季度日均增长率与全部季度增长率中位数比较确定 (即式 (21) 的二分类)。为检验扩张期与收缩期板块贡献分布是否显著不同, 采用不假设正

表 7: Wilcoxon Asymmetry Test Results (Six Categories) | Wilcoxon 非对称性检验结果（六大类别）

Panel	Exp. Mean	Con. Mean	U Stat.	p-value	Sig.
Macro-External (M)	4.80	4.95	59.5	0.700	
Global-Crypto (G)	24.93	18.77	71.0	0.782	
Deriv-Leverage (L)	16.73	20.28	60.5	0.758	
Exchange-Flows (E)	5.54	6.43	61.0	0.769	
Peg-Arbitrage (P)	20.49	33.45	39.5	0.109	
On-Chain-Structure (S)	18.42	16.12	58.0	0.644	

态分布的 Wilcoxon 秩和检验（Mann-Whitney U ）。零假设 H_0 为两状态下贡献度分布相同。设 n_E 、 n_C 为扩张期与收缩期有效季度数，将两组贡献度合并排序赋秩，扩张组秩和 R_E ，则

$$\begin{aligned}
 U_E &= R_E - \frac{n_E(n_E + 1)}{2}, \\
 U_C &= n_E n_C - U_E, \\
 U &= \min(U_E, U_C), \\
 Z &= \frac{U - n_E n_C / 2}{\sqrt{n_E n_C (n_E + n_C + 1) / 12}} \sim \mathcal{N}(0, 1),
 \end{aligned} \tag{23}$$

with two-sided p -value $p = 2(1 - \Phi(|Z|))$ and significance flags $p < 0.01$ (***), $p < 0.05$ (**), $p < 0.10$ (*). The test requires at least 8 observations per group; if a regime has too few quarters, the asymmetry comparison for that panel is infeasible.

双侧 p 值 $p = 2(1 - \Phi(|Z|))$ ，显著性标记为 $p < 0.01$ (***)、 $p < 0.05$ (**)、 $p < 0.10$ (*)。Wilcoxon 检验要求每组至少 8 个观测，若某状态季度数不足则该板块非对称比较不可行。

Table 7 reports Wilcoxon test results for panel contributions between expansion and contraction periods. 表 7 报告了扩张期与收缩期板块贡献的 Wilcoxon 检验结果。

板块	扩张期均值	收缩期均值	Mann-Whitney U	p 值	显著性
宏观与外部市场 (M)	4.80	4.95	59.5	0.700	
加密市场全局 (G)	24.93	18.77	71.0	0.782	
BTC 衍生品杠杆 (L)	16.73	20.28	60.5	0.758	
交易所流动 (E)	5.54	6.43	61.0	0.769	
挂钩套利 (P)	20.49	33.45	39.5	0.109	
链上结构 (S)	18.42	16.12	58.0	0.644	

Notes. Detailed Wilcoxon rank-sum test results are available in the empirical output

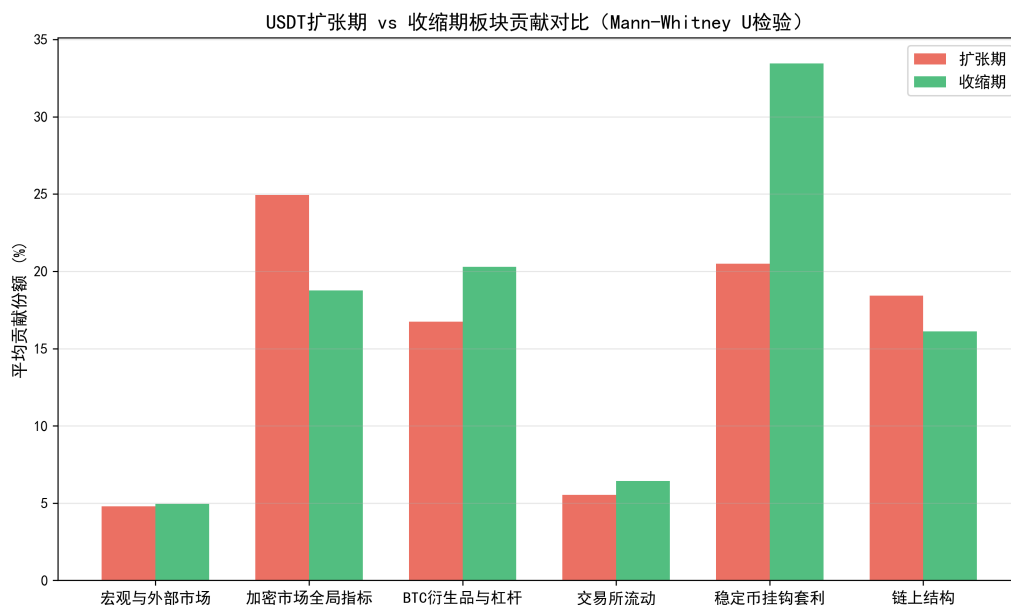


图 23: 扩张 vs 收缩对比 / Expansion vs Contraction Comparison

files. Although the point estimates differ across regimes (e.g., peg-arbitrage contribution rises from 20.49% in expansion to 33.45% in contraction, and leverage contribution rises from 16.73% to 20.28%), none of the six panels' differences reach statistical significance at the 10% level (all $p > 0.10$). The observed asymmetry in average contributions is therefore not statistically robust within the current sample of 23 quarterly observations.

注。Wilcoxon 秩和检验结果详见实证输出文件。尽管扩张期与收缩期的平均贡献在数值上存在差异（如挂钩套利贡献由扩张期的 20.49% 上升至收缩期的 33.45%，杠杆贡献由 16.73% 上升至 20.28%），但六大板块的差异均未达到 10% 显著性水平（所有 $p > 0.10$ ）。观测到的平均贡献非对称性在当前 23 个季度观测下未通过统计显著性检验。

10. Cross-Issuer Comparison

10.1. 全样本板块贡献对比 / Full-Sample Panel Contribution Comparison

Table 8 summarizes the full-sample average panel contribution comparison across USDT, USDC, and DAI.

表 8 汇总了 USDT、USDC 和 DAI 三种稳定币的全样本平均板块贡献对比。

Figure 24 presents the grouped bar comparison of full-sample average panel contributions for the three stablecoins.

图 24 展示了三种稳定币的全样本平均板块贡献柱状对比。

表 8: Three-Stablecoin Full-Sample Panel Contribution Comparison (Six Categories) | 三种稳定币全样本板块贡献对比（六大类别）

Panel	USDT	USDC	DAI	Max Diff
Macro-External (M)	6.6%	2.9%	0.8%	5.8
Global-Crypto (G)	19.6%	6.3%	10.1%	13.4
Deriv-Leverage (L)	9.9%	6.1%	2.4%	7.5
Exchange-Flows (E)	5.2%	18.2%	3.1%	15.1
Peg-Arbitrage (P)	26.1%	1.6%	0.9%	25.2
On-Chain-Structure (S)	32.5%	56.2%	69.6%	37.1

宏观与外部市场 (M)	6.6%	2.9%	0.8%	5.8
加密市场全局 (G)	19.6%	6.3%	10.1%	13.4
BTC 衍生品杠杆 (L)	9.9%	6.1%	2.4%	7.5
交易所流动 (E)	5.2%	18.2%	3.1%	15.1
挂钩套利 (P)	26.1%	1.6%	0.9%	25.2
链上结构 (S)	32.5%	56.2%	69.6%	37.1

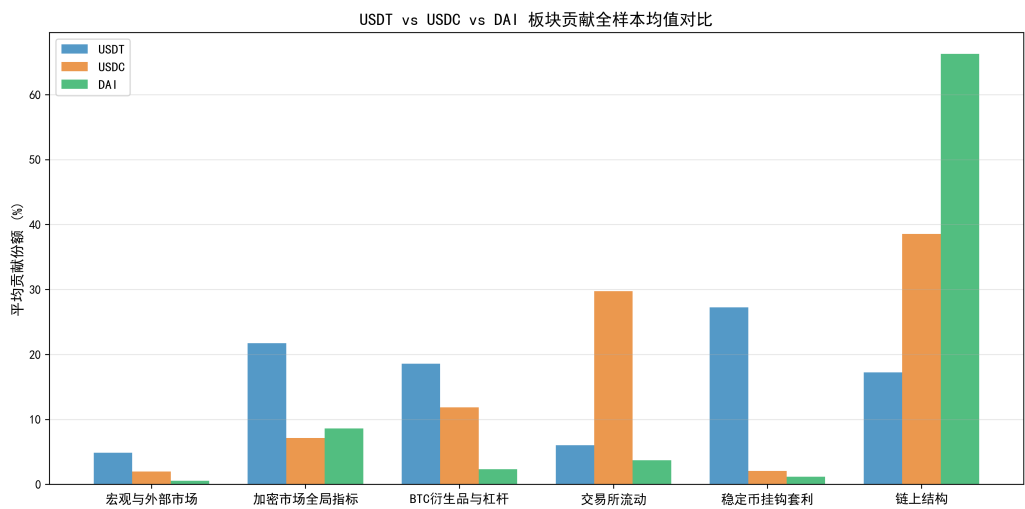


图 24: 三种稳定币全样本平均板块贡献对比 / Three-Stablecoin Full-Sample Average Panel Contribution Comparison

Figure 25 presents the radar chart comparison of panel contributions for the three stablecoins, visually demonstrating the differences in driver “profiles” across coins.

图 25展示了三种稳定币板块贡献的雷达图对比，直观呈现各币种的驱动因素“画像”差异。

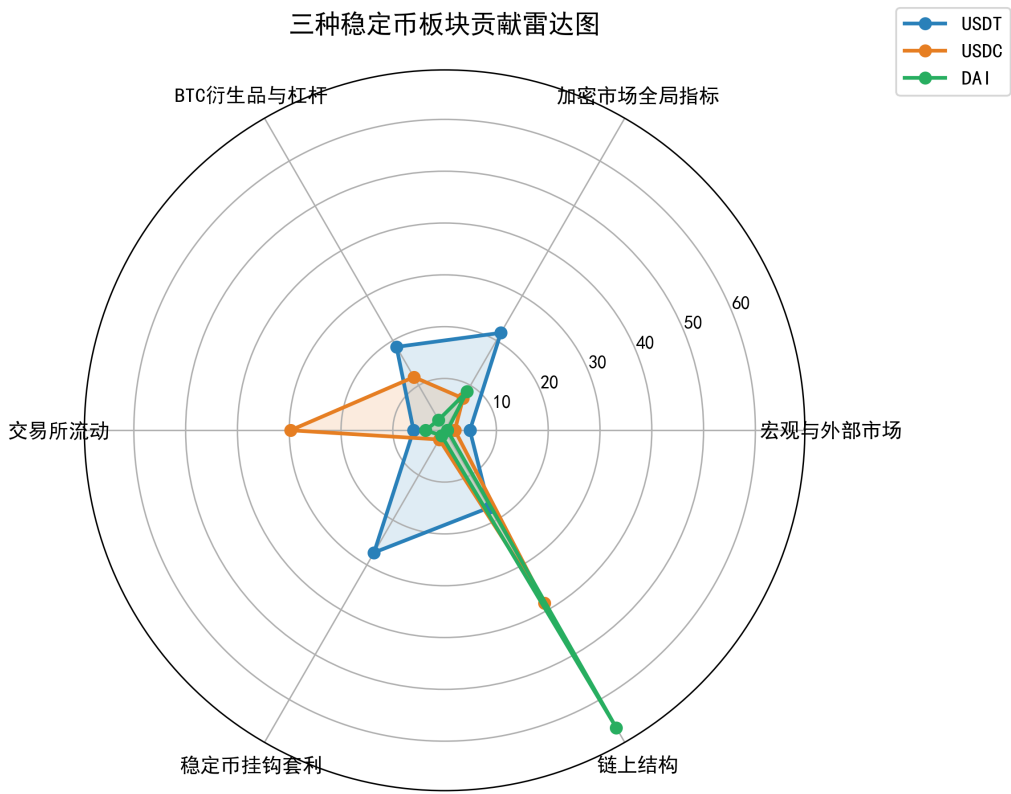


图 25: 三种稳定币板块贡献雷达图 / Three-Stablecoin Panel Contribution Radar Chart

Figure 26 presents the color-block chart of dominant panel rotation over time.

图 26展示了主导板块随时间切换的色块图。

10.2. 结构性差异的经济解释 / Economic Interpretation of Structural Differences

The three stablecoins exhibit extremely sharp structural differences, confirming the decisive impact of collateral models and regulatory environments on driving mechanisms. 三种稳定币呈现出极其鲜明的结构性差异，验证了抵押模式和监管环境对驱动机制的决定性影响。

On-chain structure dependence gradient (DAI > USDC > USDT). As a fully decentralized on-chain generated stablecoin, DAI has on-chain structure factors contributing as high as 69.6%, occupying absolute dominance; USDC, as a stablecoin issued by compliant institutions, has active large-value on-chain transfers and settlement,

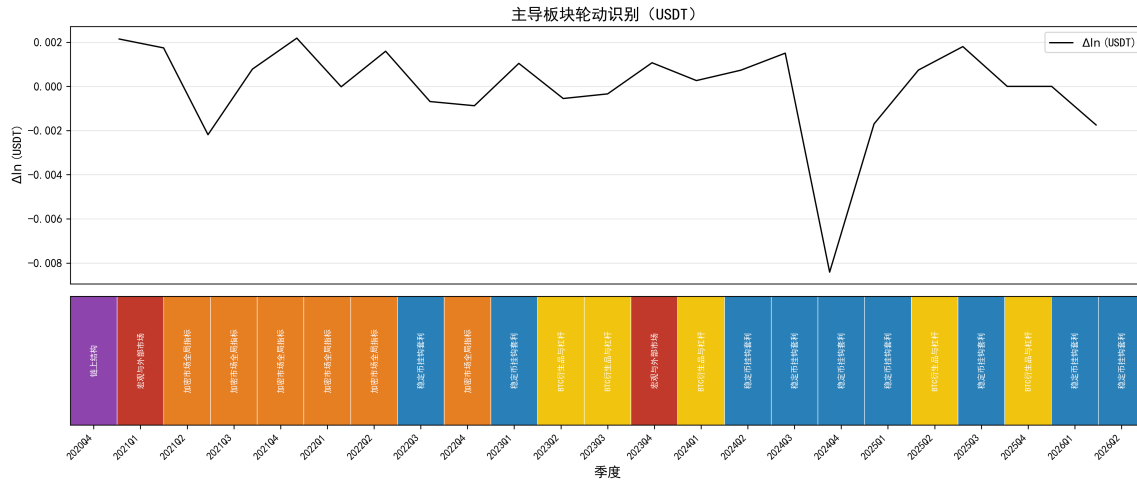


图 26: 主导板块切换色块图 / Dominant Panel Rotation Color Block Chart

with on-chain structure contributing 56.2%; USDT, due to its extensive use in OTC trading and internal exchange settlement, has relatively lower on-chain transparency, with on-chain structure contributing 32.5%. This gradient is fully consistent with differences in their issuance mechanisms.

链上结构依赖度梯度 (DAI > USDC > USDT)。DAI 作为完全去中心化的链上生成稳定币，链上结构因素贡献高达 69.6%，占据绝对主导；USDC 作为合规机构发行的稳定币，链上大额转账和结算活跃，链上结构贡献 56.2%；USDT 由于大量用于场外交易和交易所内部结算，链上透明性相对较低，链上结构贡献 32.5%。这一梯度完全符合三者发行机制上的差异。

Peg arbitrage sensitivity (USDT » USDC DAI)。USDT's peg arbitrage factor contributes as high as 26.1%, making it the second-largest driver, while USDC only contributes 1.6% and DAI only 0.9%. This directly reflects that USDT has long experienced relatively frequent price deviations from peg, with larger primary-secondary market arbitrage opportunities; while USDC and DAI have better peg stability, fewer arbitrage opportunities, and weak driving effects on issuance.

挂钩套利敏感度 (USDT » USDC DAI)。USDT 的挂钩套利因素贡献高达 26.1%，是第二大驱动因素，而 USDC 仅 1.6%，DAI 仅 0.9%。这直接反映了 USDT 长期存在较频繁的价格偏离锚定现象，一二级市场套利空间较大；而 USDC 和 DAI 锚定稳定性较好，套利机会较少，对发行量的驱动作用微弱。

Exchange flow importance (USDC » Others)。USDC's exchange flow factor contributes as high as 18.2%, far higher than USDT's 5.2% and DAI's 3.1%, reflecting that USDC is currently the primary on-ramp channel in the crypto market, with exchange

deposit demand directly driving USDC issuance; while USDT circulates more within exchanges, and DAI is mainly used within DeFi protocols, the direct driving effect of exchange flows is weaker.

交易所资金流重要性 (USDC » 其他)。 USDC 的交易所流动因素贡献高达 18.2%，远高于 USDT 的 5.2% 和 DAI 的 3.1%，反映了 USDC 是当前加密市场最主要的入金通道，交易所入金需求直接驱动 USDC 发行；而 USDT 更多在交易所内部流转，DAI 主要在 DeFi 协议内部使用，交易所资金流的直接驱动作用较弱。

Global crypto market sensitivity (USDT > DAI > USDC). USDT is most sensitive to global crypto market indicators (19.6%), reflecting USDT's high binding to crypto market conditions, bull market expansion directly drives USDT demand; DAI, as a crypto-native asset, also has some sensitivity to market conditions (10.1%); USDC is relatively least sensitive (6.3%), driven more by institutional settlement demand.

加密市场全局敏感度 (USDT > DAI > USDC)。 USDT 对加密市场全局指标最敏感 (19.6%)，反映了 USDT 与加密市场行情的高度绑定，牛市行情扩张直接带动 USDT 需求；DAI 作为加密生态原生资产，对市场行情也有一定敏感度 (10.1%)；USDC 相对最不敏感 (6.3%)，更多受机构结算需求驱动。

Macro external market sensitivity (USDT > USDC > DAI). Although macro factor contributions are generally not high for all three, USDT is relatively highest (6.6%), reflecting USDT's role as a capital flight channel for emerging markets, with macro factors such as dollar strength and RMB exchange rate having some impact on USDT issuance; DAI, being fully decentralized, is least sensitive to traditional macro factors (0.8%). **宏观外部市场敏感度 (USDT > USDC > DAI)。** 虽然三者宏观因素贡献普遍不高，但 USDT 相对最高 (6.6%)，反映了 USDT 作为新兴市场资本外逃通道的角色，美元强弱、人民币汇率等宏观因素对 USDT 发行有一定影响；DAI 完全去中心化，对传统宏观因素最不敏感 (0.8%)。

11. Robustness Checks

11.1. 窗口长度敏感性 / Window Length Sensitivity

We test three window lengths, 183 days (half-year), 365 days (baseline, 1 year), and 548 days (1.5 years). For window length W , the LASSO objective is

$$\hat{\beta}_W^{(\tau)} = \arg \min_{\beta} \left\{ \frac{1}{2W} \sum_{i=1}^W \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda_W \sum_{j=1}^p |\beta_j| \right\}, \quad (24)$$

where $W \in \{183, 365, 548\}$.

We then compare full-sample average panel contributions $\bar{C}_P^{W=183}$, $\bar{C}_P^{W=365}$, $\bar{C}_P^{W=548}$ across the three windows. If panel contribution rankings are broadly consistent across windows, core conclusions are insensitive to window length.

随后对比三种窗口下的全样本平均板块贡献 $\bar{C}_P^{W=183}$ 、 $\bar{C}_P^{W=365}$ 、 $\bar{C}_P^{W=548}$ 。若板块贡献排序在三种窗口下大体一致，则核心结论对窗口长度不敏感。

我们检验了三种窗口长度，183 天（半年）、365 天（基准，1 年）、548 天（1.5 年）。对窗口长度 W ，LASSO 目标函数为

Table 9 and Figure 27 show that on-chain structure and peg arbitrage are consistently the top two drivers, while derivative leverage and exchange flows always contribute less; core conclusions are insensitive to window length.

表 9与图 27显示，链上结构与挂钩套利始终是前两大驱动因素，衍生品杠杆和交易所流动始终贡献较小，核心结论对窗口长度不敏感。

表 9: Window Length Sensitivity Test (Six Categories M/G/L/E/P/S) | 窗口长度敏感性检验（六大类别 M/G/L/E/P/S）

Panel	183d	365d (baseline)	548d
Macro-External (M)	6.1%	6.6%	3.9%
Global-Crypto (G)	15.2%	19.6%	17.7%
Deriv-Leverage (L)	8.3%	9.9%	6.6%
Exchange-Flows (E)	3.1%	5.2%	4.2%
Peg-Arbitrage (P)	20.3%	26.1%	34.3%
On-Chain-Structure (S)	21.9%	32.5%	33.2%
Other	25.0%	0.0%	0.0%
Average R^2	0.20	0.37	0.38
Quarter Nodes	24	23	20

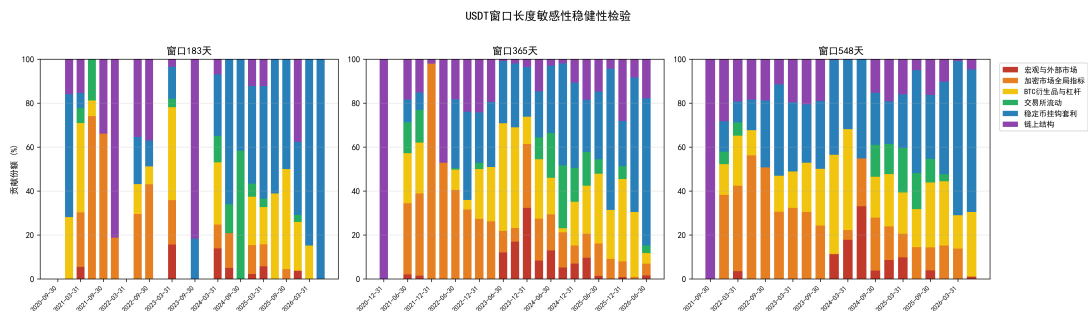


图 27: 窗口长度敏感性对比 / Window Length Sensitivity Comparison

板块	183 天	365 天（基准）	548 天
宏观与外部市场 (M)	6.1%	6.6%	3.9%
加密市场全局 (G)	15.2%	19.6%	17.7%
BTC 衍生品杠杆 (L)	8.3%	9.9%	6.6%
交易所流动 (E)	3.1%	5.2%	4.2%
挂钩套利 (P)	20.3%	26.1%	34.3%
链上结构 (S)	21.9%	32.5%	33.2%
平均 R^2	0.20	0.37	0.38
其他	25.0%	0.0%	0.0%
季度节点数	24	23	20

11.2. 目标变量替代 / *Dependent Variable Alternative*

Using $\ln(S_t)$ (log level) instead of $\Delta \ln(S_t)$ (log difference) as the dependent variable, we rerun rolling LASSO:

$$y_t^{\text{alt}} = \ln(S_t) \text{ (replacing } y_t = \Delta \ln(S_t) \text{)}. \quad (25)$$

以 $\ln(S_t)$ （对数水平值）替代 $\Delta \ln(S_t)$ （对数差分）作为被解释变量，重新运行滚动 LASSO。

The former measures the daily change rate and captures short-run marginal drivers; the latter measures the cumulative level and captures long-run stock drivers.

前者衡量日度变化率、捕捉短期边际驱动；后者衡量累积水平、捕捉长期存量驱动。

Under log levels, on-chain structure factor contribution rises to 45.1%, reflecting the long-run consistency between stock accumulation and on-chain fundamentals; peg arbitrage contribution drops substantially (2.5%) because level series contain more long-term trends and short-term arbitrage signals are relatively weaker. Overall, on-chain structure remains the most important driver, and core conclusions are robust.

对数水平值下，链上结构因素贡献上升至 45.1%，反映了存量积累与链上基本面的长期一致性；挂钩套利贡献大幅下降（2.5%），因为水平值序列包含更多长期趋势，短期套利信号相对减弱。总体而言，链上结构始终是最重要的驱动因素，核心结论稳健。

11.3. 子样本分析 / *Subsample Analysis*

We re-estimate after excluding two major depegging event periods:

$$\mathcal{D}_{\text{sub}} = \mathcal{D} \setminus \{\text{May 2022}\} \setminus \{\text{March 2023}\}. \quad (26)$$

排除理由。2022 年 5 月 UST 崩盘引发 USDT 短暂脱锚，2023 年 3 月 SVB 倒闭使 USDC 脱锚并波及 DAI。这两段窗口对三种稳定币均构成强外生冲击。

Rationale. the May 2022 UST collapse triggered a brief USDT depeg; the March 2023 SVB failure depegged USDC and 波及 DAI. Both windows constitute strong exogenous shocks for all three stablecoins. 排除两个重大脱锚事件期后重新估计。

After excluding depeg periods, peg arbitrage (Category P) contribution rises from 26.1% to 40.4%, on-chain structure (Category S) falls from 32.5% to 29.3%, and global crypto market (Category G) falls from 19.6% to 15.4%. This is because after excluding extreme events, the peg arbitrage mechanism becomes more prominent under normal market conditions, while event-driven global market shock impacts diminish. Panel importance rankings remain largely stable. 排除脱锚期后，挂钩套利（P 类）贡献从 26.1% 上升至 40.4%，链上结构（S 类）从 32.5% 下降至 29.3%，加密市场全局（G 类）从 19.6% 下降至 15.4%。这是因为排除极端事件后，正常市场状态下挂钩套利机制的作用更加突出，而事件驱动型的全局市场冲击影响降低。板块重要性排序基本稳定。

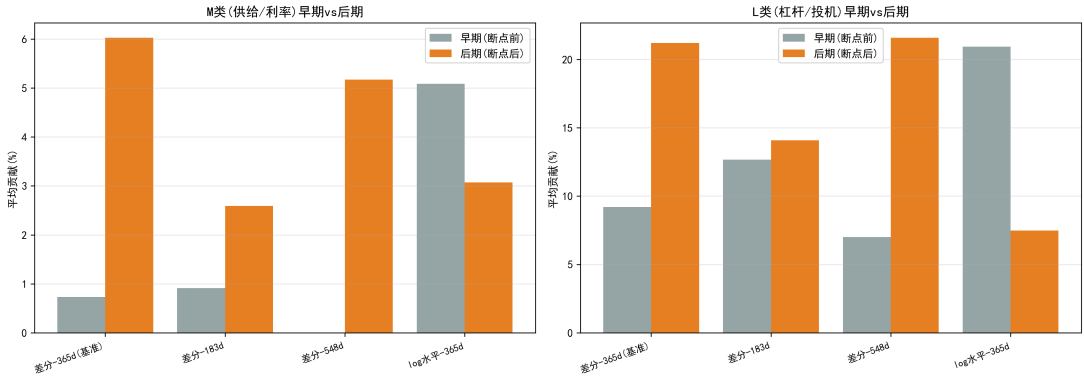
11.4. 时变规律稳健性 / *Time-Varying-Pattern Robustness*

前述稳健性检验主要验证“全样本平均板块排序”的稳定性。导师进一步要求，稳健性检验也须能证明主导因素的变化规律（而非仅平均排序）。为此，我们在四种设定（窗口 183 天、365 天（基准）、548 天、对数水平值）下，以 Quandt–Andrews 断点或加息起点为分界重做早期–后期板块贡献对比，检验“M 类（供给/利率）在断点后显著上升、L 类（杠杆/投机）在断点前显著更高”这一核心时变规律是否跨设定成立。

The preceding robustness checks mainly verify stability of the full-sample average panel ranking. The advisor further requires that robustness checks also substantiate the time-varying pattern of dominant factors (not just the average ranking). To that end, under four specifications, 183-day window, 365-day baseline, 548-day window, and log levels, we re-run the early-late panel comparison using the Quandt–Andrews break (or the rate-hike onset as fallback) and test whether the core time-varying pattern, “M (supply/rates) rises significantly post-break while L (leverage/speculation) is significantly higher pre-break,” holds across specifications.

Figure 28 shows that across all four specifications, Category-M contribution is uniformly higher in the late period and Category-L contribution uniformly higher in the early period, with most comparisons statistically significant under the Mann–Whitney U test. This direction of time variation is consistent across window lengths and dependent-variable specifications, indicating that the “demand-pull $\rightarrow$ supply-push” mechanism migration pattern is robust and does not hinge on a particular window choice or variable

图 28: 时变规律稳健性: 四设定下 M 类与 L 类的早期 vs 后期对比 / Time-Varying-Pattern Robustness across Four Specifications



specification.

图 28显示，在全部四种设定下，M 类贡献均表现为“后期高于早期”，L 类贡献均表现为“早期高于后期”，且多数对比在 Mann–Whitney U 检验下达到统计显著。这一时变方向在不同窗口长度与被解释变量形式下一致，表明“需求拉动 $\rightarrow$ 供给推动”的机制迁移规律具有稳健性，不依赖于具体的窗口选择或变量设定。

11.5. 稳健性检验汇总 / Robustness Summary

Table 10 summarizes full-sample average panel contributions and average R^2 across four specifications. Except for the significant decline in arbitrage factor contribution under the log-level specification, panel contribution rankings are largely consistent across other specifications, and core conclusions are robust. 表 10汇总了四种设定下的全样本平均板块贡献及平均 R^2 。除对数水平值设定下套利因素贡献显著下降外，其余设定下板块贡献排序基本一致，核心结论具有稳健性。

表 10: Robustness Check Summary (All Specifications)				
Panel	Window 183d	Window 365d	Log Level	Excl. Depeg
Macro-External (M)	6.1%	6.6%	5.3%	3.2%
Global-Crypto (G)	15.2%	19.6%	20.6%	15.4%
Deriv-Leverage (L)	8.3%	9.9%	6.2%	6.7%
Exchange-Flows (E)	3.1%	5.2%	2.9%	5.1%
Peg-Arbitrage (P)	20.3%	26.1%	2.5%	40.4%
On-Chain-Structure (S)	21.9%	32.5%	45.1%	29.3%
Other	25.0%	0.0%	17.4%	0.0%
Average R^2	0.20	0.37	0.51	0.32
Quarter Nodes	24	23	23	23

11.6. 全球敏感性汇总 / Global Sensitivity Summary

We summarize the three robustness checks as follows.

$$\boldsymbol{\rho} = (\bar{C}_P^{W=183}, \bar{C}_P^{W=365}, \bar{C}_P^{W=548}, \bar{C}_P^{\text{alt}}, \bar{C}_P^{\text{sub}})_{P \in \mathcal{P}}, \quad \text{Range}_P = \max(\boldsymbol{\rho}_P) - \min(\boldsymbol{\rho}_P). \quad (27)$$

对每个板块 P , 若 $\text{Range}_P < 5\%$ 则贡献估计高度稳健; 若 $\text{Range}_P \in [5\%, 10\%]$ 则适度稳健; 若 $\text{Range}_P > 10\%$ 则需谨慎解读。

For each panel P , $\text{Range}_P < 5\%$ indicates highly robust; $\text{Range}_P \in [5\%, 10\%]$ moderately robust; $\text{Range}_P > 10\%$ warrants cautious interpretation. 将上述三种稳健性检验结果汇总如下。

12. Conclusion

This paper constructs a multi-dimensional indicator framework comprising six categories (M/G/L/E/P/S) and 89 base variables spanning macroeconomic and external markets, global crypto market indicators, BTC derivative leverage, exchange flows, stablecoin depeg arbitrage, and on-chain structure, and employs a rolling LASSO model to conduct systematic time-varying analysis of the drivers behind USDT, USDC, and DAI issuance changes. 本文构建了一个包含六大类别 (M/G/L/E/P/S)、89 个基础变量的多维度分析框架, 覆盖宏观与外部市场、加密市场全局指标、BTC 衍生品杠杆、交易所资金流、稳定币脱锚套利、链上结构等维度, 利用滚动 LASSO 模型对 USDT、USDC 和 DAI 三种主流稳定币发行量变化率的驱动因素进行了系统的时变分析。

The main conclusions are as follows.

主要结论如下。

First, on-chain structure and peg arbitrage are the two most important drivers of stablecoin issuance across the full sample, with USDT average contributions of approximately 32.5% and 26.1% respectively; on-chain structure contributes even more for USDC (56.2%) and DAI (69.6%), reflecting the centrality of user-base fundamentals and arbitrage microstructure. **第一**, 全样本层面, 链上结构与挂钩套利是稳定币发行的前两大驱动因素, USDT 全样本平均贡献分别约为 32.5% 和 26.1%; USDC 和 DAI 的链上结构贡献更高, 分别达 56.2% 和 69.6%, 反映了用户基本面与套利微观结构的核心地位。

Second, and most central to this paper's contribution, the dominant driving mechanism exhibits a statistically identifiable structural break. Using Quandt–Andrews sup F and CUSUM tests on the supply-side relative-importance index R_t , we identify a highly significant break in USDT in May–June 2023 (sup $F = 412.78$ daily; quarterly Category-M break sup $F = 7.81$, $p = 0.03$), with Category-M (macro/rate) contribution surging

from 1.1% to 10.9%. The Fed hiking cycle began in March 2022, but Tether completed its reserve migration from commercial paper to T-Bills only during 2022Q4–2023Q2, so the supply-push mechanism becomes statistically detectable with a lag. Pre-break, the dominant drivers are crypto leverage and speculative demand (Categories L and G); post-break, supply-side short-rate factors (Category M) rise significantly in contribution while leverage factors recede. Mann–Whitney U tests confirm the early-late difference is statistically significant, providing direct empirical support for Hypothesis H2; the mechanism migrates from “crypto leverage and speculative demand pull” to “issuer safe-asset yield-seeking balance-sheet expansion.” Early-late panel comparisons under four alternative specifications (183/365/548-day windows, log levels) confirm the direction of this migration is robust. **第二**，也是本文最核心的贡献，稳定币发行的主导机制存在一个统计上可识别的结构性断点。通过对供给侧相对重要性指标 R_t 做 Quandt–Andrews $\sup F$ 与 CUSUM 检验，我们在 USDT 上识别到一个高度显著的断点，落在 **2023 年 5–6 月**（日度 $\sup F = 412.78$ ；季度 M 类贡献断点 $\sup F = 7.81$, $p = 0.03$ ），M 类（宏观/利率）贡献从 1.1% 跃升至 10.9%。美联储加息周期虽于 2022 年 3 月启动，但 Tether 完成储备从商业票据向短期国债的迁移发生在 2022Q4–2023Q2，供给推动机制的统计可识别时点存在约 1 年时滞。断点前，加密杠杆与投机需求（L 类、G 类）主导；断点后，供给侧短端利率因素（M 类）贡献显著上升而杠杆因素回落。Mann–Whitney U 检验确认早期后期差异统计显著，为假设 H2 提供了直接实证支持，主导机制从“加密杠杆与投机需求拉动”迁移至“发行商追逐安全资产收益的主动扩表推动”。四种替代设定（窗口 183/365/548 天、对数水平值）下的早期后期对比证实这一迁移方向稳健。

Third, transmission-channel co-movement analysis documents the chain “external environment change $\rightarrow$ demand change $\rightarrow$ channel migration.” Category-M contribution co-moves positively with the 3-month Treasury yield (rate hikes activate the reserve-yield channel), while Category-L contribution co-moves with BTC price (leverage demand dominates in bull runs). Event-vertical lines show that the first Fed hike, the LUNA/UST collapse, FTX bankruptcy, SVB/USDC depeg, BTC spot-ETF approval, and the first Fed cut align with turning points in dominant panels, confirming that exogenous macro and crypto events drive channel switching. The Rotation Intensity Index (RII) quantifies this restructuring, peaking in pivot quarters (mid-2022, Q1-2023) and staying low in calm periods. **第三**，传导渠道联动分析论证了“外部环境变化 $\rightarrow$ 需求变化 $\rightarrow$ 渠道迁移”链条，M 类贡献与 3 个月美债收益率正向联动（加息激活储备收益渠道），L 类贡献与 BTC 价格同步起落（牛市杠杆需求主导）；事件竖线显示首次加息、LUNA/UST 崩盘、FTX 破产、SVB/USDC 脱锚、BTC 现货 ETF 通过、首次降息等外部冲击与主导板块切换时点耦合，证实宏观与加密市场的外生事件驱动了渠道切换。轮动强度指数（RII）

量化了这一重构过程，在转折季度（2022 年中、2023Q1）达到峰值，平静期维持低位。

Fourth, the rolling-LASSO-selected sparse information set possesses genuine out-of-sample predictive power, in quarterly rolling OOS forecasts, LASSO outperforms random-walk, historical-mean, and full-variable OLS benchmarks in directional hit rate and OOS R^2 , confirming that the identified time-varying predictive structure is not an in-sample overfit. **第四**，滚动 LASSO 筛选出的稀疏信息集具备真实的样本外预测力，季度滚动 OOS 预测中，LASSO 在方向命中率与 OOS R^2 上优于随机游走、历史均值和全变量 OLS 基准，证实识别出的时变预测结构不是样本内过拟合。

Fifth, different frequencies are dominated by different factors, high-frequency fluctuations are dominated by derivative leverage and exchange flow shocks; mid-frequency fluctuations by peg arbitrage mechanisms; low-frequency trends by the joint action of peg arbitrage, on-chain structure fundamentals, and macro external markets. **第五**，不同频率由不同因素主导，高频波动由衍生品杠杆冲击和交易所流动主导；中频波动由挂钩套利机制主导；低频趋势由挂钩套利、链上结构基本面与宏观外部市场共同驱动。

Sixth, significant cross-issuer structural differences exist, DAI (crypto-collateralized, fully decentralized) is overwhelmingly dominated by on-chain structure (69.6%); USDC (US-regulated fiat-collateralized) is most sensitive to exchange flows (18.2%); USDT (offshore, lightly regulated) is most sensitive to peg arbitrage (26.1%) and global market factors (19.6%). These differences align with the mechanism-migration narrative, USDT, as the dominant offshore vehicle, is the primary carrier of supply-driven expansion during the high-rate environment. **第六**，跨发行商的结构差异显著，DAI（加密抵押、完全去中心化）由链上结构绝对主导（69.6%）；USDC（受美国监管法币抵押）对交易所资金流最敏感（18.2%）；USDT（离岸、监管宽松）对挂钩套利（26.1%）和全局市场因素（19.6%）最敏感。这些差异与机制迁移叙事一致，作为主导离岸载体，USDT 是高利率环境下供给驱动扩张的主要载体。

Seventh, robustness checks across window lengths (183d/365d/548d), dependent-variable specifications (log difference vs log level), subsample exclusions (depeg periods), and time-varying-pattern checks jointly confirm that both the full-sample average rankings and the early-to-late mechanism migration pattern remain qualitatively stable. **第七**，跨窗口长度（183/365/548 天）、目标变量形式（对数差分/对数水平值）、子样本排除（脱锚期）以及时变规律的稳健性检验共同表明，全样本平均排序与“早期需求拉动 → 后期供给推动”的机制迁移规律在定性层面均保持稳定。

Limitations and future research directions include the following. 本文的局限与未来研究方向包括以下几个方面。

(1) Data-wise, granular reserve-composition time series (e.g., T-bill share, CP share)

from issuer attestation reports can be integrated in future work to formally cross-validate the structural-break evidence with issuer balance-sheet data (the interface is reserved in our code); some variables (such as USDT premium rates) contain missing values; higher-quality regional demand data can further improve identification. (1) 数据方面，未来可整合发行商审计报告中的储备资产细分时序（如美债占比、商业票据占比），以发行商资产负债表数据正式交叉验证结构性断点证据（本文代码已预留接口）；部分变量（如 USDT 溢价率）存在缺失值，未来可补充更高质量的区域需求数据以提升识别力。

(2) Methodologically, online-learning frameworks and high-dimensional break-detection methods (e.g., adaptive LASSO, sparse changepoint models) can be introduced to achieve real-time factor capture and sharper break-date inference; nonlinear and interaction effects can be explored via kernel or tree-based extensions. (2) 方法方面，未来可引入在线学习（Online Learning）框架与高维断点检测方法（如 Adaptive LASSO、稀疏变点模型）以实现实时因素捕捉与更精确的断点推断；亦可通过核方法或树模型扩展探索非线性与交互效应。

(3) Application-wise, this paper’s findings, the identifiable structural break between demand-pull and supply-push regimes, the out-of-sample predictive content of the rolling-LASSO-selected information set, and the pronounced cross-issuer heterogeneity, can provide theoretical foundations for stablecoin risk monitoring and early-warning systems, and empirical evidence for regulators to design differentiated, regime-contingent regulatory policies (e.g., tighter scrutiny of reserve composition during supply-push regimes). (3) 应用方面，本文识别的“需求拉动-供给推动”体制间结构性断点、滚动 LASSO 信息集的样本外预测力、以及显著的跨发行商异质性，可为稳定币风险监测预警系统提供理论基础，并为监管机构设计差异化、体制依赖的监管政策（例如在供给推动体制下加强对储备构成的审查）提供实证依据。

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附录 A. Variable Definitions and Formulas / 变量定义与公式

This appendix provides complete variable definitions, transformation formulas, and detailed explanatory notes. The tables are presented in landscape for readability.

本附录提供完整的变量定义、变换公式和详细说明。为便于阅读，表格采用横向排版。

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ID	Variable Name	Symbol	Transformation	Formula & Explanation	Expected Sign
Panel A. Macro & External Markets (M, 9 variables)					
M1	3-Month Treasury Yield	us3m_treasury_yield	Level + 1st diff	Level: short-term Treasury yield directly corresponding to issuer reserve-asset returns; 1st diff: marginal change in interest rate environment.	± (level); + (1st diff: rate hike→reserve income↑→mint↑)
M2	SOFR–RRP Spread	sofr_rrp_spread	Level + 1st diff	Spread between Secured Overnight Financing Rate and Reverse Repo rate, measuring dollar short-end liquidity tightness; widening spread = liquidity tightening→redemption pressure.	− (spread widening→redemptions)
M3	Global Policy Risk Index	gprd_index	Level + 1st diff	Geopolitical Policy Risk Dashboard composite index based on news; level captures policy environment tension, 1st diff captures regulatory shocks.	+ (risk→safe-haven minting)
M4	Economic Policy Uncertainty Index	epu_index	Level + 1st diff	Baker-type economic policy uncertainty index; rising uncertainty→funds’ safe-haven demand.	+ (uncertainty↑→safe-haven mint↑)
M5	S&P 500 Index	sp500_index	Log-diff	US stock market performance reflecting global risk appetite; rising→risk appetite up.	+ (market up→risk appetite→mint↑)
M6	Gold Price	gold_price	Log-diff	Gold price reflecting safe-haven sentiment; rising→safe-haven demand up.	+ (gold up→safe-haven demand→mint↑)
M7	Bitcoin Price	bitcoin_price	Log-diff	Core pricing anchor of the crypto market; rising→stablecoin demand up, crash→redemption outflow. One of the core leading indicators.	+ (core variable)
M8	US Dollar Index	dxy	Level + 1st diff	Dollar strength affecting crypto valuations and emerging market capital flows; dollar strengthening→emerging market currency pressure.	±
M9	USD/CNH Exchange Rate	usdcnh_exchange_rate	Level + 1st diff	Real-time measure of RMB depreciation pressure; depreciation→capital outflow via USDT for safe haven→USDT minting demand rises (China cross-border capital channel).	+ (RMB depreciation→USDT minting)
Panel B. Global Crypto Market Indicators (G, 19 variables)					

Table A.11 (continued)

ID	Variable Name	Symbol	Transformation	Formula & Explanation	Expected Sign
G1	Total Market Cap (All Coins)	total_market_cap_all_coins	Log-diff	Equal-weight weekly-rebalanced total market cap across all coins; rising = risk appetite expansion.	+
G2	Total Market Cap (Large Cap)	total_market_cap_large_cap	Log-diff	Large-cap ($\geq \$1B$) total market cap, BTC+ETH dominated, most robust market scale proxy.	+
G3	Total Market Cap (Mid Cap)	total_market_cap_mid_cap	Log-diff	Mid-cap (\$100M–\$1B) total market cap, measuring altcoin season and mid-cap project expansion.	+
G4	Total Market Cap (Small Cap)	total_market_cap_small_cap	Log-diff	Small-cap ($< \$100M$) total market cap, most sensitive speculative heat proxy, bubble signal.	+ (reverse at extremes)
G5	Total Market Cap Index (All Coins)	total_market_cap_index_all_coins	Log-diff	Total market cap index (base=100), preferred in academic research.	+
G6	Total Open Interest (All Coins)	total_open_interest_all_coins	Log-diff	All-coin derivatives OI notional total; full-market leverage stock expanding = stablecoin margin total demand rising.	+ (core variable)
69	G7 Total Open Interest (Large Cap)	total_open_interest_large_cap	Log-diff	Large-cap OI, core leverage stock for BTC/ETH.	+
	G8 Total Open Interest (Mid Cap)	total_open_interest_mid_cap	Log-diff	Mid-cap OI, altcoin speculative leverage.	+ (reverse)
	G9 Total Open Interest (Small Cap)	total_open_interest_small_cap	Log-diff	Small-cap OI, extreme speculative leverage, surges often accompany tops.	+ (reverse)
	G10 Total OI Index (All Coins)	total_open_interest_index_all_coins	Log-diff	Total OI index (base=100), preferred in academic research.	+
	G11 Total Realized Cap (All Coins)	total_realized_cap_all_coins	Log-diff	Total market realized cap (cost basis at last on-chain move price); reflects real capital inflow.	+
	G12 Total Realized Cap (Large Cap)	total_realized_cap_large_cap	Log-diff	Large-cap realized cap, core asset cost basis.	+
	G13 Total Realized Cap (Mid Cap)	total_realized_cap_mid_cap	Log-diff	Mid-cap realized cap, mid-project capital accumulation.	+
	G14 Total Realized Cap (Small Cap)	total_realized_cap_small_cap	Log-diff	Small-cap realized cap, speculative capital accumulation.	+ (reverse)
	G15 Total Realized Cap Index (All Coins)	total_realized_cap_index_all_coins	Log-diff	Total realized cap index (base=100), preferred in academic research.	+

Table A.11 (continued)

ID	Variable Name	Symbol	Transformation	Formula & Explanation	Expected Sign
G16	Cross-Chain DeFi Total Value Locked	defi_tv1	Log-diff	$\Delta \ln(\text{TVL}_t)$; TVL rising = stablecoins locked in protocols = mint demand rising. Direct stablecoin consumption side in DeFi.	+ (core variable)
G17	Fear & Greed Index	fear_greed_index	Level	0–100 composite sentiment index; extreme fear = bottom signal, extreme greed = top signal.	+ (greed→mint↑)
G18	Spot Volume (All Assets)	spot_volume_all_assets	Log-diff	All-asset spot trading volume, overall market trading activity.	Synchronous / +
G19	USDT OTC Premium	usdt_otc_premium	Level	USDT/CNH $\div$ USD/CNH $- 1$; positive premium = OTC net demand (capital entering); negative premium (discount) = OTC capital flight, often leading supply decline. Reflects cross-border capital flow direction via USDT channel.	+ (positive premium→minting; core 1d signal)
Panel C. BTC Derivatives Leverage (L, 28 variables)					
L1	Perpetual Funding Rate (Top Exchanges)	futures_perpetual_funding_rate	Level	OI-weighted perpetual funding rate across top exchanges. Positive FR = longs pay shorts = long margin (mostly stablecoins) demand surging. Core 1-day signal.	+ (positive FR→mint↑)
L2	Perpetual Funding Rate V2	futures_perpetual_funding_rate_v2	Level	Optimized version (outlier removal + liquidity weighting + extreme-smoothing), preferred in academic research.	+ (positive FR→mint↑)
L3	Annualized Rolling Futures Basis (3M)	futures_annualized_rolling_basis_3m	Level	$\text{Basis}_t = \frac{1}{\tau} \left(\frac{F_{t,t+\tau}}{S_t} - 1 \right)$, $\tau = 3M$; positive basis = contango; cash-and-carry arbitrage needs stablecoin margin.	+ (positive basis→arbitrage→mint↑)
L4	Perpetual Reference Rate	perpetuals_reference_rate	Level	Fair price index for perpetuals, derivative pricing benchmark.	+
L5	Total Futures Open Interest	futures_open_interest	Log-diff	Total futures OI notional value across all exchanges; rising OI = leverage stock expanding = stablecoin margin demand up.	+ (core variable)
L6	Perpetual Futures OI	futures_open_interest_perpetual	Log-diff	Perpetual-only OI, 90%+ of total OI, main market capital scale.	+

Table A.11 (continued)

ID	Variable Name	Symbol	Transformation	Formula & Explanation	Expected Sign
L7	Cash-Margined Futures OI	cash_margined_futures_ open_interest	Log-diff	Stablecoin (USDT/USDC) margined futures OI. Core proxy directly measuring stablecoin stock demand in derivatives.	+ (cash-margin OI↑→mint↑)
L8	Crypto-Margined Futures OI	crypto_margined_futures_ open_interest	Log-diff	BTC/ETH-margined futures OI; declining share = market shifting toward stablecoin margin.	Indirect structural signal
L9	Perp Cash-Margin OI	futures_open_interest_ cash_margin_perpetual	Log-diff	Perpetual-only cash-margin OI (USDT-margin perps).	+
L10	Perp Crypto-Margin OI	futures_open_interest_ crypto_margin_perpetual	Log-diff	Perpetual-only crypto-margin OI (coin-margin perps).	Indirect structural signal
L11	Estimated Futures Leverage Ratio	futures_estimated_ leverage_ratio	Level + log-diff	Level: $Lev_t = OI_t/Bal_t$, market average leverage multiple; log-diff: $\Delta \ln(Lev_t)$, (de)leverage speed.	+ (leverage up→mint↑); too high→fragility↑
L12	Total Futures Liquidations	futures_total_liquidations_ total	log + rolling z-score	$z_t = (\ln(1 + Liq_t) - \mu_{t-30,t-1})/\sigma_{t-30,t-1}$; liquidations release margin→redemptions.	– (deleveraging→redemptions)
L13	Long Liquidations	futures_long_liquidations_ total	log + rolling z-score	As L12, long-side only (triggered by price drops).	– (longs squeezed→redemptions)
L14	Short Liquidations	futures_short_liquidations_ total	log + rolling z-score	As L12, short-side only (triggered by price rises). Short squeeze episodes usually accompany minting.	+ (shorts squeezed→rally→mint↑)
L15	Long-Liquidation Dominance	futures_long_liquidations_ dominance	Level	$Liq_dom_long_t = Liq_long_t/Liq_total_t$; > 50% = long squeeze dominant = selling pressure.	– (long-dominated→sell- off→redemptions)
L16	Position Openings	position_openings	Log-diff	Estimated new position size, new capital entering the market.	+
L17	Position Closures	position_closures	Log-diff	Estimated position closing size, existing capital leaving the market.	–
L18	Futures Volume	futures_volume	Log-diff	Total futures trading volume (taker basis), market activity level.	Synchronous / +
L19	Perpetual Volume	futures_volume_perpetual	Log-diff	Perpetual-only volume, main contract activity.	Synchronous / +
L20	Futures Buy Volume 24h	futures_buy_volume_24h	Log-diff	Active buy volume, long-side offensive willingness.	+
L21	Futures Sell Volume 24h	futures_sell_volume_24h	Log-diff	Active sell volume, short-side selling pressure.	–
L22	Perp Buy Volume 24h	futures_buy_volume_ perpetual_24h	Log-diff	Perpetual active buy volume, main long-short force.	+

Table A.11 (continued)

ID	Variable Name	Symbol	Transformation	Formula & Explanation	Expected Sign
L23	Perp Sell Volume 24h	futures_sell_volume__perpetual_24h	Log-diff	Perpetual active sell volume, main long-short force.	–
L24	Perp Volume Delta	futures_volume_delta__perpetual	Level	Perpetual active buy - sell net difference, net long-short direction (can be +/-).	+
L25	Options 24h Volume	options_volume_24h	Level	Options volume (premium-denominated), options market activity.	Synchronous
L26	Options Volume Put/Call Ratio	options_volume_put_call_ratio	Level	Put/Call volume ratio; > 1 bearish, sentiment contrarian indicator.	Contrarian
L27	Options OI Put/Call Ratio	options_open_interest__put_call_ratio	Level	Put/Call OI ratio; usually more stable than volume-based, reflecting medium-term expectations.	Contrarian
L28	1-Month Realized Volatility	realized_volatility_1_month	Level	1-month realized annualized volatility; surging→safe-haven minting / redemptions.	+
Panel D. Exchange Flows (E, 11 variables)					
E1	Total Exchange Inflow	exchange_inflow_volume__total	Log-diff	Total stablecoin inflow into exchanges; rising = pending selling / allocation.	+
E2	Total Exchange Outflow	exchange_outflow_volume__total	Log-diff	Total stablecoin outflow from exchanges; rising = withdrawal for hoarding / DeFi use = demand accumulation.	+
E3	Total Exchange Balance	exchange_balance_total	Log-diff	Exchange stablecoin balance; stock change reflects available margin scale.	+
E4	Depositing Addresses	depositing_addresses	Log-diff	Number of deposit addresses, inflow breadth.	+
E5	Withdrawing Addresses	withdrawing_addresses	Log-diff	Number of withdrawal addresses, outflow breadth.	+
E6	Exchange Reliance Ratio	exchange_reliance_ratio	Level	Stablecoin turnover dependence inside/outside exchanges.	Structural signal
E7	Exchange Reshuffling Ratio	exchange_reshuffling_ratio	Level	Exchange internal reallocation intensity.	Structural signal
E8	Exchange Aggregated Reliance Ratio	exchange_aggregated_reliance_ratio	Level	Aggregated cross-exchange capital dependence.	Structural signal
E9	Exchange Aggregated Reshuffling Ratio	exchange_aggregated_reshuffling_ratio	Level	Aggregated cross-exchange reallocation.	Structural signal

Table A.11 (continued)

ID	Variable Name	Symbol	Transformation	Formula & Explanation	Expected Sign
E10	In-House Exchange Volume	in_house_exchange_volume	Log-diff	Internal exchange transfer volume, internal allocation scale.	Synchronous
E11	Inter-Exchange Volume	inter_exchange_volume	Log-diff	Cross-exchange transfer volume, cross-exchange capital migration.	Leading / +
Panel E. Depeg & Arbitrage (P, 3 variables)					
P1	USDT Depeg Magnitude	usdt_depeg	Level	Positive = premium (market price > \$1), negative = discount (market price < \$1); larger deviation = bigger arbitrage space = directly triggers primary-market mint/redeem. Core 1-day signal.	+ (positive deviation→mint↑)
P2	USDC Depeg Magnitude	usdc_depeg	Level	USDC exchange price deviation from peg, fiat-reserve stablecoin arbitrage driver signal.	+ (positive deviation→mint↑)
P3	DAI Depeg Magnitude	dai_depeg	Level	DAI exchange price deviation from peg, crypto-collateralized stablecoin arbitrage signal, with larger fluctuation amplitude.	+ (positive deviation→mint↑)
Panel F. On-Chain Structure (S, 19 variables)					
S1	Total Addresses (Aggregated)	total_addresses_aggregated	Log-diff	Total stablecoin addresses, user base.	+
S2	New Addresses (Aggregated)	new_addresses_aggregated	Log-diff	New address count, breadth of new capital entry.	+
S3	Receiving Addresses (Aggregated)	receiving_addresses_aggregated	Log-diff	Receiving address count.	+
S4	Sending Addresses (Aggregated)	sending_addresses_aggregated	Log-diff	Sending address count.	+
S5	Active Addresses (Aggregated)	active_addresses_aggregated	Log-diff	Active address count, usage intensity.	+
S6	Transfer Count (Aggregated)	transfer_count_aggregated	Log-diff	Number of transfers, velocity of circulation.	Synchronous
S7	Transfer Rate (Aggregated)	transfer_rate_aggregated	Log-diff	Transfer rate.	Synchronous
S8	Transfer Volume Total (Aggregated)	transfer_volume_total_aggregated	Log-diff	Total transfer volume, on-chain settlement scale.	Synchronous
S9	Transfer Volume Mean (Aggregated)	transfer_volume_mean_aggregated	Log-diff	Average transfer size.	Synchronous

Table A.11 (continued)

ID	Variable Name	Symbol	Transformation	Formula & Explanation	Expected Sign
S10	Coin Days Destroyed	coin_days_destroyed_cdd	Level	Coin days destroyed; old coin movement = long-term holders start distributing.	– (distribution→sell pressure)
S11	Dormancy	dormancy	Level	Dormancy, average coin-days destroyed per unit volume.	Structural signal
S12	NVT Ratio	nvt_ratio	Level	Network value / transfer volume, valuation indicator.	Valuation signal
S13	Hodler Net Position Change	hodler_net_position_change	Level	Net position change of holders; positive=accumulating, negative=distributing.	+ (accumulation→mint↑)
S14	Liveliness	liveliness	Level	Liveliness, cumulative CDD / all-time CDD.	Structural signal
S15	Activity Retention Rate	activity_retention_rate	Level	Activity retention rate, user stickiness.	+
S16	Holder Accumulation Ratio	holder_accumulation_ratio	Level	Holder accumulation ratio.	+
S17	Holder Retention Rate	holder_retention_rate	Level	Holder retention rate.	+
S18	Gini Coefficient	gini_coefficient	Level	Gini coefficient; smaller = more decentralized, structural regime shift.	Regime signal
S19	Herfindahl Index	herfindahl_index	Level	Herfindahl index; smaller = more decentralized, concentration measure.	Regime signal

Notes: Stock variables (prices, OI, TVL, market cap, address counts, balances) enter as log-differences; ratio/rate/index variables (funding rates, basis, leverage ratios, P/C ratios, volatility, premiums, depeg magnitudes) enter as levels with within-window z-score standardization; spiky flow variables (liquidations, coin-days destroyed) enter as $\ln(1 + X)$ with rolling z-score standardization; regular flow variables (trading volume, transfer volume) enter as log-differences; macro interest rates and policy risk variables enter as both levels and first differences. Total feature pool: 89 base variables, 6 of which are dual-form macro indicators, yielding 95 features in total. After LASSO screening, each window retains 2–23 non-zero coefficient variables.

表 A.12: 完整变量定义与公式

ID	变量名	符号	变量形式	公式说明	预期方向
Panel A. 宏观与外部市场（M 类，9 个变量）					
M1	3 个月美债收益率	us3m_treasury_yield	原值 + 一阶差分	原值：短期美债收益率，直接对应发行商储备资产收益；一阶差分：利率环境的边际变化。	±（原值）；+（一阶差分：加息 → 储备收益 ↑→ 铸币 ↑）
M2	SOFR-逆回购利差	sofr_rrp_spread	原值 + 一阶差分	有担保隔夜融资利率与逆回购利率之差，衡量美元短端流动性松紧；利差走扩 = 流动性紧张 → 赎回压力上升。	−（利差走扩 → 赎回）
M3	全球政策风险指数	gprd_index	原值 + 一阶差分	基于新闻的地缘政治与政策风险综合指数；原值反映政策环境紧张度，一阶差分捕捉监管冲击。	+（风险 → 避险铸币）
M4	经济政策不确定性指数	epu_index	原值 + 一阶差分	Baker 型经济政策不确定性指数；不确定性上升 → 资金避险需求。	+（不确定性 ↑→ 避险铸币 ↑）
M5	标普 500 指数	sp500_index	对数差分	美股表现反映全球风险偏好；上涨 → 风险偏好上升。	+（市场涨 → 风险偏好 → 铸币 ↑）
M6	黄金价格	gold_price	对数差分	黄金反映避险情绪；上涨 → 避险需求上升。	+（黄金涨 → 避险需求 → 铸币 ↑）
M7	比特币价格	bitcoin_price	对数差分	加密市场核心定价锚；上涨 → 稳定币需求上升，崩盘 → 赎回出金。最核心领先指标之一。	+（核心变量）
M8	美元指数	dxy	原值 + 一阶差分	美元强弱影响加密估值与新兴市场资金流向；美元走强 → 新兴市场货币承压。	±
M9	美元兑离岸人民币汇率	usdcnh_exchange_rate	原值 + 一阶差分	人民币贬值压力实时度量；贬值 → 资金通过 USDT 出海避险 →USDT 铸币需求上升（中国资金跨境通道）。	+（人民币贬值 →USDT 铸币）
Panel B. 加密市场全局指标（G 类，19 个变量）					
G1	全币种总市值	total_market_cap_all_coins	对数差分	全币种总市值（等权周再平衡）；上升 → 风险偏好扩张。	+
G2	大盘股总市值	total_market_cap_large_cap	对数差分	大盘股总市值，BTC+ETH 主导，最稳健市场规模代理。	+
G3	中盘股总市值	total_market_cap_mid_cap	对数差分	中盘股总市值，衡量山寨季与中型项目扩张。	+
G4	小盘股总市值	total_market_cap_small_cap	对数差分	小盘股总市值，投机热度最敏感代理，泡沫信号。	+（反向极端）
G5	全市场总市值指数	total_market_cap_index_all_coins	对数差分	全市场总市值指数（base=100），学术研究首选。	+
G6	全币种未平仓量	total_open_interest_all_coins	对数差分	全币种衍生品未平仓名义价值总规模；全市场杠杆存量扩张 → 稳定币保证金总需求上升。	+（核心变量）
G7	大盘股未平仓量	total_open_interest_large_cap	对数差分	大盘股 OI，BTC/ETH 主导的核心杠杆存量。	+

表 A.12（续）

ID	变量名	符号	变量形式	公式说明	预期方向
G8	中盘股未平仓量	total_open_interest_mid_cap	对数差分	中盘股 OI，山寨币投机杠杆。	+（反向）
G9	小盘股未平仓量	total_open_interest_small_cap	对数差分	小盘股 OI，极端投机杠杆，飙升常伴随顶部。	+（反向）
G10	全市场未平仓量指数	total_open_interest_index_all_coins	对数差分	全市场 OI 指数（base=100），学术研究首选。	+
G11	全币种已实现市值	total_realized_cap_all_coins	对数差分	全市场已实现市值（按最后一次链上移动价格估值的总成本基础）；反映真实入场资金沉淀。	+
G12	大盘股已实现市值	total_realized_cap_large_cap	对数差分	大盘股已实现市值，核心资产成本基础。	+
G13	中盘股已实现市值	total_realized_cap_mid_cap	对数差分	中盘股已实现市值，中型项目资金沉淀。	+
G14	小盘股已实现市值	total_realized_cap_small_cap	对数差分	小盘股已实现市值，投机资金沉淀。	+（反向）
G15	全市场已实现市值指数	total_realized_cap_index_all_coins	对数差分	全市场已实现市值指数（base=100），学术研究首选。	+
G16	全链 DeFi 总锁仓价值	defi_tvl	对数差分	TVL 上升 → 稳定币大量锁入协议 → 直接拉动铸币需求；下降 → 触发赎回。稳定币在 DeFi 侧的直接消耗端。	+（核心变量）
G17	恐惧与贪婪指数	fear_greed_index	原值	0–100 综合情绪指数；极度恐惧 = 底部信号，极度贪婪 = 顶部信号。	+（贪婪 → 铸币 ↑）
G18	全资产现货成交量	spot_volume_all_assets	对数差分	全资产现货成交量，市场整体交易活跃度。	同步/+
G19	USDT OTC 溢价率	usdt_otc_premium	原值	USDT/CNH ÷ USD/CNH − 1；正溢价 = 场外净需求（资金入场）；负溢价（折价）= 场外资金出逃，往往领先发行量下降。反映以 USDT 为通道的跨境资金流方向。	+（正溢价 → 铸币；核心 1d 信号）
Panel C. BTC 衍生品杠杆（L 类，28 个变量）					
L1	永续资金费率（头部）	futures_perpetual_funding_rate	原值	头部交易所永续资金费率，正 = 多头付空头 = 多头保证金（多为 USDT）需求旺盛 → 领先铸币。最核心 1d 信号。	+
L2	永续资金费率 V2	futures_perpetual_funding_rate_v2	原值	优化版（异常值剔除 + 流动性加权 + 极端行情平滑），学术研究首选。	+
L3	期货年化滚动基差（3M）	futures_annualized_rolling_basis_3m	原值	正基差（contango）→ 套利者买现货 + 空期货需稳定币保证金。	+（正基差 → 套利 → 铸币 ↑）
L4	永续公允参考率	perpetuals_reference_rate	原值	永续公允价格指数，衍生品定价基准。	+
L5	全期货未平仓量	futures_open_interest	对数差分	全期货未平仓名义价值；上升 = 杠杆存量扩张 = 稳定币保证金需求增加。	+（核心变量）

表 A.12（续）

	ID	变量名	符号	变量形式	公式说明	预期方向
77	L6	永续未平仓量	futures_open_interest_perpetual	对数差分	仅永续 OI，占总量 90%+，主力市场资金规模。	+
	L7	现金保证金未平仓量	cash_margined_futures_open_interest	对数差分	以 USDT/USDC 为保证金的头寸，直接度量稳定币在衍生品中的存量需求。	+（核心变量）
	L8	加密保证金未平仓量	crypto_margined_futures_open_interest	对数差分	以 BTC 为保证金的头寸；占比下降 = 市场转向稳定币保证金。	间接信号
	L9	永续现金保证金未平仓量	futures_open_interest_cash_margin_perpetual	对数差分	永续口径下的现金保证金 OI（U 本位永续）。	+
	L10	永续加密保证金未平仓量	futures_open_interest_crypto_margin_perpetual	对数差分	永续口径下的加密保证金 OI（币本位永续）。	间接信号
	L11	期货估算杠杆率	futures_estimated_leverage_ratio	原值 + 对数差分	原值： $Lev_t = OI_t / Bal_t$ ，市场平均杠杆倍数；对数差分：加/去杠杆速度。	+（加杠杆 → 铸币 ↑）；过高 → 脆弱性 ↑
	L12	期货总清算量	futures_total_liquidations_total	log + 滚动 z-score	多空合计强平量；大规模清算 = 去杠杆 = 释放保证金 → 可能触发赎回。	-（去杠杆 → 赎回）
	L13	多头清算量	futures_long_liquidations_total	log + 滚动 z-score	多头强平量，价格快跌时飙升。	-
	L14	空头清算量	futures_short_liquidations_total	log + 滚动 z-score	空头强平量，价格快涨时（轧空）飙升。	+（轧空 → 行情 ↑ → 铸币 ↑）
	L15	多头清算占比	futures_long_liquidations_dominance	原值	$Liq_dom_long_t = Liq_long_t / Liq_total_t$ ；>50% 多头爆仓为主（暴跌）。	-
	L16	新开仓量	position_openings	对数差分	估算新开仓位规模，新资金入场。	+
	L17	平仓量	position_closures	对数差分	估算平仓规模，存量资金离场。	-
	L18	期货成交量	futures_volume	对数差分	全期货总成交量（taker 口径），市场活跃度。	同步/+
	L19	永续成交量	futures_volume_perpetual	对数差分	仅永续成交量，主力合约活跃度。	同步/+
	L20	期货 24h 主动买入量	futures_buy_volume_24h	对数差分	主动买入量，多头进攻意愿。	+
	L21	期货 24h 主动卖出量	futures_sell_volume_24h	对数差分	主动卖出量，空头打压意愿。	-
	L22	永续 24h 主动买入量	futures_buy_volume_perpetual_24h	对数差分	永续主动买入量，主力多空力量。	+
	L23	永续 24h 主动卖出量	futures_sell_volume_perpetual_24h	对数差分	永续主动卖出量，主力多空力量。	-
	L24	永续成交量净差额	futures_volume_delta_perpetual	原值	永续主动买入-卖出净差额，多空净方向（可正可负）。	+
	L25	期权 24h 成交量	options_volume_24h	原值	期权成交量（权利金计价），期权市场活跃度。	同步
	L26	期权成交量 P/C 比	options_volume_put_call_ratio	原值	看跌/看涨成交量比；>1 偏空，情绪反向指标。	反向

表 A.12（续）

ID	变量名	符号	变量形式	公式说明	预期方向
L27	期权未平仓量 P/C 比	options_open_interest_ put_call_ratio	原值	看跌/看涨持仓比，比成交量更稳定。	反向
L28	1 个月已实现波动率	realized_volatility_1_ month	原值	1 个月已实现年化波动率；飙升 → 避险铸币/赎回。	+
Panel D. 交易所资金流（E 类，11 个变量）					
E1	交易所流入总量	exchange_inflow_volume_ total	对数差分	稳定币流入交易所总量；上升 → 待卖出/配资。	+
E2	交易所流出总量	exchange_outflow_volume_ total	对数差分	稳定币流出交易所总量；上升 → 提币囤积/DeFi 使用 → 需求沉淀。	+
E3	交易所总余额	exchange_balance_total	对数差分	交易所稳定币余额；存量变化反映待动用保证金规模。	+
E4	充值地址数	depositing_addresses	对数差分	充值地址数，流入广度。	+
E5	提现地址数	withdrawing_addresses	对数差分	提现地址数，流出广度。	+
E6	交易所依赖比率	exchange_reliance_ratio	原值	稳定币在交易所内外的周转依赖度。	结构信号
E7	交易所重排比率	exchange_reshuffling_ratio	原值	交易所重排比率，内部资金调拨强度。	结构信号
E8	聚合依赖比率	exchange_aggregated_ reliance_ratio	原值	聚合依赖比率，跨交易所资金依赖。	结构信号
E9	聚合重排比率	exchange_aggregated_ reshuffling_ratio	原值	聚合重排比率。	结构信号
E10	交易所内部转账量	in_house_exchange_volume	对数差分	交易所内部转账量，内部调拨规模。	同步
E11	交易所间转账量	inter_exchange_volume	对数差分	交易所间转账量，跨所资金迁移 → 领先流向。	领先/+
Panel E. 脱锚与套利（P 类，3 个变量）					
P1	USDT 脱钩幅度	usdt_depeg	原值	正值 = 溢价（市场价 >1 美元），负值 = 折价（市场价 <1 美元）；偏离越大 → 套利空间越大 → 直接触发一级市场铸币/赎回。核心 1d 信号。	+（正偏离 → 铸币 ↑）
P2	USDC 脱钩幅度	usdc_depeg	原值	USDC 交易所价格偏离锚定幅度，法币储备型稳定币套利驱动信号。	+（正偏离 → 铸币 ↑）
P3	DAI 脱钩幅度	dai_depeg	原值	DAI 交易所价格偏离锚定幅度，加密抵押型（CDP 模式）套利信号，波动幅度更大。	+（正偏离 → 铸币 ↑）
Panel F. 链上结构（S 类，19 个变量）					
S1	总地址数（聚合）	total_addresses_aggregated	对数差分	稳定币地址总数，用户基数。	+
S2	新增地址数（聚合）	new_addresses_aggregated	对数差分	新增地址数，新资金入场广度。	+
S3	接收地址数（聚合）	receiving_addresses_ aggregated	对数差分	接收地址数。	+

表 A.12（续）

79	ID	变量名	符号	变量形式	公式说明	预期方向
	S4	发送地址数（聚合）	sending_addresses__aggregated	对数差分	发送地址数。	+
	S5	活跃地址数（聚合）	active_addresses__aggregated	对数差分	活跃地址数，使用热度。	+
	S6	转账笔数（聚合）	transfer_count_aggregated	对数差分	转账笔数，流通速度。	同步
	S7	转账速率（聚合）	transfer_rate_aggregated	对数差分	转账速率。	同步
	S8	转账总量（聚合）	transfer_volume_total__aggregated	对数差分	转账总量，链上结算规模。	同步
	S9	转账均值（聚合）	transfer_volume_mean__aggregated	对数差分	转账均值。	同步
	S10	币天销毁数	coin_days_destroyed_cdd	原值	币天销毁数；老币移动 = 长期持有者开始分配，脉冲型极端值主导。	-（分配 → 抛压 → 赎回）
	S11	休眠度	dormancy	原值	休眠度，平均币天销毁。	结构信号
	S12	NVT 比率	nvt_ratio	原值	网络价值/转账量，估值指标。	估值信号
	S13	持有者净头寸变化	hodler_net_position__change	原值	正 = 累积，负 = 分配（可正可负）。	+（累积 → 铸币 ↑）
	S14	活跃度	liveliness	原值	活跃度，累积币天销毁/累积币天。	结构信号
	S15	活动留存率	activity_retention_rate	原值	活动留存率，用户粘性。	+
	S16	持有者累积比率	holder_accumulation_ratio	原值	持有者累积比率。	+
	S17	持有者留存率	holder_retention_rate	原值	持有者留存率。	+
	S18	基尼系数	gini_coefficient	原值	越小越分散，结构体制切换。	体制信号
	S19	赫芬达尔指数	herfindahl_index	原值	越小越去中心化，集中度度量。	体制信号

注：存量变量（价格、OI、TVL、市值、地址数、余额等）均以对数差分进入回归；比率/利率/指数类变量（资金费率、基差、杠杆率、P/C 比、波动率、溢价率、脱钩幅度）采用原值并进行窗口内 z-score 标准化；脉冲型流量变量（清算量、币天销毁等）采用 $\ln(1 + X)$ 变换后滚动 z-score 标准化；常规流量变量（成交量、转账量等）采用对数差分；宏观利率与政策风险变量同时保留原值和一阶差分。特征池总计：89 个基础变量，其中 6 个宏观变量以原值和一阶差分双形式进入，合计 95 个特征。经 LASSO 筛选后，每个窗口通常保留 2–23 个非零系数变量。

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